

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\E-0901.pvdb

Date of Analysis : Sep 9, 2026 10:20am

PV Elite 28, January 2026



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FileName : E-0901

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Vessel Design Summary:

Step: 33 10:20am Sep 9,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 619.150 mm ID
Vessel Design Length, Tangent to Tangent 2969.78 mm
Specified Datum Line Distance 50.80 mm
Shell Side Design Temperature 82 °C
Channel Side Design Temperature 230 °C
Shell Side Design Pressure 4150.000 KPa.
Channel Side Design Pressure 3200.001 KPa.
Shop Shell Side Test Pressure 6630.006 KPa.
Field Shell Side Test Pressure 5395.000 KPa.
Shop Channel Side Test Pressure 6630.006 KPa.
Field Channel Side Test Pressure 5395.000 KPa.
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-765 II	...	...	K03047	
Flange	SA-350 LF2	2	...	K03011	+IT
Nozzle	SA-333 6	...	...	K03006	+IT
Nozzle	SA-765 II	...	...	K03047	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-334 1	...	...	K03008	+IT
Tubesheet	SA-765 II	...	...	K03047	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Dish End	3206.099	103.42	4006.000	3.0000	No	No
LHS Ch Shell	3206.099	103.42	4750.000	3.0000	N/A	No
LHS Ch Flange	3206.862	103.42	4749.000	3.0000	N/A	No
LHS Shell Flg	4156.861	0.00	5327.000	3.0000	N/A	No
Main Shell	4156.097	0.00	5593.000	3.0000	N/A	No
RHS Dish End	4156.099	0.00	4228.000	3.0000	No	No
Tube Inlet	4150.000	0.00	7443.000	3.0000	N/A	No
Shell Inlet flage	4150.000	0.00	8729.000	3.0000	N/A	No

Liquid Level: 700.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000



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Vessel Design Summary:

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Sep 9,2026

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	0.0	50.8	14.0	12.0	10.2	4.7	1.00	1.00
Cylinder	700.0	700.0	14.0	14.0	10.4	5.3	1.00	1.00
Body Flg	826.2	123.0	103.0	95.0	88.8	88.8	1.00	1.00
Body Flg	1038.2	122.0	93.0	90.0	82.8	82.8	1.00	1.00
Cylinder	2868.2	1830.0	16.0	16.0	12.6	...	1.00	1.00
Ellipse	2919.0	50.8	16.0	12.5	12.3	...	1.00	1.00
Body Flg	350.0	0.0	48.0	31.8	29.1	29.1	1.00	1.00
Body Flg	1860.0	0.0	48.0	47.8	44.6	44.6	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube inlet	WRC-297	0.696	Shell	Passed
N1-Tube Outlet	WRC-297	0.983	Nozzle	Passed
N3-Shell inlet	WRC-297	0.981	Shell	Passed
N4-Shell outle	WRC-297	0.981	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	200.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	585.000 mm
Wear Pad Width	315.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	370.000 mm

Baseplate Length	715.000 mm
Baseplate Thickness	12.000 mm
Baseplate Width	285.000 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	10.000 mm
Web Thickness	10.000 mm
Height of Center Web	275.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



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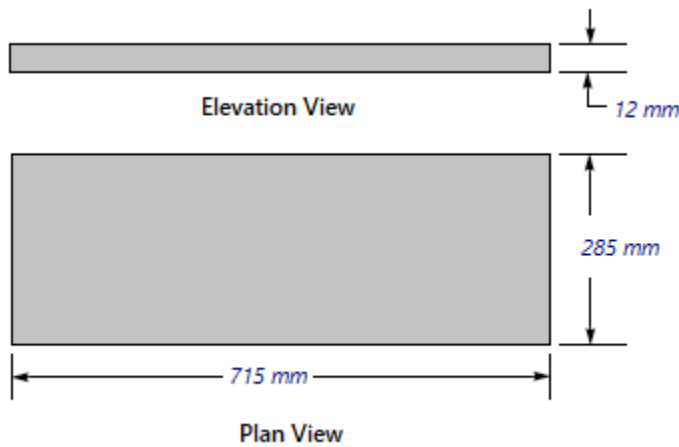
FileName : E-0901

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Vessel Design Summary:

Step: 33 10:20am

Sep 9, 2026



Maximum Tensile Bolt Load 375. Kg

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	5875.56 Kg
Maximum Transverse Saddle Shear Load	375.45 Kg
Maximum Longitudinal Saddle Shear Load	283.04 Kg

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	5022.97 Kg
Maximum Transverse Saddle Shear Load	187.72 Kg
Maximum Longitudinal Saddle Shear Load	205.31 Kg

Weights:

Fabricated - Bare w/o removable internals	3875.4 kg.
Shop Test - Fabricated + water (full)	4749.2 kg.
Shipping - Fab. + rem. intls.+ shipping app.	3875.4 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	3875.4 kg.
Empty - Fab. + intls. + details + wghts.	4102.5 kg.
Operating - Empty + operating liquid (no ca)	4969.2 kg.
Field Test - Empty weight + test medium (full)	4896.9 kg.

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Nozzle Summary:

Step: 31 10:20am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-Tube inlet	4052	OK	...	OK	8.73	OK	Passed
N1-Tube Outlet	4750	OK	...	OK	10.35	OK	Passed
N3-Shell inlet	5216	...	...	OK	11.10	OK	Passed
N4-Shell outlet	5210	...	...	OK	11.10	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 5210 Nozzle : N4-Shell outlet [Shellside]
 Minimum MAWP Nozzles : 4052 Nozzle : N1-Tube inlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	N1-Tube inlet	400.800	180.000	216.520
20	N1-Tube Outlet	450.800	0.000	484.960
50	N3-Shell inlet	2003.975	0.000	507.200
50	N4-Shell outlet	2003.975	180.000	507.200

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 30 10:20am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1-Tube inlet	102 mm	Actual	None	122.3	10.000	150.00	14.00	220	...
N1-Tube Outlet	236 mm	Actual	WNF	260.5	12.000	450.00	14.00	192	300
N3-Shell inlet	247 mm	Actual	None	273.0	12.700	...	...	319	...
N4-Shell outlet	247 mm	Actual	WNF	273.0	12.700	...	...	219	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1-Tube inl	SA-333 6	14.000	10.000	12.000	14.000	...
N1-Tube Out	SA-333 6	14.000	10.000	12.000	14.000	...
N3-Shell in	SA-765 II	14.000	10.000	...	...	...
N4-Shell ou	SA-765 II	16.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1-Tube inlet	350.000	180.0	200.00	0.00	LHS Ch Shell
N1-Tube Outlet	400.000	0.0	150.00	0.00	LHS Ch Shell
N3-Shell inlet	1953.175	0.0	250.00	0.00	Main Shell
N4-Shell outlet	1953.175	180.0	150.00	0.00	Main Shell

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Bill of Material:

Step: 29 10:20am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 14.0mm THK X 619.1mm ID X 50.8mm	SA-516 70
1	CYLINDER: 14.0mm THK X 619.1mm ID X 700.0mm	SA-516 70
1	WELD NECK FLANGE: 103.0mm THK X 800.0mm OD	SA-765 II
1	WELD NECK FLANGE: 93.0mm THK X 800.0mm OD	SA-765 II
1	CYLINDER: 16.0mm THK X 619.1mm ID X 1830.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 16.0mm THK X 619.1mm ID X 50.8mm	SA-516 70
1	WELD NECK FLANGE: 48.0mm THK X 254.0mm OD	SA-350 LF2
1	WELD NECK FLANGE: 48.0mm THK X 444.5mm OD	SA-350 LF2
2	INSULATION: 50mm X 80mm THK	Cold Insulation
1	INSULATION: 700mm X 80mm THK	Cold Insulation
1	INSULATION: 124mm X 80mm THK	Cold Insulation
3	BAFFLES: 614.0mm DIA X 8.0mm THK	SA-516 70
6	TIE RODS: 14mm DIA X 720mm	SA-193 B7M
1	TUBESHEET: 800.0mm X 90.0mm THK	SA-765 II
302	TUBES: 1830.0mm X 19.0mm DIA X 2.3mm THK	SA-334 1
1	INSULATION: 127mm X 80mm THK	Cold Insulation
2	SADDLE: 200.0mm X 120 DEG	SA-516 70
1	INSULATION: 1830mm X 80mm THK	Cold Insulation
1	GASKET: 677.2mm OD X 651.2mm ID	...
40	BODY FLANGE BOLTS: 25.4mm DIA	SA-193 B7
80	NUTS FOR BODY FLANGE BOLTS: 25.4mm DIA	...
1	GASKET: 149.4mm OD X 127.0mm ID	...
8	BODY FLANGE BOLTS: 19.0mm DIA	SA-193 B7
24	NUTS FOR BODY FLANGE BOLTS: 19.0mm DIA	...
1	NAMEPLATE	...



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MDMT Summary:

Step: 32 10:20am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
LHS Shell Flg[11]	B	-14	-22	-29	0.854	16.000	1.00	No
Main Shell [8]	B	-14	-22	-29	0.851	16.000	1.00	No
LH floating s[24]	B	-29	-37	-29	0.851	10.000	1.00	No
Sdl 2 Fr50 [24]	B	-29	-37	-29	0.851	10.000	1.00	No
N3-Shell inlet[1]	B	-14	-29	-29	0.736	16.000	1.00	No
N4-Shell outle[1]	B	-14	-29	-29	0.738	16.000	1.00	No
Nozzle Flg [5]	A	-18	-28					
Tubesheet: SS[13]	B	-4	-27	-29	0.593	22.500	1.00	No
Warmest MDMT:		-4	-22					
LHS Ch Flange[11]	B	-18	-34	-29	0.707	14.000	1.00	No
Tube Inlet [11]	!	-18	-104	-29	0.273	6.020	1.00	No
Shell Inlet f[11]	!	-18	-18	-29	0.383	12.700	1.00	No
LHS Dish End [10]	B	-23	-25	-29	0.964	12.000	1.00	No
LHS Dish End [7]	B	-18	-29	-29	0.810	14.000	1.00	No
LHS Ch Shell [8]	B	-18	-29	-29	0.810	14.000	1.00	No
RHS Dish End [10]	B	-22	-27	-29	0.913	12.500	1.00	No
RHS Dish End [7]	B	-14	-32	-29	0.686	16.000	1.00	No
N1-Tube inlet [1]	B	-18	-37	-29	0.670	14.000	1.00	No
N1-Tube Outlet[1]	B	-18	-37	-29	0.668	14.000	1.00	No
Nozzle Flg [5]	A	-18	-38					
Tubesheet: CS[14]	B	-4	-27	-29	0.593	22.500	1.00	No
Bolting [21]		-48						
Bolting [21]		-48						
Warmest MDMT:		-4	-18					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-28.0	-45.0	Fail
Channel/Tube	-29.0	-45.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



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MDMT Summary:

Step: 32 10:20am Sep 9, 2026

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 10:20am Sep 9, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning: LHS Ch Flange and LHS Shell Flg

There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.



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Warnings and Errors:

Step: 0 10:20am Sep 9, 2026

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Input Echo:

Step: 1 10:20am Sep 9,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	4150	KPa.
Channel Side Design Pressure	3200	KPa.
Shell Side Design Temperature	82.0	°C
Channel Side Design Temperature	230.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-45.0	°C
MDMT (CET), Tube Side	-45.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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Input Echo:

Step: 1 10:20am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	349.92	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	0	mm
Distance Upwind of Crest	0	mm
Distance from Crest to the Vessel	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor	1.000	
Importance Factor	1.250	
Table Value Fa	1.600	
Table Value Fv	2.400	
Max. Mapped Res. Acceleration	0.147	[Ss]
Max. Eff. Ground Acceleration	0.065	[S]
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio	0.000	z/h
Amplification Factor	0.000	Ap
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Moment Reduction Factor Tau	1.000	

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Input Echo:

Step: 1 10:20am Sep 9, 2026

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 No

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/8	
Element From Node	10	
Element To Node	20	
Element Type	Elliptical	
Description	LHS Dish End	
Distance "FROM" to "TO"	50.8	mm
Inside Diameter	619.15	mm
Element Thickness	12	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	3200	KPa.
Design Temperature Internal Pressure	230	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	230	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	220	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	619.15	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	10	
Detail Type	Insulation	



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Input Echo:

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Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	50.8 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm ³

Element#	2/8
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	LHS Ch Shell
Distance "FROM" to "TO"	700 mm
Inside Diameter	619.15 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	3200 KPa.
Design Temperature Internal Pressure	230 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	230 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	619.15 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	700 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm ³

Element From Node	20
Detail Type	Nozzle
Detail ID	N1-Tube inlet
Dist. from "FROM" Node / Offset dist	350 mm
Nozzle Diameter	102.3 mm
Nozzle Schedule	40
Nozzle Class	None
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	6.4931 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-333 6 +IT

Element From Node	20
Detail Type	Nozzle
Detail ID	N1-Tube Outlet
Dist. from "FROM" Node / Offset dist	400 mm
Nozzle Diameter	236.5 mm
Nozzle Schedule	60
Nozzle Class	300
Layout Angle	0.0



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Input Echo:

Step: 1 10:20am Sep 9, 2026

Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	73.91 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 6 +IT

Element From Node	20
Detail Type	Weight
Detail ID	Partition plate
Dist. from "FROM" Node / Offset dist	350 mm
Miscellaneous Weight	200 Kg
Offset from Element Centerline	0 mm

Element#	3/8
Element From Node	30
Element To Node	40
Element Type	Flange
Description	LHS Ch Flange
Distance "FROM" to "TO"	123 mm
Flange Inside Diameter	619.15 mm
Element Thickness	95 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	103 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	3200 KPa.
Design Temperature Internal Pressure	230 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	230 °C
Effective Diameter Multiplier	1.2
Material Name	SA-765 II
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	136.8 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	208 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03047
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	700 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	30
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	124 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm ³

Element#	4/8
Element From Node	40
Element To Node	50
Element Type	Flange



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Input Echo:

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Description	LHS Shell Flg	
Distance "FROM" to "TO"	122	mm
Flange Inside Diameter	619.15	mm
Element Thickness	90	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	93	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4150	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-765 II	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	700	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	127	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm ³

Element#	5/8	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	1830	mm
Inside Diameter	619.15	mm
Element Thickness	16	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4150	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	242.6	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	50	



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Input Echo:

Step: 1 10:20am Sep 9,2026

Detail Type	Saddle
Detail ID	LH floating sad
Dist. from "FROM" Node / Offset dist	370 mm
Width of Saddle	200 mm
Height of Saddle at Bottom	585 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	315 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.3000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm
Element From Node	50
Detail Type	Saddle
Detail ID	Sd1 2 Fr50
Dist. from "FROM" Node / Offset dist	1460 mm
Width of Saddle	200 mm
Height of Saddle at Bottom	585 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	315 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	619 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	50
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1830 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm ³
Element From Node	50
Detail Type	Nozzle
Detail ID	N3-Shell inlet
Dist. from "FROM" Node / Offset dist	915 mm
Nozzle Diameter	247.6 mm
Nozzle Schedule	60
Nozzle Class	None
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	37.599 Kg
Grade of Attached Flange	None
Nozzle Matl	SA-765 II
Element From Node	50
Detail Type	Nozzle



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Input Echo:

Step: 1 10:20am Sep 9,2026

Detail ID	N4-Shell outlet	
Dist. from "FROM" Node / Offset dist	915	mm
Nozzle Diameter	247.6	mm
Nozzle Schedule	60	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	78.916	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 II	

Element From Node	50	
Detail Type	Weight	
Detail ID	Fireproofing we	
Dist. from "FROM" Node / Offset dist	370	mm
Miscellaneous Weight	500	Kgf
Offset from Element Centerline	585	mm

Element From Node	50	
Detail Type	Weight	
Detail ID	Wght:[2 of 2]	
Dist. from "FROM" Node / Offset dist	1460	mm
Miscellaneous Weight	500	Kgf
Offset from Element Centerline	585	mm

Element#	6/8	
Element From Node	60	
Element To Node	70	
Element Type	Elliptical	
Description	RHS Dish End	
Distance "FROM" to "TO"	50.8	mm
Inside Diameter	619.15	mm
Element Thickness	12.5	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4150	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	619.15	mm
Liquid Density	0.0009984	kg./cm³

Element From Node	60	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	50.8	mm
Thickness of Insulation	80	mm
Density	0.000352	kg./cm³



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Input Echo:

Step: 1 10:20am Sep 9,2026

Element#	7/8
Element From Node	70
Element To Node	80
Element Type	Flange
Description	Tube Inlet
Distance "FROM" to "TO"	85.852 mm
Flange Inside Diameter	102.36 mm
Element Thickness	31.75 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	48 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4150 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.08 N./mm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03011
Class / Thickness / Grade	2 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	8/8
Element From Node	80
Element To Node	90
Element Type	Flange
Description	Shell Inlet flange
Distance "FROM" to "TO"	117.35 mm
Flange Inside Diameter	254.51 mm
Element Thickness	47.752 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	48 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4150 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No



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XY Coordinate Calculations:

Step: 2 10:20am

Sep 9, 2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Dish End		50.8	...	50.8	...
LHS Ch Shell		750.8	...	700	...
LHS Ch Flange		876.975	...	123	...
LHS Shell Flg		1088.97	...	122	...
Main Shell		2918.98	...	1830	...
RHS Dish End		2969.78	...	50.8	...
Tube Inlet		400.8	...	...	85.852
Shell Inlet fla		1910.8	...	...	117.348

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: LHS Ch Flg

Flng: 21 10:20am Sep 9,2026

Flange Input Data Values

Description: LHS Ch Flg :

LHS Ch Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	3206.86	KPa.
Design Temperature		230	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	619.150	mm
Flange Outside Diameter	A	800.000	mm
Flange Thickness	t	95.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	21.0000	mm
Length of Hub	h	28.0000	mm
Flange Material		SA-765 II	
Flange Material UNS number		K03047	
Flange Allowable Stress At Temperature	Sfo	136.80	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	740.000	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		40	
Flange Face Outside Diameter	Fod	688.150	mm
Flange Face Inside Diameter	Fid	619.150	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	677.150	mm
Gasket Inside Diameter	Gi	651.150	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	732.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	2.7500	
Partition Gasket Design Seating Stress	yPart	25.51	N./mm ²



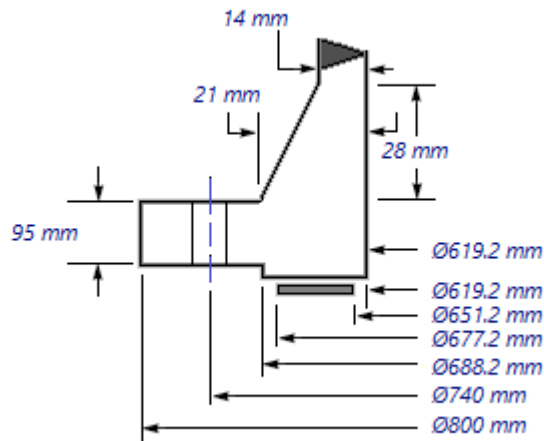
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Flg Calc [Int P]: LHS Ch Flg

Flng: 21 10:20am Sep 9, 2026



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (3206.86 \cdot (619.15/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 3206.86) + C_a$$

$$= 10.3669 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 11.0) / (312.575 + 0.6 \cdot 11.0)$$

$$= 4755.649 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (309.575 + 0.6 \cdot 14.0)$$

$$= 6075.487 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	92.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	625.150 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	18.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	11.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	39.425 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $b_o = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	664.301 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 664.3009^2 \cdot 3206.862$$

$$= 113343.445 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 664.3009 \cdot 3.0 \cdot 3206.86$$

$$+ 2 \cdot 732.0 \cdot 5.0 \cdot 2.75 \cdot 3206.8621$$

$$= 32890.867 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 625.1501^2 \cdot 3206.8621$$

$$= 100377.250 \text{ Kgf}$$



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Flg Calc [Int P]: LHS Ch Flg

Flng: 21 10:20am

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Pressure Force on Flange Face [Ht]:

$$\begin{aligned}
 &= H - H_d \\
 &= 113343 - 100377 \\
 &= 12966.198 \text{ Kgf}
 \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(W_{o,1} + W_{o,2} + H'p, 0) \\
 &= \max(113343 + 32891 + 0, 0) \\
 &= 146234.312 \text{ Kgf} \\
 &= 186712.891 \text{ Kgf, Mating Flange Load Governs}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p \\
 &= 68.95 \cdot 6.4246 \cdot \pi \cdot 664.301 + 25.51 \cdot 5.0 \cdot 732.0 \\
 &= 103789.055 \text{ Kgf}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a) \\
 &= \max((186713 + 0 + 4 \cdot 0 / 664.3) / 172, 103789 / 172) \\
 &= 106.459 \text{ cm}^2
 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned}
 &= 2a + 6t / (m + 0.5) \\
 &= 2 \cdot 25.4 + 6 \cdot 92.0 / (3.0 + 0.5) \\
 &= 208.514 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 740.0 \cdot \sin(\pi / 40) \\
 &= 58.060 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{ B_s / (2a + t) }, 1) \\
 &= \max(\sqrt{ 58.06 / (2 \cdot 25.4 + 92.0) }, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	106.459	142.193	
Radial Distance between Hub and Bolts:	34.925	39.425	
Radial Distance between Bolts and Edge:	26.987	30.000	
Circ. Spacing between the Bolts:	57.150	58.060	208.514

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= A_b \cdot S_a / (y \cdot \pi (G_o + G_i)) \\
 &= 142.193 \cdot 172.0 / (68.95 \cdot \pi (677.15 + 651.15)) \\
 &= 8.500 \text{ mm}
 \end{aligned}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= S_a \cdot A_b \\
 &= 172.0 \cdot 142.1933 \\
 &= 249385.36 \text{ Kgf}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= W_o - H \\
 &= 186713 - 113343 \\
 &= 73369.45 \text{ Kgf}
 \end{aligned}$$



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Flg Calc [Int P]: LHS Ch Flg

Flng: 21 10:20am

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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (740.0 - 664.3009) / 2$$

$$= 37.8496 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((740.0 - 625.1501) / 2 + 37.8496)$$

$$= 47.6373 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (740.0 - 625.15 - 18.0) / 2$$

$$= 48.4250 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	100377.	48.4250	1.0000	4861.
Face Pressure, Mt	12966.	47.6373	1.0000	618.
Gasket Load, Mg	73369.	37.8496	1.0000	2777.
Gasket Seating, Matm	249385.	37.8496	1.0000	9439.

Total Moment for Operation, Mo 8256. Kg-m.

Total Moment for Gasket seating, Mg 9439. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 82.926 mm

Hub Ratio, h/h0 = HL / H0 0.338

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.636

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.869

Factor V per 4.16.5 0.332

Factor f per 4.16.5 1.263

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 800.0 / 625.15$$

$$= 1.280$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.28^2 \cdot \log_{10}(1.28))) / (1.28^2 - 1)) / (1.28 - 1)$$

$$= 8.013$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.28^2 (1 + 8.55246 \cdot \log_{10}(1.28)) - 1) / ((1.0472 + 1.9448 \cdot 1.28^2) \cdot (1.28 - 1))$$

$$= 1.806$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.28^2 (1 + 8.55246 \cdot \log_{10}(1.28)) - 1) / (1.36136 (1.28^2 - 1) \cdot (1.28 - 1))$$

$$= 8.805$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.28^2 + 1) / (1.28^2 - 1)$$

$$= 4.137$$

Stress factor from Table 4.16.4 for integral flanges [d]:



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$$\begin{aligned}
 &= U \cdot g_0^2 \cdot h_0 / V \\
 &= 8.805 \cdot 11.0^2 \cdot 82.926 / 0.332 \\
 &= 265854.969 \text{ mm}^3
 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned}
 &= F / h_0 \\
 &= 0.869 / 82.926 \\
 &= 0.010 \text{ 1/mm}
 \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned}
 &= (t \cdot e + 1) / T + t^3/d \\
 &= (92.0 \cdot 0.01 + 1) / 1.806 + 92.0^3 / 265854.969 \\
 &= 4.017
 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned}
 &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\
 &= (1.2635 \cdot 8256 / 625.1501) / (4.0168 \cdot 18.0^2) \\
 &= 125.73 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned}
 &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\
 &= (1.2635 \cdot 9439 / 625.1501) / (4.0168 \cdot 18.0^2) \\
 &= 143.75 \text{ N./mm}^2
 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned}
 &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\
 &= (2.2861 \cdot 8256 / 625.1501) / (4.0168 \cdot 92.0^2) \\
 &= 8.71 \text{ N./mm}^2
 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned}
 &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\
 &= (2.2861 \cdot 9439 / 625.1501) / (4.0168 \cdot 92.0^2) \\
 &= 9.96 \text{ N./mm}^2
 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned}
 &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\
 &= (8.0127 \cdot 8256 / (92.0^2 \cdot 625.1501)) - 4.1367 \cdot 9 \\
 &= 86.58 \text{ N./mm}^2
 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned}
 &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\
 &= (8.0127 \cdot 9439 / (92.0^2 \cdot 625.1501)) - 4.1367 \cdot 10 \\
 &= 98.99 \text{ N./mm}^2
 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned}
 &= (SHo + \max(SRo, STo)) / 2 \\
 &= (126 + \max(9, 87)) / 2 \\
 &= 106.15 \text{ N./mm}^2
 \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned}
 &= (SHa + \max(SRa, STa)) / 2 \\
 &= (144 + \max(10, 99)) / 2 \\
 &= 121.37 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= Wm1 / Ab \\
 &= 186713 / 142.1933 \\
 &= 128.77 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (Wm2 / Ab) \\
 &= (103789 / 142.1933) \\
 &= 71.58 \text{ N./mm}^2
 \end{aligned}$$



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Flg Calc [Int P]: LHS Ch Flg

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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	125.73	205.20	143.75	207.00
Radial Flange	8.71	136.80	9.96	138.00
Tangential Flange	86.58	136.80	98.99	138.00
Maximum Average	106.15	136.80	121.37	138.00
Bolting	128.77	172.00	71.58	172.00

Minimum Required Flange Thickness 88.849 mm

Estimated M.A.W.P. (Operating) 4756 KPa.

Estimated Finished Weight of Flange at given Thk. 156.0 kg.

Estimated Unfinished Weight of Forging at given Thk 192.2 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	475.43	475.43	1.000	475.43	499.99	0.951
Tangential Flange	St	327.35	356.57	0.918	327.35	374.99	0.873
Tangential Flange	Sto	147.66	237.71	0.621	147.66	250.00	0.591
Combined	Sh/f + St	703.60	713.14	0.987	703.60	749.99	0.938
Combined	Sr + St	360.28	713.14	0.505	360.28	749.99	0.480
Combined	Sh + Sto	623.09	713.14	0.874	623.09	749.99	0.831
Radial Flange	Sr	32.93	475.43	0.069	32.93	499.99	0.066
Maximum Stress Ratio				1.000			0.951



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Flg Calc [Int P]: LHS Ch Flg

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Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.6456/1.3023)

= 1.2636

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.4901/1.2489) if $h/h0 < 0.35$, else max (0.25; 0.3038/0.7595)

= 0.3925

In this case, the hub ratio $h/h0$, 0.3377 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	568.8 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Flg Calc [Int P]: LHS Ch Flg Flng: 21 10:20am Sep 9, 2026

Root Area of a Single Bolt	Abs	3.5548 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	208.0 N./mm ²
Single flange rotation at Sfmax	Thetaamax	0.357 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$$

$$= \pi/4 (677.15^2 - 651.15^2) + (0.5 \cdot 732.0 \cdot 5.0)$$

$$= 289.5 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207.0 \cdot 289.544 / (40 \cdot 3.555)$$

$$= 421.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(421.508; 507.5)$$

$$= 421.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(421.508; 290.0)$$

$$= 421.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(421.508; 568.78)$$

$$= 421.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin \cdot S(Ag / (nb \cdot Abs))$$

$$:: 421.508 \geq 140.0(289.544 / (40 \cdot 3.555))$$

$$:: 421.5 \geq 285.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin \cdot O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 421.508 \geq (97.0 \cdot 289.544 + \pi/4 \cdot 3206.862 \cdot 651.15^2) / (0.7 \cdot 3.555 \cdot 40)$$

$$:: 421.5 \geq 389.5 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 421.508 \leq 276.0(289.544 / (40 \cdot 3.555))$$

$$:: 421.5 \leq 562.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetaamax})$$

$$:: 421.508 \leq 568.78(1.0/0.357)$$

$$:: 421.5 \leq 1593.3 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 421.508 \cdot 0.2 \cdot 3.555 \cdot 25.4$$

$$= 77.6 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 138 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



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Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 9439.3 / 1.0 \cdot 9806.637 \cdot 0.332 / (3.432 \cdot 202713 \cdot 11.0^2 \cdot 82.926 \cdot 0.3)$
 $= 0.766$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 8255.6 / 1.0 \cdot 9806.637 \cdot 0.332 / (3.432 \cdot 190469 \cdot 11.0^2 \cdot 82.926 \cdot 0.3)$
 $= 0.713$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 9439.3 / 1.0 \cdot 9806.637 \cdot 0.332 / (4.017 \cdot 202713 \cdot 11.0^2 \cdot 82.926 \cdot 0.3)$
 $= 0.654$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 8255.6 / 1.0 \cdot 9806.637 \cdot 0.332 / (4.017 \cdot 190469 \cdot 11.0^2 \cdot 82.926 \cdot 0.3)$
 $= 0.609$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.707, Temperature Reduction per Fig. UCS 66.1 = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-34 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	113.99	207.00	127.72	207.00
Radial Flange	10.47	138.00	11.73	138.00
Tangential Flange	72.38	138.00	81.11	138.00
Maximum Average	93.19	138.00	104.42	138.00
Bolting	128.77	172.00	71.58	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 79.908 mm

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Flg Calc [Int P]: RHS Shell Fl

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Flange Input Data Values

Description: RHS Shell Fl :

LHS Shell Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	4156.86	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	619.150	mm
Flange Outside Diameter	A	800.000	mm
Flange Thickness	t	90.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	24.0000	mm
Length of Hub	h	32.0000	mm
Flange Material		SA-765 II	
Flange Material UNS number		K03047	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	740.000	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		40	
Flange Face Outside Diameter	Fod	688.150	mm
Flange Face Inside Diameter	Fid	619.150	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	677.150	mm
Gasket Inside Diameter	Gi	651.150	mm
Gasket Factor	m	2.7500	
Gasket Design Seating Stress	y	25.51	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	732.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	2.7500	
Partition Gasket Design Seating Stress	yPart	25.51	N./mm ²



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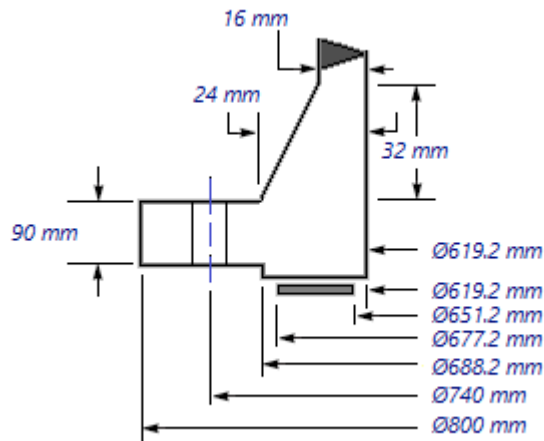
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Flg Calc [Int P]: RHS Shell F1

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4156.86 \cdot (619.15/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 4156.86) + C_a$$

$$= 12.5895 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (312.575 + 0.6 \cdot 13.0)$$

$$= 5599.261 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (309.575 + 0.6 \cdot 16.0)$$

$$= 6917.308 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	87.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	625.150 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	21.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	13.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	36.425 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $b_o = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	664.301 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 664.3009^2 \cdot 4156.861$$

$$= 146920.250 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{part} \cdot m_{part} \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 664.3009 \cdot 2.75 \cdot 4156.86$$

$$+ 2 \cdot 732.0 \cdot 5.0 \cdot 2.75 \cdot 4156.8613$$

$$= 39792.660 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 625.1501^2 \cdot 4156.8613$$

$$= 130112.945 \text{ Kgf}$$



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Flg Calc [Int P]: RHS Shell Fl Flng: 22 10:20am Sep 9, 2026

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 146920 - 130113$$

$$= 16807.297 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(146920 + 39793 + 0, 0)$$

$$= 186712.891 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 25.51 \cdot 6.4246 \cdot \pi \cdot 664.301 + 25.51 \cdot 5.0 \cdot 732.0$$

$$= 44400.262 \text{ Kgf}$$

$$= 103789.055 \text{ Kgf, Mating Flange Load Governs}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((186713 + 0 + 4 \cdot 0 / 664.3) / 172, 103789 / 172)$$

$$= 106.459 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 87.0 / (2.75 + 0.5)$$

$$= 211.415 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 740.0 \cdot \sin(\pi / 40)$$

$$= 58.060 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ (Bs / (2a + t)) }, 1)$$

$$= \max(\sqrt{ (58.06 / (2 \cdot 25.4 + 87.0)) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	106.459	142.193	
Radial Distance between Hub and Bolts:	34.925	36.425	
Radial Distance between Bolts and Edge:	26.987	30.000	
Circ. Spacing between the Bolts:	57.150	58.060	211.415

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 142.193 \cdot 172.0 / (25.51 \cdot \pi (677.15 + 651.15))$$

$$= 22.973 \text{ mm [Note: Exceeds actual gasket width, 13.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 142.1933$$

$$= 249385.36 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 186713 - 146920$$

$$= 39792.66 \text{ Kgf}$$



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Flg Calc [Int P]: RHS Shell Fl

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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (740.0 - 664.3009) / 2$$

$$= 37.8496 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((740.0 - 625.1501) / 2 + 37.8496)$$

$$= 47.6373 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (740.0 - 625.15 - 21.0) / 2$$

$$= 46.9250 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	130113.	46.9250	1.0000	6106.
Face Pressure, Mt	16807.	47.6373	1.0000	801.
Gasket Load, Mg	39793.	37.8496	1.0000	1506.
Gasket Seating, Matm	249385.	37.8496	1.0000	9439.

Total Moment for Operation, Mo 8413. Kg-m.

Total Moment for Gasket seating, Mg 9439. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 90.150 mm

Hub Ratio, h/h0 = HL / H0 0.355

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.615

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.867

Factor V per 4.16.5 0.330

Factor f per 4.16.5 1.174

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 800.0 / 625.15$$

$$= 1.280$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.28^2 \cdot \log_{10}(1.28))) / (1.28^2 - 1)) / (1.28 - 1)$$

$$= 8.013$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.28^2(1 + 8.55246 \cdot \log_{10}(1.28)) - 1) / ((1.0472 + 1.9448 \cdot 1.28^2) \cdot (1.28 - 1))$$

$$= 1.806$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.28^2(1 + 8.55246 \cdot \log_{10}(1.28)) - 1) / (1.36136(1.28^2 - 1) \cdot (1.28 - 1))$$

$$= 8.805$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.28^2 + 1) / (1.28^2 - 1)$$

$$= 4.137$$

Stress factor from Table 4.16.4 for integral flanges [d]:



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$$\begin{aligned}
 &= U \cdot g_0^2 \cdot h_0 / V \\
 &= 8.805 \cdot 13.0^2 \cdot 90.15 / 0.33 \\
 &= 406850.344 \text{ mm}^3
 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned}
 &= F / h_0 \\
 &= 0.867 / 90.15 \\
 &= 0.010 \text{ 1/mm}
 \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned}
 &= (t \cdot e + 1) / T + t^3 / d \\
 &= (87.0 \cdot 0.01 + 1) / 1.806 + 87.0^3 / 406850.344 \\
 &= 2.635
 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned}
 &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\
 &= (1.1738 \cdot 8413 / 625.1501) / (2.6353 \cdot 21.0^2) \\
 &= 133.29 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned}
 &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\
 &= (1.1738 \cdot 9439 / 625.1501) / (2.6353 \cdot 21.0^2) \\
 &= 149.56 \text{ N./mm}^2
 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned}
 &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\
 &= (2.115 \cdot 8413 / 625.1501) / (2.6353 \cdot 87.0^2) \\
 &= 13.99 \text{ N./mm}^2
 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned}
 &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\
 &= (2.115 \cdot 9439 / 625.1501) / (2.6353 \cdot 87.0^2) \\
 &= 15.70 \text{ N./mm}^2
 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned}
 &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\
 &= (8.0127 \cdot 8413 / (87.0^2 \cdot 625.1501)) - 4.1367 \cdot 14 \\
 &= 81.82 \text{ N./mm}^2
 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned}
 &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\
 &= (8.0127 \cdot 9439 / (87.0^2 \cdot 625.1501)) - 4.1367 \cdot 16 \\
 &= 91.80 \text{ N./mm}^2
 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned}
 &= (SHo + \max(SRo, STo)) / 2 \\
 &= (133 + \max(14, 82)) / 2 \\
 &= 107.55 \text{ N./mm}^2
 \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned}
 &= (SHa + \max(SRa, STa)) / 2 \\
 &= (150 + \max(16, 92)) / 2 \\
 &= 120.68 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned}
 &= Wm1 / Ab \\
 &= 186713 / 142.1933 \\
 &= 128.77 \text{ N./mm}^2
 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned}
 &= (Wm2 / Ab) \\
 &= (103789 / 142.1933) \\
 &= 71.58 \text{ N./mm}^2
 \end{aligned}$$



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Flg Calc [Int P]: RHS Shell F1

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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	133.29	207.00	149.56	207.00
Radial Flange	13.99	138.00	15.70	138.00
Tangential Flange	81.82	138.00	91.80	138.00
Maximum Average	107.55	138.00	120.68	138.00
Bolting	128.77	172.00	71.58	172.00

Minimum Required Flange Thickness 82.804 mm

Estimated M.A.W.P. (Operating) 5334 KPa.

Estimated Finished Weight of Flange at given Thk. 150.6 kg.

Estimated Unfinished Weight of Forging at given Thk 190.6 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	306.41	374.99	0.817	306.41	374.99	0.817
Tangential Flange	Sto	166.00	250.00	0.664	166.00	250.00	0.664
Combined	Sh/f + St	731.68	749.99	0.976	731.68	749.99	0.976
Combined	Sr + St	358.82	749.99	0.478	358.82	749.99	0.478
Combined	Sh + Sto	665.99	749.99	0.888	665.99	749.99	0.888
Radial Flange	Sr	52.40	499.99	0.105	52.40	499.99	0.105
Maximum Stress Ratio				1.000			1.000



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Flg Calc [Int P]: RHS Shell F1

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Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.5732/1.3381)

= 1.1757

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.5023/1.3006) if $h/h0 < 0.35$, else max (0.25; 0.296/0.7583)

= 0.3903

In this case, the hub ratio $h/h0$, 0.3550 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	574.1 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Flg Calc [Int P]: RHS Shell Fl Flng: 22 10:20am Sep 9,2026

Root Area of a Single Bolt	Abs	3.5548 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetaamax	0.390 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$$

$$= \pi/4 (677.15^2 - 651.15^2) + (0.5 \cdot 732.0 \cdot 5.0)$$

$$= 289.5 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207.0 \cdot 289.544 / (40 \cdot 3.555)$$

$$= 421.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(421.508; 507.5)$$

$$= 421.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(421.508; 290.0)$$

$$= 421.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(421.508; 574.077)$$

$$= 421.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 421.508 \geq 140.0(289.544 / (40 \cdot 3.555))$$

$$:: 421.5 \geq 285.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 421.508 \geq (97.0 \cdot 289.544 + \pi/4 \cdot 4156.861 \cdot 651.15^2) / (0.7 \cdot 3.555 \cdot 40)$$

$$:: 421.5 \geq 421.2 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 421.508 \leq 276.0(289.544 / (40 \cdot 3.555))$$

$$:: 421.5 \leq 562.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(thetagmax / thetaamax)$$

$$:: 421.508 \leq 574.077(1.0/0.39)$$

$$:: 421.5 \leq 1471.1 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 421.508 \cdot 0.2 \cdot 3.555 \cdot 25.4$$

$$= 77.6 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 138 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



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Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 9439.3 / 1.0 \cdot 9806.637 \cdot 0.33 / (2.227 \cdot 202713 \cdot 13.0^2 \cdot 90.15 \cdot 0.3)$
 $= 0.771$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 8412.5 / 1.0 \cdot 9806.637 \cdot 0.33 / (2.227 \cdot 199226 \cdot 13.0^2 \cdot 90.15 \cdot 0.3)$
 $= 0.699$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 9439.3 / 1.0 \cdot 9806.637 \cdot 0.33 / (2.635 \cdot 202713 \cdot 13.0^2 \cdot 90.15 \cdot 0.3)$
 $= 0.651$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 8412.5 / 1.0 \cdot 9806.637 \cdot 0.33 / (2.635 \cdot 199226 \cdot 13.0^2 \cdot 90.15 \cdot 0.3)$
 $= 0.591$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.854, Temperature Reduction per Fig. UCS 66.1 = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-22 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	118.34	207.00	129.36	207.00
Radial Flange	15.65	138.00	17.11	138.00
Tangential Flange	67.94	138.00	74.27	138.00
Maximum Average	93.14	138.00	101.82	138.00
Bolting	128.77	172.00	71.58	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 70.942 mm

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Flg Calc [Int P]: N1Tube Inlet

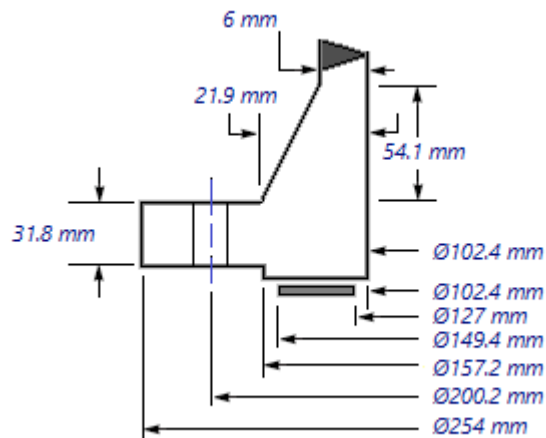
Flng: 23 10:20am Sep 9,2026

Flange Input Data Values

Description: N1Tube Inlet :

Tube Inlet

Flange Type	Integral Weld Neck		
Design Pressure	P	4150.00	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	102.362	mm
Flange Outside Diameter	A	254.000	mm
Flange Thickness	t	31.7500	mm
Thickness of Hub at Small End	go	6.0200	mm
Thickness of Hub at Large End	gl	21.8700	mm
Length of Hub	h	54.1020	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	200.152	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	8		
Flange Face Outside Diameter	Fod	157.226	mm
Flange Face Inside Diameter	Fid	102.362	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	149.352	mm
Gasket Inside Diameter	Gi	127.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: N1Tube Inlet Flng: 23 10:20am Sep 9,2026

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4150.0 \cdot (102.362/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 4150.0) + Ca$$

$$= 4.6594 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 3.02) / (54.181 + 0.6 \cdot 3.02)$$

$$= 7442.506 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 6.02) / (51.181 + 0.6 \cdot 6.02)$$

$$= 15160.635 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	28.750 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	108.362 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	18.870 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	3.020 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	27.025 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	11.176 mm
Basic Gasket Width, $b_o = N/2$	5.588 mm
Effective Gasket Width, $b = b_o$	5.588 mm
Gasket Reaction Diameter, $G = (G_o + G_i)/2$	138.176 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 138.176^2 \cdot 4150.0$$

$$= 6345.996 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 5.588 \cdot \pi \cdot 138.176 \cdot 3.0 \cdot 4150.0$$

$$= 6159.349 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 108.362^2 \cdot 4150.0$$

$$= 3902.910 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 6346 - 3903$$

$$= 2443.085 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(6346 + 6159 + 0, 0)$$

$$= 12505.344 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 5.588 \cdot \pi \cdot 138.176 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 17054.717 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((12505 + 0 + 4 \cdot 0 / 138.18) / 172, 17055 / 172)$$

$$= 9.724 \text{ cm}^2$$



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Flg Calc [Int P]: N1Tube Inlet

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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 19.05 + 6 \cdot 28.75/(3.0 + 0.5)$$

$$= 87.386 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 200.152 \cdot \sin(\pi / 8)$$

$$= 76.595 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{76.595/(2 \cdot 19.05 + 28.75)}, 1)$$

$$= 1.0704$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	9.724	15.587	
<i>Radial Distance between Hub and Bolts:</i>	<i>28.575</i>	<i>27.025</i>	
Radial Distance between Bolts and Edge:	20.637	26.924	
Circ. Spacing between the Bolts:	44.450	76.595	87.386

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 15.587 \cdot 172.0 / (68.95 \cdot \pi (149.352 + 127.0))$$

$$= 4.479 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 15.5871$$

$$= 27337.34 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 12505 - 6346$$

$$= 6159.35 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (200.152 - 138.176) / 2$$

$$= 30.9880 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((200.152 - 108.362) / 2 + 30.988)$$

$$= 38.4415 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (200.152 - 108.362 - 18.87) / 2$$

$$= 36.4600 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	3903.	36.4600	1.0704	152.
Face Pressure, Mt	2443.	38.4415	1.0704	101.
Gasket Load, Mg	6159.	30.9880	1.0704	204.
Gasket Seating, Matm	27337.	30.9880	1.0704	907.



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Flg Calc [Int P]: N1Tube Inlet Flng: 23 10:20am Sep 9, 2026

Total Moment for Operation, Mo 457. Kg-m.
Total Moment for Gasket seating, Mg 907. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 18.090 mm
Hub Ratio, h/h0 = HL / H0 2.991
Thickness Ratio, g1/g0 = (g1Cor/goCor) 6.248

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.469
Factor V per 4.16.5 0.012
Factor f per 4.16.5 1.000

Warning: The hub ratio Xh was greater than 2 and has been set to 2.

Stress factor from Table 4.16.4 [K]:

= A / B
= 254.0/108.362
= 2.344

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K)) / (K² - 1)) / (K - 1))
= (0.66845 + (5.7169(2.344² • log10(2.344)) / (2.344² - 1)) / (2.344 - 1))
= 2.421

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (2.344²(1 + 8.55246 • log10(2.344)) - 1) / ((1.0472 + 1.9448 • 2.344²) • (2.344 - 1))
= 1.387

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (2.344²(1 + 8.55246 • log10(2.344)) - 1) / (1.36136(2.344² - 1) • (2.344 - 1))
= 2.661

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (2.344² + 1) / (2.344² - 1)
= 1.445

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 2.661 • 3.02² • 18.09 / 0.012
= 35838.348 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.469 / 18.09
= 0.026 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (28.75 • 0.026 + 1) / 1.387 + 28.75³/35838.348
= 1.921

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.0•457/108.362)/(1.9205•18.87²)
= 60.50 N/mm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)



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Flg Calc [Int P]: N1Tube Inlet

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$$= (1.0 \cdot 907 / 108.362) / (1.9205 \cdot 18.87^2)$$

$$= 120.00 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.9929 \cdot 457 / 108.362) / (1.9205 \cdot 28.75^2)$$

$$= 51.94 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.9929 \cdot 907 / 108.362) / (1.9205 \cdot 28.75^2)$$

$$= 103.03 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$

$$= (2.4212 \cdot 457 / (28.75^2 \cdot 108.362)) - 1.445 \cdot 52$$

$$= 46.13 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (2.4212 \cdot 907 / (28.75^2 \cdot 108.362)) - 1.445 \cdot 103$$

$$= 91.51 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRO, STo)) / 2$$

$$= (60 + \max(52, 46)) / 2$$

$$= 56.22 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRA, STa)) / 2$$

$$= (120 + \max(103, 92)) / 2$$

$$= 111.51 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 12505 / 15.5871$$

$$= 78.68 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (17055 / 15.5871)$$

$$= 107.30 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	60.50	207.00	120.00	207.00
Radial Flange	51.94	138.00	103.03	138.00
Tangential Flange	46.13	138.00	91.51	138.00
Maximum Average	56.22	138.00	111.51	138.00
Bolting	78.68	172.00	107.30	172.00

Minimum Required Flange Thickness

29.083 mm

Estimated M.A.W.P. (Operating)

7443 KPa.

Estimated Finished Weight of Flange at given Thk.

12.6 kg.

Estimated Unfinished Weight of Forging at given Thk

28.2 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 925.4 / 1.0924 \cdot 9806.637 \cdot 0.012 / (1.701 \cdot 202713 \cdot 3.02^2 \cdot 18.09 \cdot 0.3)$$



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Flg Calc [Int P]: N1Tube Inlet

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= 0.310 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

= $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$

= $52.14 \cdot 466.6 / 1.0924 \cdot 9806.637 \cdot 0.012 / (1.701 \cdot 199226 \cdot 3.02^2 \cdot 18.09 \cdot 0.3)$

= 0.159 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$

= $52.14 \cdot 906.8 / 1.0704 \cdot 9806.637 \cdot 0.012 / (1.921 \cdot 202713 \cdot 3.02^2 \cdot 18.09 \cdot 0.3)$

= 0.275 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

= $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$

= $52.14 \cdot 457.2 / 1.0704 \cdot 9806.637 \cdot 0.012 / (1.921 \cdot 199226 \cdot 3.02^2 \cdot 18.09 \cdot 0.3)$

= 0.141 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Calculated Minimum Design Metal Temperature -104 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	52.35	207.00	101.39	207.00
Radial Flange	47.87	138.00	92.71	138.00
Tangential Flange	32.37	138.00	62.69	138.00
Maximum Average	50.11	138.00	97.05	138.00
Bolting	78.68	172.00	107.30	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 27.534 mm

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Flg Calc [Int P]: N3Shel Inlet

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Warnings for Flange : N3Shel Inlet :

Note: Flange Hub Taper 0.3448 is > 1:3. Check Fig. 2-4(6) of VIII-1 Appendix 2.
This warning should not be considered for Standard Flanges as permitted by ASME VIII-1 in Table U-3 or for non-Code vessels.

Flange Input Data Values

Description: N3Shel Inlet :

Shell Inlet flange

Flange Type	Integral Weld Neck		
Design Pressure	P	4150.00	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	254.508	mm
Flange Outside Diameter	A	444.500	mm
Flange Thickness	t	47.7520	mm
Thickness of Hub at Small End	go	12.7000	mm
Thickness of Hub at Large End	g1	36.7000	mm
Length of Hub	h	69.5960	mm
Flange Material		SA-350 LF2 +IT	
Flange Material UNS number		K03011	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	387.350	mm
Nominal Bolt Diameter	a	25.4000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		16	
Flange Face Outside Diameter	Fod	323.850	mm
Flange Face Inside Diameter	Fid	254.508	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	317.500	mm
Gasket Inside Diameter	Gi	287.274	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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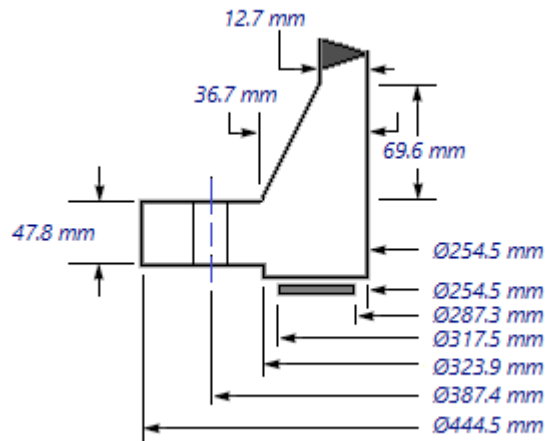
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Flg Calc [Int P]: N3Shel Inlet

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4150.0 \cdot (254.508/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 4150.0) + Ca$$

$$= 6.9893 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.7) / (130.254 + 0.6 \cdot 9.7)$$

$$= 9836.545 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.7) / (127.254 + 0.6 \cdot 12.7)$$

$$= 12993.360 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	44.752 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	260.508 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	33.700 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	9.700 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	29.721 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	15.113 mm
Basic Gasket Width, $bo = N/2$	7.557 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	6.927 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	303.646 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 303.6459^2 \cdot 4150.0$$

$$= 30645.693 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.927 \cdot \pi \cdot 303.6459 \cdot 3.0 \cdot 4150.0$$

$$= 16778.785 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 260.508^2 \cdot 4150.0$$

$$= 22556.756 \text{ Kgf}$$



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Flg Calc [Int P]: N3Shel Inlet Flng: 24 10:20am Sep 9, 2026

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 30646 - 22557$$

$$= 8088.937 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(30646 + 16779 + 0, 0)$$

$$= 47424.480 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 6.927 \cdot \pi \cdot 303.646 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 46459.035 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((47424 + 0 + 4 \cdot 0 / 303.65) / 172, 46459 / 172)$$

$$= 27.040 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 44.752 / (3.0 + 0.5)$$

$$= 127.518 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 387.35 \cdot \sin(\pi / 16)$$

$$= 75.568 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ B_s / (2a + t) }, 1)$$

$$= \max(\sqrt{ 75.568 / (2 \cdot 25.4 + 44.752) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	27.040	56.877	
<i>Radial Distance between Hub and Bolts:</i>	<i>34.925</i>	<i>29.721</i>	
Radial Distance between Bolts and Edge:	26.987	28.575	
Circ. Spacing between the Bolts:	57.150	75.568	127.518

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 56.877 \cdot 172.0 / (68.95 \cdot \pi (317.5 + 287.27))$$

$$= 7.468 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 56.8773$$

$$= 99754.14 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 47424 - 30646$$

$$= 16778.79 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (387.35 - 303.6459) / 2$$



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= 41.8520 mm

Distance to Face Pressure Reaction [ht]:

= $0.5 (C - B) / 2 + hg$

= $0.5 (387.35 - 260.508) / 2 + 41.852$

= 52.6365 mm

Distance to End Pressure Reaction [hd]:

= $(C - B - g1) / 2$

= $(387.35 - 260.508 - 33.7) / 2$

= 46.5710 mm

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	22557.	46.5710	1.0000	1051.
Face Pressure, Mt	8089.	52.6365	1.0000	426.
Gasket Load, Mg	16779.	41.8520	1.0000	702.
Gasket Seating, Matm	99754.	41.8520	1.0000	4175.

Total Moment for Operation, Mo 2179. Kg-m.

Total Moment for Gasket seating, Mg 4175. Kg-m.

Effective Hub Length, ho = $\sqrt{Bcor \cdot goCor}$ 50.269 mm

Hub Ratio, h/h0 = HL / H0 1.384

Thickness Ratio, g1/g0 = (g1Cor/goCor) 3.474

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.622

Factor V per 4.16.5 0.049

Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B

= $444.5 / 260.508$

= 1.706

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$

= $(0.66845 + (5.7169 (1.706^2 \cdot \log_{10}(1.706))) / (1.706^2 - 1)) / (1.706 - 1)$

= 3.807

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$

= $(1.706^2(1 + 8.55246 \cdot \log_{10}(1.706)) - 1) / ((1.0472 + 1.9448 \cdot 1.706^2) \cdot (1.706 - 1))$

= 1.623

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$

= $(1.706^2(1 + 8.55246 \cdot \log_{10}(1.706)) - 1) / (1.36136(1.706^2 - 1) \cdot (1.706 - 1))$

= 4.184

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$

= $(1.706^2 + 1) / (1.706^2 - 1)$

= 2.046

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g0^2 \cdot h0 / V$

= $4.184 \cdot 9.7^2 \cdot 50.269 / 0.049$

= 402983.062 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0



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$$= 0.622/50.269$$

$$= 0.012 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (44.752 \cdot 0.012 + 1) / 1.623 + 44.752^3/402983.062$$

$$= 1.180$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g1^2)$$

$$= (1.0 \cdot 2179/260.508)/(1.1798 \cdot 33.7^2)$$

$$= 61.20 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g1^2)$$

$$= (1.0 \cdot 4175/260.508)/(1.1798 \cdot 33.7^2)$$

$$= 117.29 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.7382 \cdot 2179/260.508)/(1.1798 \cdot 44.752^2)$$

$$= 60.33 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm/Bcor) / (L \cdot t^2)$$

$$= (1.7382 \cdot 4175/260.508)/(1.1798 \cdot 44.752^2)$$

$$= 115.61 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (3.8074 \cdot 2179/(44.752^2 \cdot 260.508)) - 2.0464 \cdot 60$$

$$= 32.46 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (3.8074 \cdot 4175/(44.752^2 \cdot 260.508)) - 2.0464 \cdot 116$$

$$= 62.21 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (61 + \max(60, 32))/2$$

$$= 60.76 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (117 + \max(116, 62))/2$$

$$= 116.45 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 47424/56.8773$$

$$= 81.77 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (46459/56.8773)$$

$$= 80.11 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

Operating		Gasket Seating	
Actual	Allowed	Actual	Allowed



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Flg Calc [Int P]: N3Shel Inlet Flng: 24 10:20am Sep 9,2026

Longitudinal Hub	61.20	207.00	117.29	207.00
Radial Flange	60.33	138.00	115.61	138.00
Tangential Flange	32.46	138.00	62.21	138.00
Maximum Average	60.76	138.00	116.45	138.00
Bolting	81.77	172.00	80.11	172.00

Minimum Required Flange Thickness 44.577 mm

Estimated M.A.W.P. (Operating) 8729 KPa.

Estimated Finished Weight of Flange at given Thk. 50.3 kg.

Estimated Unfinished Weight of Forging at given Thk 94.9 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 4175.0 / 1.0 \cdot 9806.637 \cdot 0.049 / (1.111 \cdot 202713 \cdot 9.7^2 \cdot 50.269 \cdot 0.3)$$

$$= 0.327 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2178.5 / 1.0 \cdot 9806.637 \cdot 0.049 / (1.111 \cdot 199226 \cdot 9.7^2 \cdot 50.269 \cdot 0.3)$$

$$= 0.174 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 4175.0 / 1.0 \cdot 9806.637 \cdot 0.049 / (1.18 \cdot 202713 \cdot 9.7^2 \cdot 50.269 \cdot 0.3)$$

$$= 0.308 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2178.5 / 1.0 \cdot 9806.637 \cdot 0.049 / (1.18 \cdot 199226 \cdot 9.7^2 \cdot 50.269 \cdot 0.3)$$

$$= 0.164 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	53.69	207.00	100.49	207.00
Radial Flange	55.38	138.00	103.65	138.00
Tangential Flange	28.35	138.00	53.07	138.00
Maximum Average	54.54	138.00	102.07	138.00
Bolting	81.77	172.00	80.11	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 42.316 mm

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Internal Pressure Calculations: Step: 7 10:20am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Dish End		3206.1	14	3	619.15	138	1
LHS Ch Shell		3206.1	14	3	619.15	138	2
LHS Ch Flange		3206.9	103	3	619.15	136.8	3
LHS Shell Flg		4156.9	93	3	619.15	138	4
Main Shell		4156.1	16	3	619.15	138	5
RHS Dish End		4156.1	16	3	619.15	138	6
Tube Inlet		4150	48	3	102.36	138	7
Shell Inlet flag		4150	48	3	254.51	138	8

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Dish End		3200	4006.11	5328.21	12	10.1876	1
LHS Ch Shell		3200	4749.55	6075.49	14	10.3652	2
LHS Ch Flange		3200	4748.79	5468.93	95	88.8492	3
LHS Shell Flg		4150	5326.66	5552.16	90	82.804	4
Main Shell		4150	5593.17	6917.31	16	12.5877	5
RHS Dish End		4150	4228.32	5549.32	12.5	12.3237	6
Tube Inlet		4150	7442.51	9072.12	31.75	29.083	7
Shell Inlet flag		4150	8729.24	8729.24	47.752	44.577	8

Warning: Shell Side MAWP governed by a standard Flange!

Warning: Channel Side MAWP governed by a standard Flange!

Warning: Tube side flange MAWP was less than the design pressure!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 4782 KPa.
 Shell Side MAPnc = 5100 KPa.
 Channel Side MAWP = 3869 KPa., < design pressure: 4150.000 KPa.
 Channel Side MAPnc = 5100 KPa.

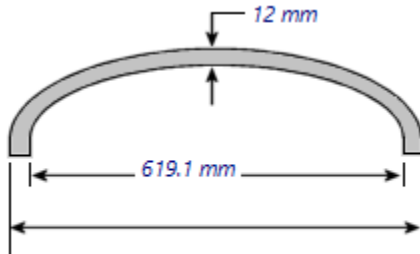
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Dish End

Elliptical Head Design Information:							
Design Pressure	P	3206.099 KPa.		Int. Design Temperature		230 °C	
Material		SA-516 70		Ext. Design Temperature		230 °C	
Length		50.800 mm		External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.	c	3.0000 mm		Allowable Stress (ope)	S	138.0 N./mm ²	
Ext. Corr. All.	ce	0.0000 mm		Allowable Stress (amb)	Sa	138.0 N./mm ²	
Inside Diameter	Di	619.150 mm		Actual Stress		110.3 N./mm ²	
Spec. Min. Thk t,e, or sa		12.0000 mm		Spec. Nominal Thk.		14.000 mm	
Surface Area		5572.6 cm ²		SG of Contents		1.000	
Element Volume		46372.1 cm ³		Weight of Contents		46.3 Kg	
Empty Weight		63.0 Kg		Operating Weight		109.4 Kg	
				Comp. All. Ope. Stress		113.5 N./mm ²	
				Aspect Ratio		2.00	
Radiography Information:							
User Defined				User Defined			
Circ. Joint Efficiency		100 %		Long. Joint Eff.		E or v 100 %	

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.739 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.870 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (3206.099 \cdot 625.15 \cdot 0.987) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 3206.099)$$

$$= 7.1876 + 3.0000 = 10.1876 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 6.098 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 9.0) / (0.987 \cdot 625.15 + 0.2 \cdot 9.0)$$

$$= 4012 - 6 = 4006 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 12.0) / (1.0 \cdot 619.15 + 0.2 \cdot 12.0)$$

$$= 5328 \text{ KPa.}$$



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Element: LHS Dish End

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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (3206.099 \cdot (0.987 \cdot 625.15 + 0.2 \cdot 9.0)) / (2 \cdot 1.0 \cdot 9.0)$$

$$= 110.272 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (3206.099 \cdot 312.575) / (138.0 \cdot 1.0 - 0.6 \cdot 3206.099) + 3.0$$

$$= 10.365 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 6.098 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 11.0) / (312.575 + 0.6 \cdot 11.0)$$

$$= 4756 - 6 = 4750 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (625.15 / (2 \cdot 157.787))^2) / 6$$

$$= 0.987385$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 9.354 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 12.0, tr = 8.677, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.964$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-25 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 14.0, tr = 8.913, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.81$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-29 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3868.516 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-45 °C
Governing Thk.	12.000 mm	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.964	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-25 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.739 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.870 mm

EXTERNAL PRESSURE RESULTS for: LHS Dish End

Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 230 °C



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Element: LHS Dish End

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Elastic Modulus from Chart: CS-2 at 230 °C: 189879744 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
9.000	643.15	71.46	0.0019436	85.23

$$MAEP = B / (K0 * Do / t) = 85.2291 / (0.9 * 71.4611) = 1325.1058 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
1.709	643.15	376.38	0.0003690	35.04

$$MAEP = B / (K0 * Do / t) = 35.0364 / (0.9 * 376.3759) = 103.4261 \text{ KPa.}$$
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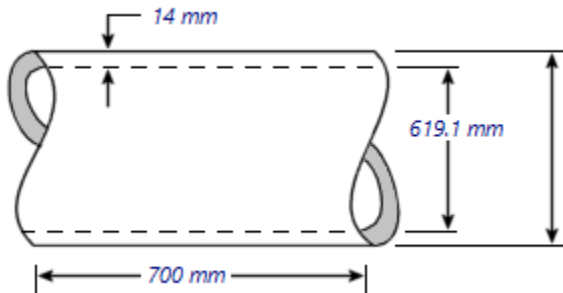
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Element: LHS Ch Shell

Step: 9 10:20am Sep 9,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	3206.099 KPa.	Int. Design Temperature	230 °C
Material	SA-516 70	Ext. Design Temperature	230 °C
Length	700.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	619.150 mm	Actual Stress	93.0 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	14231.6 cm ²	SG of Contents	1.000
Element Volume	210793.6 cm ³	Weight of Contents	210.7 Kg
Empty Weight	151.1 Kg	Operating Weight	361.7 Kg
		Comp. All. Ope. Stress	116.9 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (3206.099 \cdot 312.575) / (138.0 \cdot 1.0 - 0.6 \cdot 3206.099) \\
 &= 7.3652 + 3.0000 = 10.3652 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 6.098 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 11.0) / (312.575 + 0.6 \cdot 11.0) \\
 &= 4756 - 6 = 4750 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 14.0) / (309.575 + 0.6 \cdot 14.0) \\
 &= 6075 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (3206.099 \cdot (312.575 + 0.6 \cdot 11.0)) / (1.0 \cdot 11.0) \\
 &= 93.033 \text{ N./mm}^2
 \end{aligned}$$



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Element: LHS Ch Shell

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.211 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14.0$, $t_r = 8.913$, $c = 3.0$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.81$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -29 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3868.516 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-45 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.810	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	-29 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: LHS Ch Shell

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 230 °C

Elastic Modulus from Chart: CS-2 at 230 °C: 189879744 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	647.15	802.40	58.83	1.2399	0.0023863	88.64

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 88.6381) / (3 \cdot 58.8318) = 2008.731 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
2.317	647.15	802.40	279.36	1.2399	0.0002282	21.67

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 21.6706) / (3 \cdot 279.3611) = 103.4234 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	647.15	28665.70	58.83	44.2953	0.0003199	30.38

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 30.3767) / (3 \cdot 58.8318) = 688.4009 \text{ KPa.}$$



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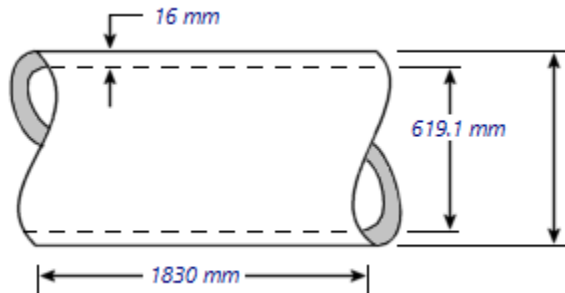
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Element: Main Shell

Step: 10 10:20am Sep 9,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	4156.097 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	1830.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	619.150 mm	Actual Stress	102.4 N./mm ²
Spec. Min. Thk t,e, or sa	16.0000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	37435.4 cm ²	SG of Contents	1.000
Element Volume	373361.2 cm ³	Weight of Contents	373.1 Kg
Empty Weight	452.8 Kg	Operating Weight	826.0 Kg
		Comp. All. Ope. Stress	143.2 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4156.097 \cdot 312.575) / (138.0 \cdot 1.0 - 0.6 \cdot 4156.097) \\
 &= 9.5877 + 3.0000 = 12.5877 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 6.097 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 13.0) / (312.575 + 0.6 \cdot 13.0) \\
 &= 5599 - 6 = 5593 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 16.0) / (309.575 + 0.6 \cdot 16.0) \\
 &= 6917 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4156.097 \cdot (312.575 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0) \\
 &= 102.430 \text{ N./mm}^2
 \end{aligned}$$



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Element: Main Shell

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.519 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 16.0$, $t_r = 11.061$, $c = 3.0$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.851$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp. at Required thickness (UCS 66.1) -22 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4781.667 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-45 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.851	Temperature Difference	8 °C
UCS-66(c) red.	No	Computed Min. Temp.	-22 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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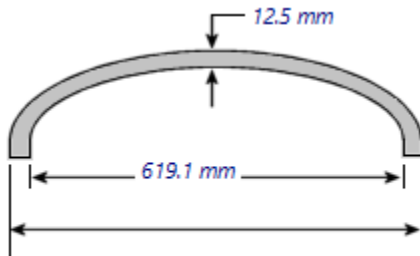
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Element: RHS Dish End

Step: 11 10:20am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: RHS Dish End

Elliptical Head Design Information:					
Design Pressure	P	4156.099 KPa.	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		50.800 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	619.150 mm	Actual Stress		135.4 N./mm ²
Spec. Min. Thk t,e, or sa		12.5000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		5635.2 cm ²	SG of Contents		1.000
Element Volume		46372.1 cm ³	Weight of Contents		46.3 Kg
Empty Weight		72.4 Kg	Operating Weight		118.8 Kg
			Comp. All. Ope. Stress		143.2 N./mm ²
			Aspect Ratio		2.00

Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.739 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.870 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (4156.099 \cdot 625.15 \cdot 0.987) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 4156.099)$$

$$= 9.3237 + 3.0000 = 12.3237 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 6.098 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 9.5) / (0.987 \cdot 625.15 + 0.2 \cdot 9.5)$$

$$= 4234 - 6 = 4228 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 12.5) / (1.0 \cdot 619.15 + 0.2 \cdot 12.5)$$

$$= 5549 \text{ KPa.}$$



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Element: RHS Dish End

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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (4156.099 \cdot (0.987 \cdot 625.15 + 0.2 \cdot 9.5)) / (2 \cdot 1.0 \cdot 9.5)$$

$$= 135.445 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (4156.099 \cdot 312.575) / (138.0 \cdot 1.0 - 0.6 \cdot 4156.099) + 3.0$$

$$= 12.588 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 6.098 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 13.0) / (312.575 + 0.6 \cdot 13.0)$$

$$= 5599 - 6 = 5593 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (625.15 / (2 \cdot 157.787)))^2 / 6$$

$$= 0.987385$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 10.596 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 12.5, tr = 8.677, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.913$, Temp. Reduction = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 16.0, tr = 8.913, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.686$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-32 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3868.516 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-45 °C
Governing Thk.	12.500 mm	Basic MDMT	-22 °C
Stress/Thk. Ratio	0.913	Temperature Difference	5 °C
UCS-66(c) red.	No	Computed Min. Temp.	-27 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.739 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.870 mm



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Shop Pressure Test Results: Step: 12 10:20am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6216.167 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5395.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6630.006 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5259.833 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	5934.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5934.500 KPa.
App 27-4	= M.A.W.P.	4781.667 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5029.071 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5395.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6630.006 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	4255.368 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	5934.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5934.500 KPa.
App 27-4	= M.A.W.P.	3868.516 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Dish End	228.2	234.0	0.975	6636.08
LHS Ch Shell	192.6	234.0	0.823	6636.08
Main Shell	163.6	234.0	0.699	6636.08
RHS Dish End	216.3	234.0	0.924	6636.08

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube inlet	Nozzle	118.00	118.00	1.000
N1-Tube inlet	Pad	138.00	138.00	1.000
N1-Tube Outlet	Nozzle	118.00	118.00	1.000
N1-Tube Outlet	Pad	138.00	138.00	1.000
N3-Shell inlet	Nozzle	138.00	138.00	1.000
N4-Shell outlet	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Dish End	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
LHS Ch Flange	138.00	136.80	1.009
LHS Shell Flg	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
RHS Dish End	138.00	138.00	1.000
Tube Inlet	138.00	138.00	1.000
Shell Inlet flange	138.00	138.00	1.000



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Shop Pressure Test Results:

Step: 12 10:20am

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Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	108.00	108.00	1.000
Tubesheet Material	138.00	136.80	1.009

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube inlet	55.30	216.00	0.256
N1-Tube Outlet	93.38	216.00	0.432
N3-Shell inlet	90.73	225.00	0.403
N4-Shell outlet	90.73	225.00	0.403

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 13 10:20am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6216.167 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5395.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6630.006 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5259.833 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	5934.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5934.500 KPa.
App 27-4	= M.A.W.P.	4781.667 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	5029.071 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5395.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6630.006 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	4255.368 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	5934.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	5934.500 KPa.
App 27-4	= M.A.W.P.	3868.516 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 4150.0 • 1.0

= 5395.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 4150.0 • 1.0

= 5395.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Dish End	185.8	234.0	0.794	5401.07
LHS Ch Shell	156.7	234.0	0.670	5401.07
Main Shell	133.1	234.0	0.569	5401.07
RHS Dish End	176.0	234.0	0.752	5401.07

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube inlet	Nozzle	118.00	118.00	1.000
N1-Tube inlet	Pad	138.00	138.00	1.000
N1-Tube Outlet	Nozzle	118.00	118.00	1.000
N1-Tube Outlet	Pad	138.00	138.00	1.000
N3-Shell inlet	Nozzle	138.00	138.00	1.000
N4-Shell outlet	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
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Field Pressure Test Results: Step: 13 10:20am Sep 9, 2026

LHS Dish End	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
LHS Ch Flange	138.00	136.80	1.009
LHS Shell Flg	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
RHS Dish End	138.00	138.00	1.000
Tube Inlet	138.00	138.00	1.000
Shell Inlet flange	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	108.00	108.00	1.000
Tubesheet Material	138.00	136.80	1.009

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube inlet	45.01	216.00	0.208
N1-Tube Outlet	76.00	216.00	0.352
N3-Shell inlet	73.85	225.00	0.328
N4-Shell outlet	73.85	225.00	0.328

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 10:20a Sep 9, 2026

Input Echo, Tubesheet Number 1, Description: U-Tube

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		5395.00 KPa.
Shell Thickness	ts	16.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	619.150 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		5395.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	619.150 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	302
Tube Wall Thickness	et	2.3200 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	1830.00 mm
Straight Tube Length (bet. inner tubsht faces) L		1740.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		108.00 N./mm ²
Tube Allowable Stress At Ambient		108.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	205.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	28.5750 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.3200 mm
Groove Weld Leg	ag	2.3200 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	284.072 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	2089.6 cm ²
Perimeter of Tube Layout	Cp	1714.3 mm



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ASME TS Calc: Pressure Test

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Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-765 II

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number	K03047
Tubesheet Allowable Stress at Temperature	S 138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt 138.00 N./mm ²
Thickness of Tubesheet	h 90.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc 0.0000 mm
Tubesheet Outside Diameter	A 800.000 mm
Dimension G for the Channel Side	Gc 664.301 mm
Area of the Untubed Lanes	AL 0.0 cm ²
Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes
Channel Side Min. Design Pressure	Ptd,min 103.4195 KPa.

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side	Both
Flange Outside Diameter	A 800.000 mm
Flange Inside Diameter	B 619.150 mm
Flange Face Outside Diameter	Fod 688.150 mm
Flange Face Inside Diameter	Fid 619.150 mm
Gasket Outside Diameter	Go 677.150 mm
Gasket Inside Diameter	Gi 651.150 mm
Small end Hub thk.	g0 16.0000 mm
Large end Hub thk.	g1 24.0000 mm
Gasket Factor,	m 2.75
Gasket Design Seating Stress	y 25.51 N./mm ²
Flange Facing Sketch	Code Sketch 1a
Gasket Thickness	tg 3.1750 mm
Full face Gasket Flange Option	Program Selects
Length of Partition Gasket	lp 732.000 mm
Width of Partition Gasket	wp 10.0000 mm
Partition Gasket Factor,	mPart 2.7500
Partition Gasket Design Seating Stress	yPart 25.51 N./mm ²
Partition Gasket Facing Sketch	Code Sketch 1a
Partition Gasket Column for Gasket Seating	Code Column II

Bolting Information:

Diameter of Bolt Circle	C 740.000 mm
Nominal Bolt Diameter	dB 25.4000 mm
Thread Series	TEMA Thread Series
Number of Bolts	n 40
Bolt Material	SA-193 B7
Bolt Allowable Stress At Temperature	Sb 172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa 172.00 N./mm ²
Weld between Flange and Shell/Channel	0.0000 mm
Alternate Flange Operating Bolt Load, Wm1	246001.17 Kgf
Alternate Flange Seating Bolt Load, Wm2	103789.05 Kgf
Alternate Flange Design Bolt Load, W	249385.36 Kgf

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with None



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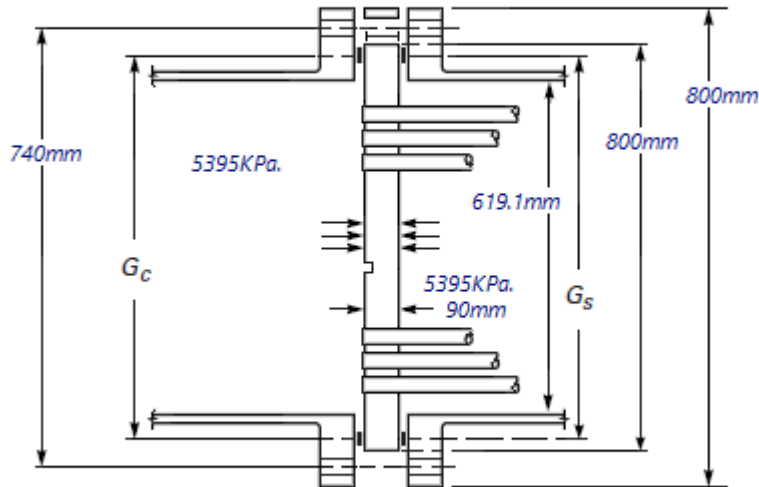
ASME TS Calc: Pressure Test

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Tubesheet Extended as Flange

No

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubesheet Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

$$\begin{aligned}
 &\text{Thickness Due to Internal Pressure:} \\
 &= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)} \\
 &= (5395.0 \cdot (19.05/2 - 0)) / (108.0 \cdot 1.0 + 0.4 \cdot 5395.0) \\
 &= 0.4665 + 0.0000 = 0.4665 \text{ mm}
 \end{aligned}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	CS-2	at	21.11 °C
Elastic Modulus for Material			199943392.00 KPa.



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Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.3200	19.05	2024.07	8.21	50.0000	0.0163147	122.73
EMAWP = (2.167/(D/T)-0.0833)*B = 22164.8672 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.7739	19.05	2024.07	24.61	50.0000	0.0018155	101.52
EMAWP = (4*B)/(3*(D/T)) = (4 *101.522)/(3 *24.615) = 5498.8809 KPa.						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.7739 mm
Allowable Internal Pressure at Corroded thickness	29142.83 KPa.
Required Internal Design Pressure	5395.00 KPa.
Allowable External Pressure at Corroded thickness	22164.87 KPa.
Required External Design Pressure	5498.42 KPa.
Required Thickness due to Shell Side pressure	0.7739 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	664.306 mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 284.072 + 19.05$$

$$= 587.194 \text{ mm}$$

Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (28.575 - 19.05) / 28.575$$

$$= 0.333$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$= Gs / Do \text{ (Configurations d, e, f)}$$

$$= 664.3056 / 587.194$$

$$= 1.1313$$

Ratio [pc]:

$$= Gc / Do \text{ (Configurations d)}$$

$$= 664.3009 / 587.194$$

$$= 1.1313$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= ltx / h$$



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$$= 0/90.0$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2tt)$$

$$= \max(19.05 - 2 \cdot 2.32 (0.20270E+09 / 2.0270E+09) \cdot$$

$$(107/137) \cdot (0); 19.05 - 2 \cdot 2.32)$$

$$= 19.0500 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A / 4 \cdot Do \cdot p); (\pi \cdot Do^2)) }$$

$$= 28.575 / \sqrt{ (1 - 4 \cdot \min(0 ; 4 \cdot 587.194 \cdot 28.575); (\pi \cdot 587.194^2)) }$$

$$= 28.5750 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (28.575 - 19.05) / 28.575$$

$$= 0.3333$$

$h/p = 3.149606$; $\mu^* = 0.333333$; $E^* = 65896428 \text{ KPa}$.

$E^*/E = 0.325091$; $\nu^* = 0.334167$

Note: As h/p (3.150) is > 2 , data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / Do$$

$$= 800.0/587.194$$

$$= 1.3624$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (E \cdot \ln(K))$$

$$= (1 - 0.33) / 65896428 \cdot (0.20270E+09 \cdot \ln(1.36))$$

$$= 0.6334$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$$

$$= 587.194^2/16 \cdot [(1.131 - 1) \cdot (1.131^2 + 1) \cdot 5395.0 -$$

$$(1.131 - 1) \cdot (1.131^2 + 1) \cdot 5395.0]$$

$$= 2394.0669 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]

$$= Mts + W^* \cdot (Gc - Gs) / (2 \cdot \pi \cdot Do)$$

$$= 2394.1 + 249385.4 \cdot (664.301 - 664.301) / (2 \cdot \pi \cdot 587.194)$$

$$= 2394.0669 \text{ KPa} \cdot \text{mm}^2$$

Note: W^* is the maximum of the bolt loads between the shell and channel sides.

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M^*) - Do^2/32 \cdot F(Ps - Pt)) / (1 + F)$$

$$= ((2394.07) - 587.194^2/32 \cdot 0.633 (5395.0 - 5395.0)) / (1 + 0.63)$$

$$= 1465.6943 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu^*) \cdot (Ps - Pt)$$

$$= 1465.69 + 587.194^2/64 \cdot (3 + 0.334) \cdot (5395.0 - 5395.0)$$

$$= 1465.6943 \text{ KPa} \cdot \text{mm}^2$$



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Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|1465.694|; |1465.694|)$$

$$= 1465.6943 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 1465.694 / (0.3333 \cdot (90.0 - 0)^2)$$

$$= 0.0033 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta_T \cdot S_y$$

$$= 2 \cdot 0.95 \cdot 250.0$$

$$= 474.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - h_g')^2)$$

$$= 6 \cdot 1464.851 / (0.3333 \cdot (0.508 - 0)^2)$$

$$= 102.1795 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 0.508 + 0 + 0$$

$$= 0.5080 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(0.508; 0)$$

$$= 0.5080 \text{ mm}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$= |P_s - P_t|$$

$$= |5395.0 - 5395.0|$$

$$= 0.000 \text{ KPa.}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta_T \cdot S_y$$

$$= 4 \cdot 0.333 \cdot 90.0 / 587.19 \cdot 0.8 \cdot 0.95 \cdot 249.995$$

$$= 38.828 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta_T \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.333 \cdot 189.997) \cdot (4 \cdot 2089.6 / 1714.3) \cdot |5395.0 - 5395.0| + 0$$

$$= 0.0000 \text{ mm}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19.05 - 2.32) 2.32$$

$$= 1.219 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.32 (19.05 - 2.32) 194.746$$

$$= 2421.446 \text{ Kg}$$



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ASME TS Calc: Pressure Test

Case: 1 10:20a Sep 9, 2026

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.32 (19.05 + 0.67 \cdot 2.32) 194.75$$

$$= 2421.4458 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 194.746 / 194.746$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.32 (19.05 + 0.67 \cdot 2.32) 194.75$$

$$= 1640.2175 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{\left((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d \right)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{\left((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.32 \cdot (19.05 - 2.32) \cdot 1.0 \cdot 1.0 \right)} - 0.75 \cdot 19.05 \right)$$

$$= 2.772 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3920 mm

Minimum Required Groove Weld Leg [agr]: 1.3920 mm

Tube-Tubesheet Jt allowable, 2421.45 is $\geq$ tube strength 2421.45 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	0.0 $\leq$	475.0 N./mm ²	Ok
Tubesheet shear stress	0.0 $\leq$	190.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	0.508	90.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.392	2.320	Ok
Tube-Tubesheet Groove Weld Leg :	1.392	2.320	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

	--Reqd. Thk. + CA		----	Tubesheet	Stresses		Case	Pass/
Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Type	Fail
D1uc	60.215	7.956	213	475	22	190	Ps+Pt D1	Ok
D2uc	60.790	8.108	217	475	22	190	Ps+Pt D2	Ok
D3uc	0.508	...	...	475	...	190	Ps+Pt D3	Ok
Max:	60.7901	8.108	mm	0.456		0.118	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures



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ASME TS Calc: Pressure Test

Case: 1 10:20a Sep 9, 2026

derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[c]:
With or Without Corrosion Allowance

[D1, D2, D3]:
Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:
Because there was no net pressure differential across the tubesheet due to
the specified pressures psd,min and ptd,min; load case D4 was not processed as
the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	60.790	90.000	Ok
Tube thickness :	0.774	2.320	Ok
Tube-tubesheet fillet weld leg :	1.392	2.320	Ok
Tube-tubesheet groove weld leg :	1.392	2.320	Ok

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculations: Step: 14 10:20am Sep 9,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	643.15	9	0.0019436	85.2291
20	30	802.396	647.15	11	0.0023863	88.6381
30	40	No Calc	...	92	No Calc	No Calc
40	50	No Calc	...	87	No Calc	No Calc
50	60	1932.4	651.15	13	0.001168	89.931
60	70	No Calc	644.15	9.5	0.0020483	104.29
70	80	No Calc	...	28.75	No Calc	No Calc
80	90	No Calc	...	44.752	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	12	4.7088	103.419	1325.11
20	30	14	5.31654	103.419	2008.73
30	40	95	88.8492	103.419	No Calc
40	50	90	82.804	...	No Calc
50	60	16	No Calc	...	2393.79
60	70	12.5	No Calc	...	1708.89
70	80	31.75	29.083	...	No Calc
80	90	47.752	44.577	...	No Calc

Minimum

1325

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	802.396	28665.7	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	1932.4	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	63.0251	46372.1	49.5198	47582.2	6.30251
20	30	151.082	210794	119.269	214899	15.1082
30	40	168.514	40052.6	162.56	40216.8	16.8514
40	50	155.255	38187.1	149.574	38374.8	15.5255
50	60	452.821	373361	369.655	384094	45.2821
60	70	72.4448	46372.1	58.8614	47582.2	7.24448
70	80	17.9258	888.061	17.14	941.794	1.79258
80	90	50.4908	6100.12	48.2788	6269.06	5.04908
Total		1131	762126.94	974	779959.38	113



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	46.3438	-51.5958	0.72670E-04	...	Liquid: 10
10	Insl	21.8897	25.4	...	...	Insulation
20	Liqd	210.665	350	...	...	Liquid: 20
20	Insl	45.083	350	...	...	Insulation
20	Nozl	26.8608	350	-360.705	...	N1-Tube inlet
20	Nozl	81.3011	400	427.815	...	N1-Tube Outlet
20	Wght	200	350	...	...	Partition plate
30	Liqd	40.0282	61.5	-40.425	...	Liquid: 30
30	Insl	8.83691	62	...	...	Insulation
40	Liqd	38.1638	61	-40.425	...	Liquid: 40
40	Insl	8.55718	63.5	...	...	Insulation
50	Sadl	77.817	370	434.287	...	LH floating sad
50	Sadl	77.817	1460	434.287	...	Sdl 2 Fr50
50	Liqd	373.133	915	...	...	Liquid: 50
50	Insl	118.508	915	...	...	Insulation
50	Nozl	96.8986	915	433.375	...	N3-Shell inlet
50	Nozl	86.8078	915	-433.375	...	N4-Shell outlet
50	Wght	500	370	-585	...	Fireproofing we
50	Wght	500	1460	-585	...	Wght:[2 of 2]
60	Liqd	46.3438	102.396	0.72670E-04	...	Liquid: 60
60	Insl	24.1691	25.4	...	...	Insulation
30	FTsh	319.64	171.175	...	...	U-Tube
30	Tube	672.834	1041.18	...	...	Tubes
40	Bafl	21.8958	541.05	...	...	Baffles
40	Bafl	21.8958	1082.1	...	...	Baffles
40	Bafl	22.2142	1623.15	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	155.6
Liquid	754.7
Insulation	227.0
Nozzles	291.9
Weights	1200.0
Exchanger Components	1058.5



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Element and Detail Weights: Step: 15 10:20am Sep 9,2026

Liquid in Tubes 112.1

Sum of the Detail Weights 3799.8 kg.

Note:

Because some nozzles were attached to a head or flange, their weights include the attached head or flange.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	1169.5	1169.5	1169.5	1169.5	1169.5	1169.5
Saddles	155.6	155.6	155.6	155.6	155.6	155.6
Nozzles	291.9	291.9	291.9	291.9	291.9	291.9
Wld Weights	1200.0	1200.0	1200.0	1200.0	1200.0	1200.0
Exchanger	1058.5	1058.5	1058.5	1058.5	1058.5	1058.5
Insulation	...	...	...	...	227.0	227.0 *
Ope. Liquid	...	...	...	...	...	754.7
Tube Ope Lqd	...	...	...	...	...	112.1
Test Liquid	...	761.7	...	...	...	...
Tube Tst Lqd	...	112.1	...	...	...	...
Totals	3875.4	4749.2	3875.4	3875.4	4102.5	4969.2

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	3875.4 kg.
Shop Test Wt.	- Fabricated weight + water (full)	4749.2 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	3875.4 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	3875.4 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	4102.5 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	4969.2 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	4800.1 kg.
Field Test Wt.	- Empty weight + test medium (full)	4896.9 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	112128.7 cm ³
Weight of Ope Liq in tubes :	112.1 kg.
Weight of Water in tubes :	112.1 kg.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	5572.57	5143.54
20	30	14231.6	13615.8
30	40	5186	2548.1
40	50	5023.79	2431.4
50	60	37435.4	35595.6
60	70	5635.25	5143.54
70	80	1055.76	328.339



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Element and Detail Weights:

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80	90	2430.29	940.251
----	----	---------	---------

Total	76570.6	65746.6 cm ²
Total	82.4	70.8 ft ²

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Nozzle Flange MAWP:

Step: 16 10:20am Sep 9,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N1-Tube Outlet	4272.0	5100.0	230	300	GR 1.1	2539.5	3868.5	4272.0	3868.5
N4-Shell outle	4781.7	5100.0	82	300	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 4781.667 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 5100.005 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 3868.516 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 5100.005 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N1-Tube Outlet	302.387	0.50	1261	1912	3868.5	3868.5	Pass

Evaluating Flange on Nozzle [N1-Tube Outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(1261) + 4 \cdot \text{abs}(1912) 302.387 \leq \pi \cdot 302.387^3 [(4272.041 - 3868.516) + 500 \cdot 4272.041]$$

$$= 0.22496E+05 \leq 0.22496E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Tube Outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4272.041, 0.500 \cdot 4272.041 + 4272.041 - 22495.7 / (\pi \cdot 302.387^3))$$

$$= \min(4272.041, 3868.516)$$

$$= 3868.516 \text{ KPa.}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation:

Step: 17 10:20am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	349.92 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$\begin{aligned}
 &= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar}))) \\
 &= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.992)/(1 + 1.7 \cdot 3.4 \cdot 0.228))) \\
 &= \min(0.85, 0.921) \\
 &= 0.850
 \end{aligned}$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.509
Structure Height to Diameter ratio	1.547

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Zg = 274320.0$ mm

Effective Height [z]:

$$\begin{aligned}
 &= \text{Centroid Height} + \text{Vessel Base Elevation} \\
 &= 585.0 + 300.0 = 885.0 \text{ mm} \\
 &= 2.904 \text{ ft. Imperial Units}
 \end{aligned}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\begin{aligned}
 &\text{Because } z (2.904 \text{ ft.}) < 15 \text{ ft.} \\
 &= 2.01(15 / Zg)^{2/\alpha} \\
 &= 2.01(15/900.0)^{2/9.5} \\
 &= 0.849
 \end{aligned}$$

Type of Hill: No Hill



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Wind Load Calculation:

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Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0.0 • 0.0 • 0.0)²

= 1.0

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 0.849 • 1.0 • 1.0 • 1.0 • 217.435²)

= 102.7 psf [501.637] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 102.742 • 0.85 • 0.509 • 2.203

= 97.9 lbs. [44.4] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Dish End	885.0	0.000	0.000	0.000	0.849	1.000	501.637
LHS Ch Shell	885.0	0.000	0.000	0.000	0.849	1.000	501.637
LHS Ch Flange	885.0	0.000	0.000	0.000	0.849	1.000	501.637
LHS Shell Flg	885.0	0.000	0.000	0.000	0.849	1.000	501.637
Main Shell	885.0	0.000	0.000	0.000	0.849	1.000	501.637
RHS Dish End	885.0	0.000	0.000	0.000	0.849	1.000	501.637
Tube Inlet	342.9	0.000	0.000	0.000	0.849	1.000	501.637
Shell Inlet fla	358.7	0.000	0.000	0.000	0.849	1.000	501.637

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Partition pla	0.00	300.00	501.64	0.00
Fireproofing	0.00	300.00	501.64	0.00
Wght:[2 of 2]	0.00	300.00	501.64	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	885	963.78	2046.32	501.637	48.8639
20	30	885	968.58	6780.06	501.637	161.901
30	40	885	934.98	1150.03	501.637	27.4614
40	50	885	934.98	1140.68	501.637	27.2381
50	60	885	973.38	17812.9	501.637	425.352
60	70	885	964.98	2052.02	501.637	49
70	80	342.926	122.834	105.456	501.637	2.51817
80	90	358.674	305.41	358.392	501.637	8.55802

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation: Step: 18 10:20am Sep 9,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	0.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_n^x$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(2.9350.75)$
 $= 0.045$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33.0)$
 $= 0.030$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.045, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.157 \cdot 4814 \cdot 1.25$
 $= 283.039$ Kg

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.157 \cdot 4814$
 $= 151.0$ Kg

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight kgf	Element Ope Load kgf
10	20	309.575	601.698	35.3799



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Earthquake Load Calculation: Step: 18 10:20am Sep 9, 2026

20	30	309.575	601.698	35.3799
30	40	309.575	601.698	35.3799
40	50	309.575	601.698	35.3799
50	Sad1	309.575	601.698	35.3799
Sad1	60	309.575	601.698	35.3799
50	60	309.575	601.698	35.3799
60	70	309.575	601.698	35.3799

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 19 10:20am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	2003.975 mm
Center of Gravity of the Liquid	1380.309 mm
Center of Gravity of the Insulation	1512.060 mm
Center of Gravity of the Nozzles	1423.791 mm
Center of Gravity of the Added Weights (Operating)	1736.779 mm
Center of Gravity of the Added Weights (Empty)	1736.779 mm
Center of Gravity of the Tubesheet(s)	921.975 mm
Center of Gravity of the Tubes	1791.975 mm
Center of Gravity of the Baffles	2051.685 mm
Center of Gravity of Bare Shell New and Cold	1396.797 mm
Center of Gravity of Bare Shell Corroded	1380.435 mm
Vessel CG in the Operating Condition	1542.148 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	1565.805 mm
Vessel CG in the Test Condition	1542.180 mm

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Saddle Calcs: Bundle Pull New: Step: 20 10:20am Sep 9,2026

ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	317.57	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	1952.00	mm
Distance from Saddle to Vessel tangent	a1 or a	370.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	315.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	161.04	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	585.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		1200.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		309.39	Kgf
Pressure used in Analysis	P	0.000	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-0.05	-234.00
Long. Stress at Top of Midspan		0.05	138.00
Long. Stress at Bottom of Midspan		0.05	138.00
Long. Stress at Top of Saddles		0.46	138.00
Long. Stress at Bottom of Saddles		-0.26	-234.00
Long. Stress at Bottom of Saddles		0.26	138.00
Tangential Shear in Shell		0.39	110.40
Circ. Stress at Horn of Saddle		0.56	207.00
Circ. Compressive Stress in Shell		0.03	138.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
= Bundle Pull Out Force • Impact Factor • B / Ls
= 1200 • 1.5 • 585.0/1090.0
= 966.1 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:



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Saddle Calcs: Bundle Pull New: Step: 20 10:20am Sep 9, 2026

$$\begin{aligned}
 &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\
 &= -657 + \max(966, 0, 0, 0) \\
 &= 309.4 \text{ Kg}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	387.20	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	1800.00	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2(619.15/2 + 16.0 + 10.0)\sin(120.0/2), 2 \cdot 317.575) \\
 &= 581.233 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)] \\
 &= -309 \cdot 370.0 [1 - (1 - 370.0/1952.0 + (317.575^2 - 161.037^2)/(2 \cdot 370.0 \cdot 1952.0))/(1 + (4 \cdot 161.04)/(3 \cdot 1952.0))] \\
 &= -25.5 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L) \\
 &= 309 \cdot 1952/4((1 + 2(318^2 - 161^2)/(1952^2))/(1 + (4 \cdot 161)/(3 \cdot 1952)) - 4 \cdot 370/1952) \\
 &= 26.9 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t) \\
 &= 0.0 \cdot 317.575/(2 \cdot 16.0) - 26.9/(\pi \cdot 317.6^2 \cdot 16.0) \\
 &= -0.05 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t) \\
 &= 0.0 \cdot 317.575/(2 \cdot 16.0) + 26.9/(\pi \cdot 317.6^2 \cdot 16.0) \\
 &= 0.05 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t) \\
 &= 0.0 \cdot 317.575/(2 \cdot 16.0) - 25.5/(0.1066 \cdot \pi \cdot 317.6^2 \cdot 16.0) \\
 &= 0.46 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 0.0 \cdot 317.575/(2 \cdot 16.0) + 25.5/(0.1923 \cdot \pi \cdot 317.6^2 \cdot 16.0) \\
 &= -0.26 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot h_m/3)) \\
 &= 309(1952.0 - 2 \cdot 370.0)/(1952.0 + (4 \cdot 161.04/3)) \\
 &= 173.1 \text{ Kg}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$



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$$= 1.1707 \cdot 173.06 / (317.575 \cdot 16.0)$$

$$= 0.39 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(317.575 \cdot 16.0)}$$

$$= 55.600 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{(317.575 \cdot 16.0)}, 2 \cdot 370.0)$$

$$= 311.20 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, bundle pull case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 309 \cdot 0.1 / (311.20(16.0 + 1.0 \cdot 10.0))$$

$$= -0.03 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -309 / (4(16.0 + 1.0 \cdot 10.0)311.20) -$$

$$12 \cdot 0.053 \cdot 309 \cdot 317.575 / (1952.0(16.0 + 1.0 \cdot 10.0)^2)$$

$$= -0.56 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	715.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	285.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	275.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(309.575 \cdot 16.0)} + 315.0)^3) / 12$$

$$= 14.500 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 14.5 + 67.967(88.022 - 8.0)^2$$

$$= 4366.709 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	424.8	16.0	8.0	68.0	54373.3	0.437E+04
Wearplate	315.0	10.0	21.0	31.5	66150.0	0.142E+04
Web	10.0	237.4	144.7	23.7	343583.7	0.188E+04
BasePlate	285.0	12.0	269.4	34.2	921433.4	0.113E+05
Totals	...	...	...	157.4	1385540.4	0.189E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1385540 / 157.409$$

$$= 88.022 \text{ mm}$$



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Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120.0/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.204 \cdot 309.386 \\
 &= 62.9668 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 0.0690 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 585.0 - 309.57 \cdot \sin(120.0/2) / 1.0472 \\
 &= 328.984 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 20.7155 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 0.0945 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{3(309 + 78) \cdot 285.0 / (4 \cdot 715.0 \cdot 173.33)} \\
 &= 2.559 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120.0/2) (309.575 + 16.0 + 10.0) \\
 &= 581.233 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 581.2328 / (3 - 1) \\
 &= 290.616 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 290.6164 \cdot 285.0 / 2 \\
 &= 414.128 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 387.203 / (715.0 \cdot 285.0) \\
 &= 0.190 \text{ Kgf/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 414.1 \cdot 0.19 \\
 &= 78.691 \text{ Kgf}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 19.0 + 14.531 \\
 &= 33.531 \text{ cm}^2
 \end{aligned}$$



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Saddle Calcs: Bundle Pull New: Step: 20 10:20am Sep 9, 2026

Compressive Stress [Sc]:

= P/Ar

= 78.7/33.5308

= 0.230 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

= Saddle Width - Web Thickness

= 200.0 - 10.0

= 190.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/33.531

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 200.0/2

= 100.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{666.7/33.531}$

= 44.589 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 585.0 - cos(120.0/2)(309.575 + 16.0 + 10.0) - 12.0

= 405.213 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 405.213/44.589

= 9.088

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 1800.0 / (2 • 715.0) • 290.616 • 405.21/2

= 74.117 Kg-m.

Compressive Allowable, KL/r < Cc (9.0876 < 123.2105) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (9.09)^2 / (2 \cdot 123.21^2)) 260 /$

$(5/3 + 3 \cdot (9.09) / (8 \cdot 123.21) - (9.09^3) / (8 \cdot 123.21^3))$

= 153 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 0.23/153.04 + (74.12 • 100.0/6666667) / 173.33

= 0.064



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Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	290.6	10.0	0.0	29.1	0.0	2.42
Totals	...	...	...	48.1	...	669.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/48.062$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 585.0 - \cos(335.575/2)(12.0 + 0.0 + 0.0) - 0.0$$

$$= 237.425 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(669.0/48.062)}$$

$$= 37.309 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 237.425/37.309$$

$$= 6.364$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 1800.0 / (2 \cdot 715.0)$$

$$= 1.259 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.259 \cdot 290.616 \cdot 237.425$$

$$= 86.854 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 86.854/66900.5$$

$$= 12.732 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($6.3637 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.36)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (6.36) / (8 \cdot 123.21) - (6.36^3) / (8 \cdot 123.21^3))$$

$$= 154 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 0.29/154.0 + (86.85 \cdot 100.0/669.004) / 173.33$$

$$= 0.075$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A



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Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6476 cm ²
Saddle Load Q0 (Weight)	Q0	387.2 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1800.0 Kgf
Maximum Transverse Force	Ft	0.0 Kgf
Maximum Longitudinal Force	F1	1800.0 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

= $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $1800.0 / (1.648 \cdot 4)$

= 26.8 N./mm². Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $0.0 / (1.648 \cdot 4)$

= 0.0 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= $(QL - Q0) / (Stba \cdot Rnbolts)$

= $(1800.0 - 387.2) / (48.3 \cdot 4.0)$

= 0.7171 cm²

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= $1800.0 / (28.98 \cdot 4.0)$

= 1.5228 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$

= $585.0 \cdot 0.0 + 0.0$

= 0.00 Kg-m.

Eccentricity (e):

= $Rmom / Q0$

= $0.0 / 387.2$

= 0.00 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$

= $\max[0.7171, 1.5228, 0.0]$

= 1.5228 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	319.07	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	1952.00	mm
Distance from Saddle to Vessel tangent	a1 or a	370.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	315.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	162.54	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	585.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		1200.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		223.20	Kgf
Pressure used in Analysis	P	0.000	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-0.05	-234.00
Long. Stress at Top of Midspan	0.05	138.00
Long. Stress at Bottom of Midspan	0.05	138.00
Long. Stress at Top of Saddles	0.41	138.00
Long. Stress at Bottom of Saddles	-0.23	-234.00
Long. Stress at Bottom of Saddles	0.23	138.00
Tangential Shear in Shell	0.35	110.40
Circ. Stress at Horn of Saddle	0.51	207.00
Circ. Compressive Stress in Shell	0.02	138.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 1200 • 1.5 • 585.0/1090.0
 = 966.1 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:



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$$\begin{aligned}
 &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\
 &= -743 + \max(966, 0, 0, 0) \\
 &= 223.2 \text{ Kg}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	301.02	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	1800.00	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2(619.15/2 + 16.0 + 10.0)\sin(120.0/2), 2 \cdot 319.075) \\
 &= 581.233 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)] \\
 &= -223 \cdot 370.0 [1 - (1 - 370.0/1952.0 + (319.075^2 - 162.537^2)/(2 \cdot 370.0 \cdot 1952.0))/(1 + (4 \cdot 162.54)/(3 \cdot 1952.0))] \\
 &= -18.5 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L) \\
 &= 223 \cdot 1952/4((1 + 2(319^2 - 163^2)/(1952^2))/(1 + (4 \cdot 163)/(3 \cdot 1952)) - 4 \cdot 370/1952) \\
 &= 19.3 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t) \\
 &= 0.0 \cdot 319.075/(2 \cdot 13.0) - 19.3/(\pi \cdot 319.1^2 \cdot 13.0) \\
 &= -0.05 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\
 &= 0.0 \cdot 319.075/(2 \cdot 13.0) + 19.3/(\pi \cdot 319.1^2 \cdot 13.0) \\
 &= 0.05 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 0.0 \cdot 319.075/(2 \cdot 13.0) - 18.5/(0.1066 \cdot \pi \cdot 319.1^2 \cdot 13.0) \\
 &= 0.41 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M1/(K1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 0.0 \cdot 319.075/(2 \cdot 13.0) + 18.5/(0.1923 \cdot \pi \cdot 319.1^2 \cdot 13.0) \\
 &= -0.23 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a)/(L + (4 \cdot h_m/3)) \\
 &= 223(1952.0 - 2 \cdot 370.0)/(1952.0 + (4 \cdot 162.54/3)) \\
 &= 124.7 \text{ Kg}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$



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$$= 1.1707 \cdot 124.74 / (319.075 \cdot 13.0)$$

$$= 0.35 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(319.075 \cdot 13.0)}$$

$$= 50.236 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{(319.075 \cdot 13.0)}, 2 \cdot 370.0)$$

$$= 300.47 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, bundle pull case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 223 \cdot 0.1 / (300.471(13.0 + 1.0 \cdot 10.0))$$

$$= -0.02 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -223 / (4(13.0 + 1.0 \cdot 10.0)300.471) -$$

$$12 \cdot 0.053 \cdot 223 \cdot 319.075 / (1952.0(13.0 + 1.0 \cdot 10.0)^2)$$

$$= -0.51 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	715.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	285.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	275.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(312.575 \cdot 13.0)} + 315.0)13.0^3) / 12$$

$$= 7.588 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 7.588 + 53.878(93.452 - 6.5)^2$$

$$= 4081.080 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	414.4	13.0	6.5	53.9	35020.4	0.408E+04
Wearplate	315.0	10.0	18.0	31.5	56700.0	0.180E+04
Web	10.0	237.4	141.7	23.7	336460.9	0.167E+04
BasePlate	285.0	12.0	266.4	34.2	911173.5	0.102E+05
Totals	...	...	...	143.3	1339354.8	0.178E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1339355 / 143.32$$

$$= 93.452 \text{ mm}$$



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Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120.0/2 \\
 &= 120.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\
 &= 0.2035
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.204 \cdot 223.201 \\
 &= 45.4263 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 0.0498 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 585.0 - 312.57 \cdot \sin(120.0/2) / 1.0472 \\
 &= 326.503 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 14.8321 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 0.0764 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{(3(223 + 78) \cdot 285.0) / (4 \cdot 715.0 \cdot 173.33)} \\
 &= 2.256 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120.0/2) (309.575 + 16.0 + 10.0) \\
 &= 581.233 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 581.2328 / (3 - 1) \\
 &= 290.616 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 290.6164 \cdot 285.0 / 2 \\
 &= 414.128 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 301.018 / (715.0 \cdot 285.0) \\
 &= 0.148 \text{ Kgf/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 414.1 \cdot 0.15 \\
 &= 61.175 \text{ Kgf}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 19.0 + 14.531 \\
 &= 33.531 \text{ cm}^2
 \end{aligned}$$



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Compressive Stress [Sc]:

= P/Ar

= 61.2/33.5308

= 0.179 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

= Saddle Width - Web Thickness

= 200.0 - 10.0

= 190.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/33.531

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 200.0/2

= 100.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{666.7/33.531}$

= 44.589 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 585.0 - cos(120.0/2)(309.575 + 16.0 + 10.0) - 12.0

= 405.213 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 405.213/44.589

= 9.088

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 1800.0 / (2 • 715.0) • 290.616 • 405.21/2

= 74.117 Kg-m.

Compressive Allowable, KL/r < Cc (9.0876 < 123.2105) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (9.09)^2 / (2 \cdot 123.21^2)) 260 /$

$(5/3 + 3 \cdot (9.09) / (8 \cdot 123.21) - (9.09^3) / (8 \cdot 123.21^3))$

= 153 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 0.18/153.04 + (74.12 • 100.0/6666667) / 173.33

= 0.064



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Saddle Calcs: Bundle Pull Cor: Step: 21 10:20am Sep 9, 2026

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	290.6	10.0	0.0	29.1	0.0	2.42
Totals	...	...	...	48.1	...	669.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/48.062$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 585.0 - \cos(335.575/2)(12.0 + 0.0 + 0.0) - 0.0$$

$$= 237.425 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(669.0/48.062)}$$

$$= 37.309 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 237.425/37.309$$

$$= 6.364$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 1800.0 / (2 \cdot 715.0)$$

$$= 1.259 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.259 \cdot 290.616 \cdot 237.425$$

$$= 86.854 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 86.854/66900.5$$

$$= 12.732 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($6.3637 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.36)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (6.36) / (8 \cdot 123.21) - (6.36^3) / (8 \cdot 123.21^3))$$

$$= 154 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 0.22/154.0 + (86.85 \cdot 100.0/669.004) / 173.33$$

$$= 0.075$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A



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Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6476 cm ²
Saddle Load Q0 (Weight)	Q0	301.0 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	1800.0 Kgf
Maximum Transverse Force	Ft	0.0 Kgf
Maximum Longitudinal Force	F1	1800.0 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1800.0 / (1.648 \cdot 4)$$

$$= 26.8 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0.0 / (1.648 \cdot 4)$$

$$= 0.0 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= (QL - Q0) / (Stba \cdot Rnbolts)$$

$$= (1800.0 - 301.02) / (48.3 \cdot 4.0)$$

$$= 0.7609 \text{ cm}^2$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 1800.0 / (28.98 \cdot 4.0)$$

$$= 1.5228 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 585.0 \cdot 0.0 + 0.0$$

$$= 0.00 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 0.0 / 301.02$$

$$= 0.00 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.7609, 1.5228, 0.0]$$

$$= 1.5228 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Road Transport:

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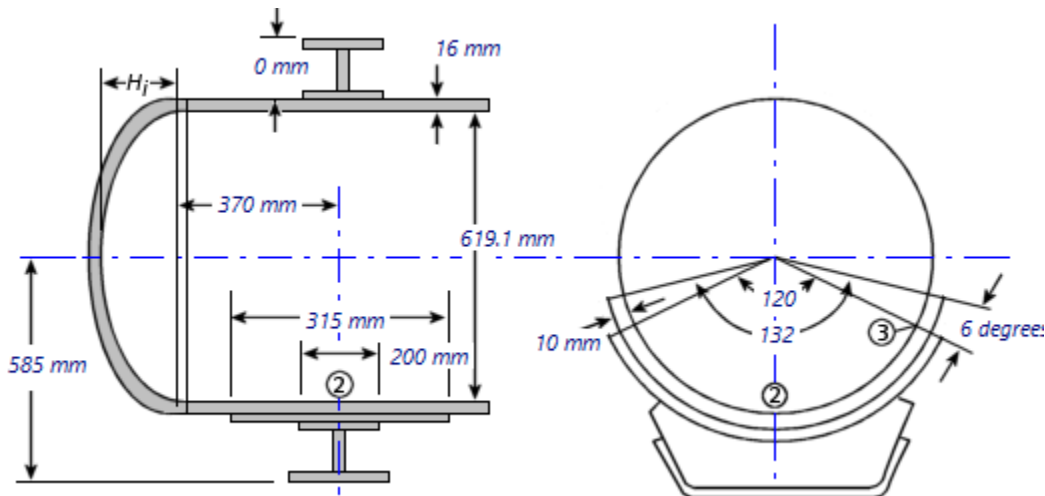
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	317.57	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	1952.00	mm
Distance from Saddle to Vessel tangent	a1 or a	370.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	315.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	585.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		3912.79	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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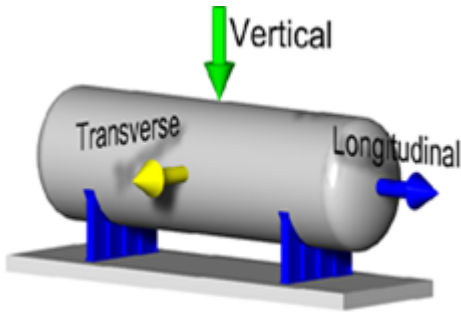
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Saddle Calcs: Road Transport:

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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-1.09	-234.00
Long. Stress at Top of Midspan	1.09	138.00
Long. Stress at Bottom of Midspan	1.09	138.00
Long. Stress at Top of Saddles	3.15	138.00
Long. Stress at Bottom of Saddles	-1.74	-234.00
Long. Stress at Bottom of Saddles	1.74	138.00
Tangential Shear in Shell	5.49	110.40
Circ. Stress at Horn of Saddle	7.04	207.00
Circ. Compressive Stress in Shell	0.36	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 33.85 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \\
 &= 287.5 \text{ lbf} \\
 &= 130.416 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 2.661^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \cdot 1.2 \\
 &= 56.7 \text{ lbf} \\
 &= 25.7 \text{ Kg}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 3719.8 \cdot 1.0 + 130.42 \\
 &= 502.4 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 3719.8 \cdot 1.0 + 25.72 \\
 &= 397.7 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 502.4/2 \cdot 585.0/581.2329 \\
 &= 252.8 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 398 \cdot 585.0/1090.0 \\
 &= 213.4 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 3485.7 \cdot 0.05
 \end{aligned}$$



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= 174.284 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= 3486 + 174 + max(213, 253)

= 3912.8 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 3990.60 Kgf

Transverse Shear Load 251.20 Kgf

Longitudinal Shear Load 397.70 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(619.15/2 + 16.0 + 10.0)sin(120.0/2), 2•317.575)

= 581.233 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -3913•370.0[1-(1-370.0/1952.0+(317.575²-0.0²)/

(2•370.0•1952.0))/(1+(4•0.0)/(3•1952.0))]

= -173.3 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 3913•1952/4((1+2(318²-0²)/(1952²))/(1+(4•0)/

(3•1952))-4•370/1952)

= 562.8 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0.0 • 317.575/(2•16.0) - 562.8/(π•317.6²•16.0)

= -1.09 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0.0 • 317.575/(2 • 16.0) + 562.8/(π • 317.6² • 16.0)

= 1.09 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)

= 0.0•317.575/(2•16.0)--173.3/(0.1066•π•317.6²•16.0)

= 3.15 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 0.0•317.575/(2•16.0)+-173.3/(0.1923•π•317.6²•16.0)

= -1.74 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 3913(1952.0 - 2 • 370.0)/(1952.0 + (4 • 0.0/3))

= 2429.5 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 2429.46 / (317.575 \cdot 16.0)$$

$$= 5.49 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(317.575 \cdot 16.0)}$$

$$= 55.600 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{(317.575 \cdot 16.0)}, 2 \cdot 370.0)$$

$$= 311.20 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 3913 \cdot 0.1 / (311.201(16.0 + 1.0 \cdot 10.0))$$

$$= -0.36 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -3913 / (4(16.0 + 1.0 \cdot 10.0)311.201) -$$

$$12 \cdot 0.053 \cdot 3913 \cdot 317.575 / (1952.0(16.0 + 1.0 \cdot 10.0)^2)$$

$$= -7.04 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	715.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	285.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	275.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(309.575 \cdot 16.0)} + 315.0)16.0^3) / 12$$

$$= 14.500 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 14.5 + 67.967(88.022 - 8.0)^2$$

$$= 4366.709 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	424.8	16.0	8.0	68.0	54373.3	0.437E+04
Wearplate	315.0	10.0	21.0	31.5	66150.0	0.142E+04
Web	10.0	237.4	144.7	23.7	343583.7	0.188E+04
BasePlate	285.0	12.0	269.4	34.2	921433.4	0.113E+05
Totals	...	...	...	157.4	1385540.4	0.189E+05

Distance to Centroid [C1]:

$$= AY / A$$



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$$= 1385540/157.409$$

$$= 88.022 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 3912.785$$

$$= 796.3367 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 0.8731 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 585.0 - 309.57 \cdot \sin(120.0/2) / 1.0472$$

$$= 328.984 \text{ mm}$$

Bending Moment, M = Fh * d = 261.9868 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 1.1952 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(3913 + 78)285.0) / (4 \cdot 715.0 \cdot 173.33)}$$

$$= 8.216 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (309.575 + 16.0 + 10.0)$$

$$= 581.233 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 581.2328 / (3 - 1)$$

$$= 290.616 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 290.6164 \cdot 285.0/2$$

$$= 414.128 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3990.602 / (715.0 \cdot 285.0)$$

$$= 1.958 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 414.1 \cdot 1.96$$

$$= 811.003 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



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$$= 19.0 + 14.531$$

$$= 33.531 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 811.0/33.5308$$

$$= 2.372 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200.0 - 10.0$$

$$= 190.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/33.531$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0/2$$

$$= 100.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(666.7/33.531)}$$

$$= 44.589 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 585.0 - \cos(120.0/2) (309.575 + 16.0 + 10.0) - 12.0$$

$$= 405.213 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 405.213/44.589$$

$$= 9.088$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 397.7 / (2 \cdot 715.0) \cdot 290.616 \cdot 405.21/2$$

$$= 16.376 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($9.0876 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.09)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (9.09) / (8 \cdot 123.21) - (9.09)^3 / (8 \cdot 123.21^3))$$

$$= 153 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.37/153.04 + (16.38 \cdot 100.0/6666667) / 173.33$$



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= 0.029

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	290.6	10.0	0.0	29.1	0.0	2.42
Totals	...	...	...	48.1	...	669.

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/48.062

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 200.0/2

= 100.000 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 585.0 - cos(335.575/2)(12.0 + 0.0 + 0.0) - 0.0

= 237.425 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{669.0/48.062}$

= 37.309 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 237.425/37.309

= 6.364

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 397.699/(2 • 715.0)

= 0.278 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.278 • 290.616 • 237.425

= 19.190 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σB,base,c]:

= Bending Moment / Section Modulus

= 19.19/66900.5

= 2.813 N./mm²

Compressive Allowable, KL/r < Cc (6.3637 < 123.2105) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (6.36)^2 / (2 \cdot 123.21^2)) 260 /$

$(5/3 + 3 \cdot (6.36) / (8 \cdot 123.21) - (6.36^3) / (8 \cdot 123.21^3))$

= 154 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 2.94/154.0 + (19.19 • 100.0/669.004)/173.33

= 0.035

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



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Vessel Mean Radius	Rm	317.57	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	1952.00	mm
Distance from Saddle to Vessel tangent	a1 or a	370.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	315.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	161.04	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	585.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		577.68	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-0.10	-234.00
Long. Stress at Top of Midspan	0.10	138.00
Long. Stress at Bottom of Midspan	0.10	138.00
Long. Stress at Top of Saddles	0.87	138.00
Long. Stress at Bottom of Saddles	-0.48	-234.00
Long. Stress at Bottom of Saddles	0.48	138.00
Tangential Shear in Shell	0.73	110.40
Circ. Stress at Horn of Saddle	1.04	207.00
Circ. Compressive Stress in Shell	0.05	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 33.85 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \\
 &= 287.5 \text{ lbf} \\
 &= 130.416 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 2.661^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \cdot 1.2 \\
 &= 56.7 \text{ lbf} \\
 &= 25.7 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:



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$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + F_{tw} \\
 &= 0.1 \cdot 3719.8 \cdot 1.0 + 130.42 \\
 &= 502.4 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [FI]:

$$\begin{aligned}
 &= gx \cdot \text{Shipping Weight} \cdot gw + F_{lw} \\
 &= 0.1 \cdot 3719.8 \cdot 1.0 + 25.72 \\
 &= 397.7 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [Fwt]:

$$\begin{aligned}
 &= F_{t}/\text{Num of Saddles} \cdot B / E \\
 &= 502.4/2 \cdot 585.0/581.2329 \\
 &= 252.8 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [Fwl]:

$$\begin{aligned}
 &= F_{l} \cdot B / L_s \\
 &= 398 \cdot 585.0/1090.0 \\
 &= 213.4 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 309.4 \cdot 0.05 \\
 &= 15.469 \text{ Kgf}
 \end{aligned}$$

Load Combination Results due to Transport Loading [Q]:

$$\begin{aligned}
 &= \text{Saddle Load} + F_{rv} + \max(F_{wl}, F_{wt}) \\
 &= 309 + 15 + \max(213, 253) \\
 &= 577.7 \text{ Kgf}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	655.50	Kgf
Transverse Shear Load	251.20	Kgf
Longitudinal Shear Load	397.70	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2(619.15/2 + 16.0 + 10.0)\sin(120.0/2), 2 \cdot 317.575) \\
 &= 581.233 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m^2)/(3L))] \\
 &= -578 \cdot 370.0 [1 - (1 - 370.0/1952.0 + (317.575^2 - 161.037^2)/(2 \cdot 370.0 \cdot 1952.0))/(1 + (4 \cdot 161.04)/(3 \cdot 1952.0))] \\
 &= -47.7 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L) \\
 &= 578 \cdot 1952/4((1 + 2(318^2 - 161^2)/(1952^2))/(1 + (4 \cdot 161)/(3 \cdot 1952)) - 4 \cdot 370/1952) \\
 &= 50.2 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$



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$$= 0.0 \cdot 317.575 / (2 \cdot 16.0) - 50.2 / (\pi \cdot 317.6^2 \cdot 16.0)$$

$$= -0.10 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 317.575 / (2 \cdot 16.0) + 50.2 / (\pi \cdot 317.6^2 \cdot 16.0)$$

$$= 0.10 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 317.575 / (2 \cdot 16.0) - 47.7 / (0.1066 \cdot \pi \cdot 317.6^2 \cdot 16.0)$$

$$= 0.87 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 317.575 / (2 \cdot 16.0) + 47.7 / (0.1923 \cdot \pi \cdot 317.6^2 \cdot 16.0)$$

$$= -0.48 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 578(1952.0 - 2 \cdot 370.0) / (1952.0 + (4 \cdot 161.04 / 3))$$

$$= 323.1 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 323.14 / (317.575 \cdot 16.0)$$

$$= 0.73 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(317.575 \cdot 16.0)}$$

$$= 55.600 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{(317.575 \cdot 16.0)}, 2 \cdot 370.0)$$

$$= 311.20 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 578 \cdot 0.1 / (311.201(16.0 + 1.0 \cdot 10.0))$$

$$= -0.05 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -578 / (4(16.0 + 1.0 \cdot 10.0)311.201) -$$

$$12 \cdot 0.053 \cdot 578 \cdot 317.575 / (1952.0(16.0 + 1.0 \cdot 10.0)^2)$$

$$= -1.04 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	715.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	285.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	275.0	mm
Friction Coefficient	mu	0.000	



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Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(309.575 \cdot 16.0)} + 315.0)^3) / 12$$

$$= 14.500 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 14.5 + 67.967 (88.022 - 8.0)^2$$

$$= 4366.709 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	424.8	16.0	8.0	68.0	54373.3	0.437E+04
Wearplate	315.0	10.0	21.0	31.5	66150.0	0.142E+04
Web	10.0	237.4	144.7	23.7	343583.7	0.188E+04
BasePlate	285.0	12.0	269.4	34.2	921433.4	0.113E+05
Totals	...	...	...	157.4	1385540.4	0.189E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1385540 / 157.409$$

$$= 88.022 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 577.682$$

$$= 117.5708 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.1289 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 585.0 - 309.57 \cdot \sin(120.0 / 2) / 1.0472$$

$$= 328.984 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 38.6796 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.1765 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(578 + 78)285.0) / (4 \cdot 715.0 \cdot 173.33)}$$

$$= 3.330 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0 / 2) (309.575 + 16.0 + 10.0)$$

$$= 581.233 \text{ mm}$$



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Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 581.2328 / (3 - 1)$$

$$= 290.616 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 290.6164 \cdot 285.0 / 2$$

$$= 414.128 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 655.499 / (715.0 \cdot 285.0)$$

$$= 0.322 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 414.1 \cdot 0.32$$

$$= 133.216 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19.0 + 14.531$$

$$= 33.531 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 133.2 / 33.5308$$

$$= 0.390 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200.0 - 10.0$$

$$= 190.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0 / 33.531$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0 / 2$$

$$= 100.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (666.7 / 33.531) }$$

$$= 44.589 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 585.0 - \cos(120.0 / 2) (309.575 + 16.0 + 10.0) - 12.0$$

$$= 405.213 \text{ mm}$$

Intermediate Term [Cc]:



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$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 405.213 / 44.589$$

$$= 9.088$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 397.7 / (2 \cdot 715.0) \cdot 290.616 \cdot 405.21 / 2$$

$$= 16.376 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($9.0876 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.09)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (9.09) / (8 \cdot 123.21) - (9.09)^3 / (8 \cdot 123.21^3))$$

$$= 153 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 0.39 / 153.04 + (16.38 \cdot 100.0 / 6666667) / 173.33$$

$$= 0.016$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	290.6	10.0	0.0	29.1	0.0	2.42
Totals	...	...	...	48.1	...	669.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 48.062$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0 / 2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 585.0 - \cos(335.575/2) (12.0 + 0.0 + 0.0) - 0.0$$

$$= 237.425 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(669.0 / 48.062)}$$

$$= 37.309 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 237.425 / 37.309$$

$$= 6.364$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 397.699 / (2 \cdot 715.0)$$

$$= 0.278 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:



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$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 0.278 \cdot 290.616 \cdot 237.425 \\
 &= 19.190 \text{ Kg-m.}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B, base, c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 19.19 / 66900.5 \\
 &= 2.813 \text{ N./mm}^2
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($6.3637 < 123.2105$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (6.36)^2 / (2 \cdot 123.21^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (6.36) / (8 \cdot 123.21) - (6.36^3) / (8 \cdot 123.21^3)) \\
 &= 154 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{\text{base}, c} \cdot C_{1/I}) / S_{ba} \\
 &= 0.48 / 154.0 + (19.19 \cdot 100.0 / 669.004) / 173.33 \\
 &= 0.019
 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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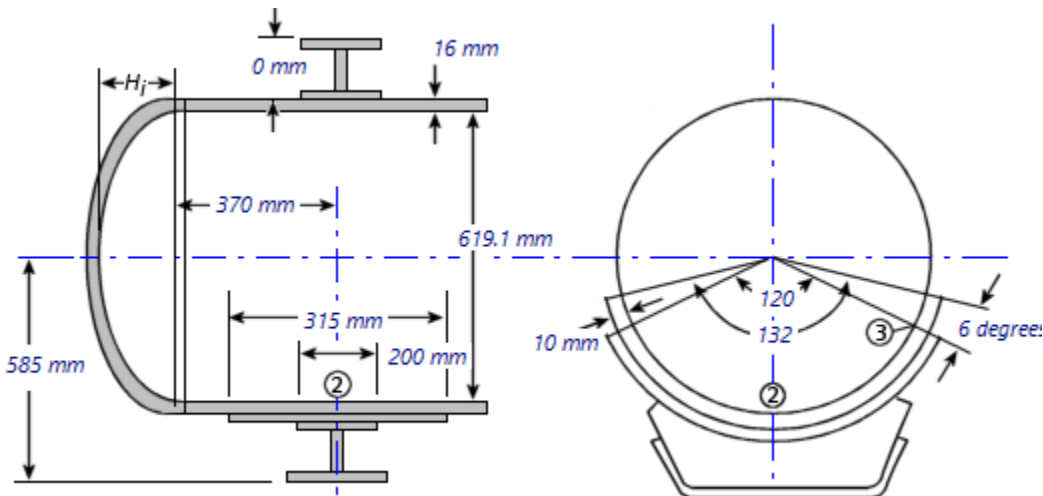
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	319.07	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	1952.00	mm
Distance from Saddle to Vessel tangent	a1 or a	370.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	315.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	585.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	5797.75	Kgf
Pressure used in Analysis	P 4153.033	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	49.00	138.00
Long. Stress at Bottom of Midspan	52.94	138.00
Long. Stress at Top of Saddles	56.62	138.00
Long. Stress at Bottom of Saddles	47.84	138.00
Tangential Shear in Shell	9.96	110.40
Circ. Stress at Horn of Saddle	13.20	172.50
Circ. Compressive Stress in Shell	0.63	138.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(750.9/2 + 0) \cdot 585.0/581.2329$$

$$= 1133.6 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(410.62, 0) \cdot 585.0/1090.0$$

$$= 220.4 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(283.04, 0) \cdot 585.0/1090.0$$

$$= 151.9 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(283/2 + 0) \cdot 585.0/581.2329$$

$$= 427.3 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 4664 + \max(220, 1134, 152, 427)$$

$$= 5797.7 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 4919.274(0.0 + \pi/4(0.811 \cdot 1.2)^2)$$

$$= 4026.721 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	5875.56	Kgf
Transverse Shear Load	375.45	Kgf
Longitudinal Shear Load	410.62	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(619.15/2 + 16.0 + 10.0)\sin(120.0/2), 2 \cdot 319.075)$$

$$= 581.233 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -5798 \cdot 370.0 [1 - (1 - 370.0/1952.0 + (319.075^2 - 0.0^2)/(2 \cdot 370.0 \cdot 1952.0))/(1 + (4 \cdot 0.0)/(3 \cdot 1952.0))]$$

$$= -255.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 5798 \cdot 1952/4((1 + 2(319^2 - 0^2)/(1952^2))/(1 + (4 \cdot 0)/$$



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$$(3 \cdot 1952) - 4 \cdot 370 / 1952 \\ = 835.3 \text{ Kg-m.}$$

$$\text{Longitudinal Stress at Top of Shell (4.15.4) } [\sigma_1]: \\ = P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t) \\ = 4153.033 \cdot 319.075 / (2 \cdot 13.0) - 835.3 / (\pi \cdot 319.1^2 \cdot 13.0) \\ = 49.00 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Bottom of Shell (4.15.5) } [\sigma_2]: \\ = P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 4153.033 \cdot 319.075 / (2 \cdot 13.0) + 835.3 / (\pi \cdot 319.1^2 \cdot 13.0) \\ = 52.94 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Top of Shell at Support (4.15.8) } [\sigma^*3]: \\ = P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t) \\ = 4153.033 \cdot 319.075 / (2 \cdot 13.0) - 255.4 / (0.1066 \cdot \pi \cdot 319.1^2 \cdot 13.0) \\ = 56.62 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Bottom of Shell at Support (4.15.9) } [\sigma^*4]: \\ = P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 4153.033 \cdot 319.075 / (2 \cdot 13.0) + 255.4 / (0.1923 \cdot \pi \cdot 319.1^2 \cdot 13.0) \\ = 47.84 \text{ N./mm}^2$$

$$\text{Maximum Shear Force in the Saddle (4.15.3) } [T]: \\ = Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 5798(1952.0 - 2 \cdot 370.0) / (1952.0 + (4 \cdot 0.0 / 3)) \\ = 3599.8 \text{ Kgf}$$

$$\text{Shear Stress in the shell no rings, not stiffened (4.15.12) } [\tau_2]: \\ = K_2 \cdot T / (R_m \cdot t) \\ = 1.1707 \cdot 3599.83 / (319.075 \cdot 13.0) \\ = 9.96 \text{ N./mm}^2$$

$$\text{Decay Length (4.15.20) } [x_1, x_2]: \\ = 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{319.075 \cdot 13.0} \\ = 50.236 \text{ mm}$$

$$\text{Effective reinforcing plate width (4.15.24) } [B_1]: \\ = \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(200.0 + 1.56 \cdot \sqrt{319.075 \cdot 13.0}, 2 \cdot 370.0) \\ = 300.47 \text{ mm}$$

$$\text{Wear Plate/Shell Stress ratio (4.15.26) } [\eta]: \\ = \min(S_r / S, 1) \\ = \min(137.997 / 137.997, 1) \\ = 1.0000$$

$$\text{Circumferential Stress at Saddle Base with Wear Plate (4.15.25) } [\sigma_6, r]: \\ = -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr)) \\ = -0.7603 \cdot 5798 \cdot 0.1 / (300.471(13.0 + 1.0 \cdot 10.0)) \\ = -0.63 \text{ N./mm}^2$$

$$\text{Circ. Comp. Stress at Horn of Saddle, } L < 8R_m \text{ (4.15.29) } [\sigma_7, r^*]: \\ = -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2) \\ = -5798 / (4(13.0 + 1.0 \cdot 10.0)300.471) - \\ 12 \cdot 0.053 \cdot 5798 \cdot 319.075 / (1952.0(13.0 + 1.0 \cdot 10.0)^2) \\ = -13.20 \text{ N./mm}^2$$

$$\text{Distance between Saddle Supports } [L_s]: \\ = 1090.0 \text{ mm}$$

$$\text{Free Un-Restrained Thermal Expansion between the Saddles } [\text{Exp}]: \\ = \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.000012 \cdot 1090.0(82.0 - 21.1)$$



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= 0.795 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	715.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	285.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	275.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(312.575 \cdot 13.0)} + 315.0)13.0^3)/12$$

$$= 7.588 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 7.588 + 53.878(93.452 - 6.5)^2$$

$$= 4081.080 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	414.4	13.0	6.5	53.9	35020.4	0.408E+04
Wearplate	315.0	10.0	18.0	31.5	56700.0	0.180E+04
Web	10.0	237.4	141.7	23.7	336460.9	0.167E+04
BasePlate	285.0	12.0	266.4	34.2	911173.5	0.102E+05
Totals	...	...	...	143.3	1339354.8	0.178E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1339355/143.32$$

$$= 93.452 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 5797.745$$

$$= 1179.9668 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 1.2938 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 585.0 - 312.57 \cdot \sin(120.0/2) / 1.0472$$

$$= 326.503 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 385.2698 \text{ Kg-m.}$$



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Bending Stress, $S_b = (M * C_1 / I) = 1.9856 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(5798 + 78)285.0 / (4 * 715.0 * 174.663))}$
 $= 9.931 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120.0/2) (309.575 + 16.0 + 10.0)$
 $= 581.233 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 581.2328 / (3 - 1)$
 $= 290.616 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 290.6164 * 285.0 / 2$
 $= 414.128 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 5875.562 / (715.0 * 285.0)$
 $= 2.883 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 414.1 * 2.88$
 $= 1194.080 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 19.0 + 14.531$
 $= 33.531 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 1194.1 / 33.5308$
 $= 3.492 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 200.0 - 10.0$
 $= 190.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0 / 33.531$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



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= Saddle Width / 2
= 200.0/2
= 100.000 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{666.7/33.531}$
= 44.589 mm

Length of Outer Rib [L]:
= Saddle Height - $\cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
= 585.0 - $\cos(120.0/2) (309.575 + 16.0 + 10.0) - 12.0$
= 405.213 mm

Intermediate Term [Cc]:
= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
= $\sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/262.0)}$
= 122.739

Slenderness ratio [KL/r]:
= KL/r
= 1 • 405.213/44.589
= 9.088

Bending Moment [Rm]:
= $Fl / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
= 410.6/(2 • 715.0) • 290.616 • 405.21/2
= 16.908 Kg-m.

Compressive Allowable, $KL/r < Cc$ (9.0876 < 122.7394) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2/(2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r)/(8 \cdot Cc) - (KL/r)^3/(8 \cdot Cc^3))$
= $(1 - (9.09)^2/(2 \cdot 122.74^2)) 262 / (5/3 + 3 \cdot (9.09)/(8 \cdot 122.74) - (9.09^3)/(8 \cdot 122.74^3))$
= 154.2 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
= $Sc/Sca + (Rm \cdot C1 / I) / Sba$
= 3.49/154.2 + (16.91 • 100.0/6666667)/174.66
= 0.037

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	290.6	10.0	0.0	29.1	0.0	2.42
Totals	...	...	...	48.1	...	669.

Distance to Centroid from Datum [ytot]:
= AY / A
= 0.0/48.062
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 200.0/2
= 100.000 mm

Length of Inner Rib [L]:
= Saddle Height - Outside Radius - Bpthk
= 585.0 - $\cos(335.575/2) (12.0 + 0.0 + 0.0) - 0.0$
= 237.425 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{669.0 / 48.062}$$

$$= 37.309 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 237.425 / 37.309$$

$$= 6.364$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 410.617 / (2 \cdot 715.0)$$

$$= 0.287 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.287 \cdot 290.616 \cdot 237.425$$

$$= 19.813 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 19.813 / 66900.5$$

$$= 2.904 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($6.3637 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.36)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (6.36) / (8 \cdot 122.74) - (6.36)^3 / (8 \cdot 122.74^3))$$

$$= 155.2 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 4.34 / 155.18 + (19.81 \cdot 100.0 / 669.004) / 174.66$$

$$= 0.045$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6476 cm ²
Saddle Load Q0 (Weight)	Q0	4741.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	220.4 Kg
Maximum Transverse Force	Ft	375.4 Kg
Maximum Longitudinal Force	F1	410.6 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 410.6 / (1.648 \cdot 4)$$

$$= 6.1 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 375.4 / (1.648 \cdot 4)$$

$$= 5.6 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (BoltShearAllowable \cdot Nbolts)$
= $410.62 / (82.8 \cdot 4.0)$
= 0.1216 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $585.0 \cdot 375.45 + 0.0$
= 219.64 Kg-m.

Eccentricity (e):
= $Rmom / QO$
= $219.64 / 4741.92$
= 46.32 mm < $B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.0, 0.1216, 0.0]$
= 0.1216 cm²

MDMT Calculations for: LH floating sad, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10.0, c = 3.0 mm, $E^* = 1.0$
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.851$, Temp. Reduction = 8 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	319.07	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	1952.00	mm
Distance from Saddle to Vessel tangent	a1 or a	370.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	315.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	162.54	mm
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	585.00	mm
Coefficient of Friction	mu	0.00	



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Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 1509.33 Kgf
Pressure used in Analysis P 4153.033 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	50.66	138.00
Long. Stress at Bottom of Midspan	51.28	138.00
Long. Stress at Top of Saddles	53.73	138.00
Long. Stress at Bottom of Saddles	49.44	138.00
Tangential Shear in Shell	2.33	110.40
Circ. Stress at Horn of Saddle	3.44	172.50
Circ. Compressive Stress in Shell	0.16	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3.0(750.9/2 + 0) • 585.0/581.2329
= 1133.6 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
= max(Fl,wind; Sum of X Forces) • B / Ls
= max(410.62, 0) • 585.0/1090.0
= 220.4 Kgf

Saddle Reaction Force due to Earthquake Fl [Fsl]:
= max(Fl,eq; Sum of X Forces) • B / Ls
= max(283.04, 0) • 585.0/1090.0
= 151.9 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:
= Ftr(Ft/Num of Saddles + Z Force Load) • B / E
= 3.0(283/2 + 0) • 585.0/581.2329
= 427.3 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
= 376 + max(220, 1134, 152, 427)
= 1509.3 Kgf

Longitudinal Wind Force [Fl]:
= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
= 1.1 • 4919.274(0.0 + $\pi/4$ (0.811 • 1.2)²)
= 4026.721 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 1587.15 Kgf
Transverse Shear Load 375.45 Kgf
Longitudinal Shear Load 410.62 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
= min(2(619.15/2 + 16.0 + 10.0)sin(120.0/2), 2•319.075)
= 581.233 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405



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K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -1509 \cdot 370.0 \left[1 - \left(1 - \frac{370.0}{1952.0} + \frac{(319.075^2 - 162.537^2)}{(2 \cdot 370.0 \cdot 1952.0)} \right) / \left(1 + \frac{(4 \cdot 162.54)}{(3 \cdot 1952.0)} \right) \right]$$

$$= -124.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L}{(3 \cdot 1952)} \right) - 4 \cdot 370 / 1952$$

$$= 1509 \cdot 1952 / 4 \left(\frac{(1 + 2(319^2 - 163^2)/(1952^2)) / (1 + (4 \cdot 163)/(3 \cdot 1952)) - 4 \cdot 370 / 1952}{(3 \cdot 1952)} \right) - 4 \cdot 370 / 1952$$

$$= 130.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 4153.033 \cdot 319.075 / (2 \cdot 13.0) - 130.7 / (\pi \cdot 319.1^2 \cdot 13.0)$$

$$= 50.66 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 4153.033 \cdot 319.075 / (2 \cdot 13.0) + 130.7 / (\pi \cdot 319.1^2 \cdot 13.0)$$

$$= 51.28 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 4153.033 \cdot 319.075 / (2 \cdot 13.0) - 124.8 / (0.1066 \cdot \pi \cdot 319.1^2 \cdot 13.0)$$

$$= 53.73 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 4153.033 \cdot 319.075 / (2 \cdot 13.0) + 124.8 / (0.1923 \cdot \pi \cdot 319.1^2 \cdot 13.0)$$

$$= 49.44 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 1509(1952.0 - 2 \cdot 370.0) / (1952.0 + (4 \cdot 162.54 / 3))$$

$$= 843.5 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 843.5 / (319.075 \cdot 13.0)$$

$$= 2.33 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(319.075 \cdot 13.0)}$$

$$= 50.236 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{(319.075 \cdot 13.0)}, 2 \cdot 370.0)$$

$$= 300.47 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(137.997/137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:



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$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 1509 \cdot 0.1 / (300.471(13.0 + 1.0 \cdot 10.0))$$

$$= -0.16 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -1509 / (4(13.0 + 1.0 \cdot 10.0)300.471) -$$

$$12 \cdot 0.053 \cdot 1509 \cdot 319.075 / (1952.0(13.0 + 1.0 \cdot 10.0)^2)$$

$$= -3.44 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	715.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	285.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	275.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(312.575 \cdot 13.0)} + 315.0)13.0^3) / 12$$

$$= 7.588 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 7.588 + 53.878(93.452 - 6.5)^2$$

$$= 4081.080 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	414.4	13.0	6.5	53.9	35020.4	0.408E+04
Wearplate	315.0	10.0	18.0	31.5	56700.0	0.180E+04
Web	10.0	237.4	141.7	23.7	336460.9	0.167E+04
BasePlate	285.0	12.0	266.4	34.2	911173.5	0.102E+05
Totals	...	...	...	143.3	1339354.8	0.178E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1339355 / 143.32$$

$$= 93.452 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 1509.328$$

$$= 307.1810 \text{ Kgf}$$



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Tension Stress, $St = (Fh/As) = 0.3368 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 157.1972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 585.0 - 312.57 \cdot \sin(120.0/2) / 1.0472$
 $= 326.503 \text{ mm}$

Bending Moment, $M = Fh \cdot d = 100.2974 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 0.5169 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(1509 + 78) \cdot 285.0) / (4 \cdot 715.0 \cdot 174.663)}$
 $= 5.161 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120.0/2) \cdot (309.575 + 16.0 + 10.0)$
 $= 581.233 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 581.2328 / (3 - 1)$
 $= 290.616 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 290.6164 \cdot 285.0 / 2$
 $= 414.128 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 1587.145 / (715.0 \cdot 285.0)$
 $= 0.779 \text{ Kg/cm}^2$

Axial Load [P]:
 $= Ap \cdot Bp$
 $= 414.1 \cdot 0.78$
 $= 322.553 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 19.0 + 14.531$
 $= 33.531 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 322.6/33.5308$
 $= 0.943 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$



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Saddle Calcs: Operating Case: Step: 23 10:20am Sep 9,2026

= 200.0 - 10.0
= 190.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
= 0.0/33.531
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
= 200.0/2
= 100.000 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$
= $\sqrt{(666.7/33.531)}$
= 44.589 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
= 585.0 - cos(120.0/2)(309.575 + 16.0 + 10.0) - 12.0
= 405.213 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
= $\sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/262.0)}$
= 122.739

Slenderness ratio [KL/r]:

= KL/r
= 1 • 405.213/44.589
= 9.088

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
= 410.6 / (2 • 715.0) • 290.616 • 405.21/2
= 16.908 Kg-m.

Compressive Allowable, KL/r < Cc (9.0876 < 122.7394) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (9.09)^2 / (2 \cdot 122.74^2)) 262 /$
 $(5/3 + 3 \cdot (9.09) / (8 \cdot 122.74) - (9.09^3) / (8 \cdot 122.74^3))$
= 154.2 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba
= 0.94/154.2 + (16.91 • 100.0/6666667)/174.66
= 0.020

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	290.6	10.0	0.0	29.1	0.0	2.42
Totals	...	...	...	48.1	...	669.

Distance to Centroid from Datum [ytot]:

= AY / A
= 0.0/48.062
= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2



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Saddle Calcs: Operating Case: Step: 23 10:20am Sep 9, 2026

$$= 200.0/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 585.0 - \cos(335.575/2) (12.0 + 0.0 + 0.0) - 0.0$$

$$= 237.425 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{669.0/48.062}$$

$$= 37.309 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 237.425/37.309$$

$$= 6.364$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 410.617 / (2 \cdot 715.0)$$

$$= 0.287 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.287 \cdot 290.616 \cdot 237.425$$

$$= 19.813 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 19.813/66900.5$$

$$= 2.904 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($6.3637 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.36)^2 / (2 \cdot 122.74^2)) 262 / (5/3 + 3 \cdot (6.36) / (8 \cdot 122.74) - (6.36^3) / (8 \cdot 122.74^3))$$

$$= 155.2 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$$

$$= 1.17/155.18 + (19.81 \cdot 100.0/669.004)/174.66$$

$$= 0.024$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6476 cm ²
Saddle Load Q0 (Weight)	Q0	453.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	220.4 Kg
Maximum Transverse Force	Ft	375.4 Kg
Maximum Longitudinal Force	F1	410.6 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

$$= 1422.6 \text{ Kg}$$



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Saddle Calcs: Operating Case: Step: 23 10:20am Sep 9, 2026

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= \max(F_f; F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1422.6 / (1.648 \cdot 4)$$

$$= 21.2 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 375.4 / (1.648 \cdot 4)$$

$$= 5.6 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 410.62 / (82.8 \cdot 4.0)$$

$$= 0.1216 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 585.0 \cdot 375.45 + 0.0$$

$$= 219.64 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 219.64 / 453.51$$

$$= 484.31 \text{ mm} > B_{plen}/6 \text{ --> Uplift in Transverse direction}$$

f = Bplen / 2 - Edgedis

$$= 715.0/2 - 50.8$$

$$= 306.70 \text{ mm}$$

Modular Ratio Of Steel/Concrete (n1):

$$= E_S / E_C$$

$$= 203402.5 / 21526.32$$

$$= 9.45$$

K1 = 3 (e - 0.5 • Bplen)

$$= 3 (484.31 - 0.5 \cdot 715.0)$$

$$= 380.42 \text{ mm}$$

K2 = 6 • n1 • At / Bpwid • (f + e)

$$= 6 \cdot 9.45 \cdot 3.3 / 285.0 \cdot (306.7 + 484.31)$$

$$= 51849.40 \text{ mm}^2$$

K3 = -K2 • (0.5 • Bplen + f)

$$= -51849.4 \cdot (0.5 \cdot 715.0 + 306.7)$$

$$= -34438371.70 \text{ mm}^3$$

Iteratively Solving for the Effective Bearing Length:

$$Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$$

$$Y^3 + 380.42 \cdot Y^2 + 51849.4 \cdot Y + -0.3E+08 = 0$$

$$Y = 202.61 \text{ mm}$$

Num = (Bplen / 2 - Y / 3 - e)

$$= (715.0/2 - 202.61/3 - 484.31)$$

$$= -194.34$$

Denom = (Bplen / 2 - Y / 3 + f)

$$= (715.0/2 - 202.61/3 + 306.7)$$

$$= 596.66$$



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Total Bolt Tension Force [Tforce]:

$$= - QO \cdot Num / Denom$$

$$= - 453.51 \cdot -194.34 / 596.66$$

$$= 147.71 \text{ Kg}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$= Tforce / (Stba \cdot Nbt)$$

$$= 147.71 / (138.0 \cdot 2.0)$$

$$= 0.0525 \text{ cm}^2$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.1216, 0.0525]$$

$$= 0.1216 \text{ cm}^2$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$= 2(QO + Tforce) / (Y \cdot Bpwid)$$

$$= 2(453.51 + 147.71) / (202.61 \cdot 285.0)$$

$$= 204.20 \text{ KPa.}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$= (Bplen - WebIngh) / 2$$

$$= (715.0 - 664.2) / 2$$

$$= 25.4000 \text{ mm}$$

Overturning Moment due To Bolt Tension [Mt]:

$$= Tforce \cdot Adist$$

$$= 147.71 \cdot 25.4$$

$$= 3.75 \text{ Kg-m.}$$

Equivalent Bearing Pressure (f1):

$$= fc \cdot (Y - Adist) / Y$$

$$= 204.2 \cdot (202.61 - 25.4) / 202.61$$

$$= 178.60 \text{ KPa.}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc)$$

$$= (25.4^2 \cdot 285.0 / 6) \cdot (178.6 + 2 \cdot 204.2)$$

$$= 1.83 \text{ Kg-m.}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$$

$$= \sqrt{(6 \cdot \max(3.75, 1.83) / (285.0 \cdot 207.0))}$$

$$= 1.9345 \text{ mm}$$

MDMT Calculations for: Sdl 2 Fr50, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10.0, c = 3.0 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.851, Temp. Reduction = 8 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

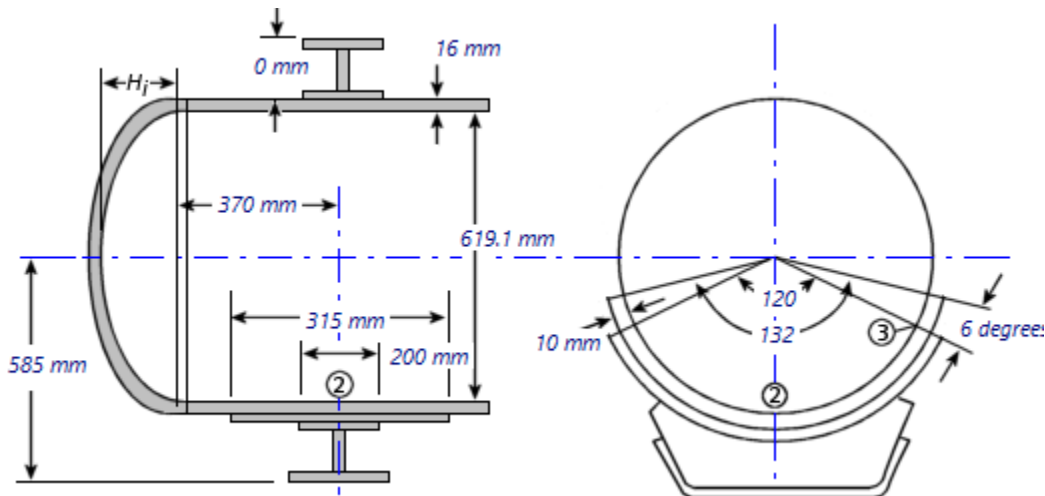
Step: 24 10:20am Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	319.07	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	1952.00	mm
Distance from Saddle to Vessel tangent	a1 or a	370.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	315.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	585.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	4945.15	Kgf
Pressure used in Analysis	P	5398.127 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	64.57	234.00
Long. Stress at Bottom of Midspan	67.93	234.00
Long. Stress at Top of Saddles	71.07	234.00
Long. Stress at Bottom of Saddles	63.58	234.00
Tangential Shear in Shell	8.50	138.58
Circ. Stress at Horn of Saddle	11.26	350.99
Circ. Compressive Stress in Shell	0.53	234.00



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Saddle Calcs: Test Case:

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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(375.4/2 + 0) \cdot 585.0/581.2329$$

$$= 566.8 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(205.31, 0) \cdot 585.0/1090.0$$

$$= 110.2 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 4378 + \max(110, 567, 0, 0)$$

$$= 4945.2 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 4919.274(0.0 + \pi/4(0.811 \cdot 1.2)^2)$$

$$= 4026.721 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	5022.97	Kgf
Transverse Shear Load	187.72	Kgf
Longitudinal Shear Load	205.31	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(619.15/2 + 16.0 + 10.0)\sin(120.0/2), 2 \cdot 319.075)$$

$$= 581.233 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -4945 \cdot 370.0 [1 - (1 - 370.0/1952.0 + (319.075^2 - 0.0^2)/(2 \cdot 370.0 \cdot 1952.0))/(1 + (4 \cdot 0.0)/(3 \cdot 1952.0))]$$

$$= -217.9 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 4945 \cdot 1952/4((1 + 2(319^2 - 0^2)/(1952^2))/(1 + (4 \cdot 0)/(3 \cdot 1952)) - 4 \cdot 370/1952)$$

$$= 712.5 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 5398.127 \cdot 319.075/(2 \cdot 13.0) - 712.5/(\pi \cdot 319.1^2 \cdot 13.0)$$

$$= 64.57 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$



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$$= 5398.127 \cdot 319.075 / (2 \cdot 13.0) + 712.5 / (\pi \cdot 319.1^2 \cdot 13.0)$$

$$= 67.93 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 5398.127 \cdot 319.075 / (2 \cdot 13.0) - 217.9 / (0.1066 \cdot \pi \cdot 319.1^2 \cdot 13.0)$$

$$= 71.07 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 5398.127 \cdot 319.075 / (2 \cdot 13.0) + 217.9 / (0.1923 \cdot \pi \cdot 319.1^2 \cdot 13.0)$$

$$= 63.58 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 4945(1952.0 - 2 \cdot 370.0) / (1952.0 + (4 \cdot 0.0 / 3))$$

$$= 3070.5 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 3070.45 / (319.075 \cdot 13.0)$$

$$= 8.50 \text{ N./mm}^2$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(319.075 \cdot 13.0)}$$

$$= 50.236 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{(319.075 \cdot 13.0)}, 2 \cdot 370.0)$$

$$= 300.47 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma6, r$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 4945 \cdot 0.1 / (300.471(13.0 + 1.0 \cdot 10.0))$$

$$= -0.53 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [$\sigma7, r^*$]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -4945 / (4(13.0 + 1.0 \cdot 10.0)300.471) -$$

$$12 \cdot 0.053 \cdot 4945 \cdot 319.075 / (1952.0(13.0 + 1.0 \cdot 10.0)^2)$$

$$= -11.26 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	715.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	285.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	275.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(312.575 \cdot 13.0)} + 315.0)^3) / 12$$

$$= 7.588 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 7.588 + 53.878(93.452 - 6.5)^2$$

$$= 4081.080 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	414.4	13.0	6.5	53.9	35020.4	0.408E+04
Wearplate	315.0	10.0	18.0	31.5	56700.0	0.180E+04
Web	10.0	237.4	141.7	23.7	336460.9	0.167E+04
BasePlate	285.0	12.0	266.4	34.2	911173.5	0.102E+05
Totals	...	...	...	143.3	1339354.8	0.178E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1339355/143.32$$

$$= 93.452 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 4945.15$$

$$= 1006.4453 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.1035 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 585.0 - 312.57 \cdot \sin(120.0/2) / 1.0472$$

$$= 326.503 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 328.6134 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.6936 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(4945 + 78)285.0) / (4 \cdot 715.0 \cdot 173.33)}$$

$$= 9.217 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (309.575 + 16.0 + 10.0)$$

$$= 581.233 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 581.2328 / (3 - 1)$$

$$= 290.616 \text{ mm}$$



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Saddle Calcs: Test Case:

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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 290.6164 \cdot 285.0 / 2$$

$$= 414.128 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 5022.967 / (715.0 \cdot 285.0)$$

$$= 2.465 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 414.1 \cdot 2.46$$

$$= 1020.809 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 19.0 + 14.531$$

$$= 33.531 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 1020.8 / 33.5308$$

$$= 2.986 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 200.0 - 10.0$$

$$= 190.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 33.531$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 200.0 / 2$$

$$= 100.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (666.7 / 33.531) }$$

$$= 44.589 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 585.0 - \cos(120.0 / 2) (309.575 + 16.0 + 10.0) - 12.0$$

$$= 405.213 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.19994E+09 / 260.0) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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Saddle Calcs: Test Case: Step: 24 10:20am Sep 9, 2026

$$= 1 \cdot 405.213/44.589$$

$$= 9.088$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 205.3 / (2 \cdot 715.0) \cdot 290.616 \cdot 405.21/2$$

$$= 8.454 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($9.0876 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (9.09)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (9.09) / (8 \cdot 123.21) - (9.09^3) / (8 \cdot 123.21^3))$$

$$= 153 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.99/153.04 + (8.45 \cdot 100.0/6666667) / 173.33$$

$$= 0.027$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	290.6	10.0	0.0	29.1	0.0	2.42
Totals	...	...	...	48.1	...	669.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/48.062$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 585.0 - \cos(335.575/2) (12.0 + 0.0 + 0.0) - 0.0$$

$$= 237.425 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (669.0/48.062) }$$

$$= 37.309 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 237.425/37.309$$

$$= 6.364$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 205.309 / (2 \cdot 715.0)$$

$$= 0.144 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.144 \cdot 290.616 \cdot 237.425$$

$$= 9.907 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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Saddle Calcs: Test Case:

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= 9.907/66900.5
= 1.452 N./mm²

Compressive Allowable, $KL/r < C_c$ (6.3637 < 123.2105) per AISC E2-1 [Sca]:

= $(1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$
= $(1 - (6.36)^2 / (2 \cdot 123.21^2)) 260 /$
 $(5/3 + 3 \cdot (6.36) / (8 \cdot 123.21) - (6.36)^3 / (8 \cdot 123.21^3))$
= 154 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$
= $3.71 / 154.0 + (9.91 \cdot 100.0 / 669.004) / 173.33$
= 0.032

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6476 cm ²
Saddle Load QO (Weight)	QO	4456.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	110.2 Kg
Maximum Transverse Force	Ft	187.7 Kg
Maximum Longitudinal Force	F1	205.3 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $F1 / (Bolt\ Area \cdot Number\ of\ Bolts)$
= $205.3 / (1.648 \cdot 4)$
= 3.1 N./mm². Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $Ft / (Bolt\ Area \cdot Number\ of\ Bolts)$
= $187.7 / (1.648 \cdot 4)$
= 2.8 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (BoltShearAllowable \cdot Nbolts)$
= $205.31 / (28.98 \cdot 4.0)$
= 0.1737 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + Sum\ of\ X\ Moments$
= $585.0 \cdot 187.72 + 0.0$
= 109.82 Kg-m.

Eccentricity (e):

= $Rmom / QO$
= $109.82 / 4456.15$
= 24.64 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)



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Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 0.1737, 0.0]

= 0.1737 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	319.07	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	1952.00	mm
Distance from Saddle to Vessel tangent	a1 or a	370.00	mm
Saddle Width	b1 or b	200.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	315.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	162.54	mm
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	585.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		857.27	Kgf
Pressure used in Analysis	P	5398.127	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	66.08	234.00
Long. Stress at Bottom of Midspan	66.43	234.00
Long. Stress at Top of Saddles	67.82	234.00
Long. Stress at Bottom of Saddles	65.38	234.00
Tangential Shear in Shell	1.33	138.58
Circ. Stress at Horn of Saddle	1.95	350.99
Circ. Compressive Stress in Shell	0.09	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3.0(375.4/2 + 0) • 585.0/581.2329

= 566.8 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl,wind; Sum of X Forces) • B / Ls

= max(205.31, 0) • 585.0/1090.0

= 110.2 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:



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$$\begin{aligned}
 &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\
 &= 290 + \max(110, 567, 0, 0) \\
 &= 857.3 \text{ Kg}
 \end{aligned}$$

Longitudinal Wind Force [F_L]:

$$\begin{aligned}
 &= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2)) \\
 &= 1.1 \cdot 4919.274 (0.0 + \pi/4 (0.811 \cdot 1.2)^2) \\
 &= 4026.721 \text{ N}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	935.09	Kgf
Transverse Shear Load	187.72	Kgf
Longitudinal Shear Load	205.31	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$\begin{aligned}
 &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m) \\
 &= \min(2(619.15/2 + 16.0 + 10.0)\sin(120.0/2), 2 \cdot 319.075) \\
 &= 581.233 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to R_m/2.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned}
 &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)] \\
 &= -857 \cdot 370.0 [1 - (1 - 370.0/1952.0 + (319.075^2 - 162.537^2)/(2 \cdot 370.0 \cdot 1952.0))/(1 + (4 \cdot 162.54)/(3 \cdot 1952.0))] \\
 &= -70.9 \text{ Kg-m.}
 \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned}
 &= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L) \\
 &= 857 \cdot 1952/4 ((1 + 2(319^2 - 163^2)/(1952^2))/(1 + (4 \cdot 163)/(3 \cdot 1952)) - 4 \cdot 370/1952) \\
 &= 74.3 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ₁]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t) \\
 &= 5398.127 \cdot 319.075/(2 \cdot 13.0) - 74.3/(\pi \cdot 319.1^2 \cdot 13.0) \\
 &= 66.08 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ₂]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\
 &= 5398.127 \cdot 319.075/(2 \cdot 13.0) + 74.3/(\pi \cdot 319.1^2 \cdot 13.0) \\
 &= 66.43 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*₃]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 5398.127 \cdot 319.075/(2 \cdot 13.0) - 70.9/(0.1066 \cdot \pi \cdot 319.1^2 \cdot 13.0) \\
 &= 67.82 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*₄]:

$$\begin{aligned}
 &= P \cdot R_m/(2t) + M1/(K1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 5398.127 \cdot 319.075/(2 \cdot 13.0) + 70.9/(0.1923 \cdot \pi \cdot 319.1^2 \cdot 13.0) \\
 &= 65.38 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$



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$$= 857(1952.0 - 2 \cdot 370.0) / (1952.0 + (4 \cdot 162.54/3))$$

$$= 479.1 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 479.09 / (319.075 \cdot 13.0)$$

$$= 1.33 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(319.075 \cdot 13.0)}$$

$$= 50.236 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200.0 + 1.56 \cdot \sqrt{(319.075 \cdot 13.0)}, 2 \cdot 370.0)$$

$$= 300.47 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 857 \cdot 0.1 / (300.471(13.0 + 1.0 \cdot 10.0))$$

$$= -0.09 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -857 / (4(13.0 + 1.0 \cdot 10.0)300.471) -$$

$$12 \cdot 0.053 \cdot 857 \cdot 319.075 / (1952.0(13.0 + 1.0 \cdot 10.0)^2)$$

$$= -1.95 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	715.0000	mm
Baseplate Thickness	Bpthk	12.0000	mm
Baseplate Width	Bpwid	285.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	275.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(312.575 \cdot 13.0)} + 315.0)13.0^3) / 12$$

$$= 7.588 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 7.588 + 53.878(93.452 - 6.5)^2$$

$$= 4081.080 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	414.4	13.0	6.5	53.9	35020.4	0.408E+04
Wearplate	315.0	10.0	18.0	31.5	56700.0	0.180E+04
Web	10.0	237.4	141.7	23.7	336460.9	0.167E+04
BasePlate	285.0	12.0	266.4	34.2	911173.5	0.102E+05
Totals	...	...	...	143.3	1339354.8	0.178E+05



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Distance to Centroid [C1]:

$$= AY / A$$

$$= 1339355 / 143.32$$

$$= 93.452 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 857.271$$

$$= 174.4733 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.1913 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 585.0 - 312.57 \cdot \sin(120.0/2) / 1.0472$$

$$= 326.503 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 56.9671 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.2936 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(857 + 78)285.0 / (4 \cdot 715.0 \cdot 173.33))}$$

$$= 3.977 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (309.575 + 16.0 + 10.0)$$

$$= 581.233 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 581.2328 / (3 - 1)$$

$$= 290.616 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 290.6164 \cdot 285.0 / 2$$

$$= 414.128 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 935.088 / (715.0 \cdot 285.0)$$

$$= 0.459 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 414.1 \cdot 0.46$$

$$= 190.036 \text{ Kgf}$$



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Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 19.0 + 14.531

= 33.531 cm²

Compressive Stress [Sc]:

= P/Ar

= 190.0/33.5308

= 0.556 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

= Saddle Width - Web Thickness

= 200.0 - 10.0

= 190.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/33.531

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 200.0/2

= 100.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{666.7/33.531}$

= 44.589 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 585.0 - cos(120.0/2)(309.575 + 16.0 + 10.0) - 12.0

= 405.213 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 405.213/44.589

= 9.088

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 205.3 / (2 • 715.0) • 290.616 • 405.21/2

= 8.454 Kg-m.

Compressive Allowable, KL/r < Cc (9.0876 < 123.2105) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))

= (1 - (9.09)² / (2 • 123.21²)) 260 /

(5/3 + 3 • (9.09) / (8 • 123.21) - (9.09³) / (8 • 123.21³))

= 153 N./mm²



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Saddle Calcs: Test Case:

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AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 0.56/153.04 + (8.45 \cdot 100.0/6666667) / 173.33$$

$$= 0.011$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	290.6	10.0	0.0	29.1	0.0	2.42
Totals	...	...	...	48.1	...	669.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/48.062$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200.0/2$$

$$= 100.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 585.0 - \cos(335.575/2) (12.0 + 0.0 + 0.0) - 0.0$$

$$= 237.425 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (669.0/48.062) }$$

$$= 37.309 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 237.425/37.309$$

$$= 6.364$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 205.309 / (2 \cdot 715.0)$$

$$= 0.144 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.144 \cdot 290.616 \cdot 237.425$$

$$= 9.907 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 9.907/66900.5$$

$$= 1.452 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ (6.3637 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (6.36)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (6.36) / (8 \cdot 123.21) - (6.36^3) / (8 \cdot 123.21^3))$$

$$= 154 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1 / I) / Sba$$

$$= 0.69/154.0 + (9.91 \cdot 100.0/669.004) / 173.33$$

$$= 0.013$$



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Saddle Calcs: Test Case:

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Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6476 cm ²
Saddle Load Q0 (Weight)	Q0	368.3 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	110.2 Kgf
Maximum Transverse Force	Ft	187.7 Kgf
Maximum Longitudinal Force	Fl	205.3 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 205.3 / (1.648 \cdot 4)$$

$$= 3.1 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 187.7 / (1.648 \cdot 4)$$

$$= 2.8 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 205.31 / (28.98 \cdot 4.0)$$

$$= 0.1737 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 585.0 \cdot 187.72 + 0.0$$

$$= 109.82 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 109.82 / 368.27$$

$$= 298.20 \text{ mm} > Bplen / 6 \text{ --> Uplift in Transverse direction}$$

f = Bplen / 2 - Edgedis

$$= 715.0 / 2 - 50.8$$

$$= 306.70 \text{ mm}$$

Modular Ratio Of Steel/Concrete (n1):

$$= ES / EC$$

$$= 203402.5 / 21526.32$$

$$= 9.45$$

K1 = 3 (e - 0.5 • Bplen)

$$= 3 (298.2 - 0.5 \cdot 715.0)$$

$$= -177.90 \text{ mm}$$

K2 = 6 • n1 • At / Bpwid • (f + e)

$$= 6 \cdot 9.45 \cdot 3.3 / 285.0 \cdot (306.7 + 298.2)$$

$$= 39650.46 \text{ mm}^2$$



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Saddle Calcs: Test Case:

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$$\begin{aligned} K3 &= -K2 \cdot (0.5 \cdot Bplen + f) \\ &= -39650.46 \cdot (0.5 \cdot 715.0 + 306.7) \\ &= -26335833.51 \text{ mm}^3 \end{aligned}$$

Iteratively Solving for the Effective Bearing Length:

$$\begin{aligned} Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 &= 0 \\ Y^3 + -177.9 \cdot Y^2 + 39650.46 \cdot Y + -0.3E+08 &= 0 \\ Y &= 316.07 \text{ mm} \end{aligned}$$

$$\begin{aligned} Num &= (Bplen / 2 - Y / 3 - e) \\ &= (715.0/2 - 316.07/3 - 298.2) \\ &= -46.06 \end{aligned}$$

$$\begin{aligned} Denom &= (Bplen / 2 - Y / 3 + f) \\ &= (715.0/2 - 316.07/3 + 306.7) \\ &= 558.84 \end{aligned}$$

$$\begin{aligned} \text{Total Bolt Tension Force [Tforce]:} \\ &= -QO \cdot Num / Denom \\ &= -368.27 \cdot -46.06/558.84 \\ &= 30.35 \text{ Kg} \end{aligned}$$

$$\begin{aligned} \text{Bolt Area Required due to Transverse Load [Bltareart]:} \\ &= Tforce / (Stba \cdot Nbt) \\ &= 30.35 / (48.3 \cdot 2.0) \\ &= 0.0308 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Required Area of a Single Bolt [Bltarear]:} \\ &= \max[Bltarearl, Bltarears, Bltareart] \\ &= \max[0.0, 0.1737, 0.0308] \\ &= 0.1737 \text{ cm}^2 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

$$\begin{aligned} \text{Bearing Pressure (fc)} \\ &= 2(QO + Tforce) / (Y \cdot Bpwid) \\ &= 2(368.27 + 30.35) / (316.07 \cdot 285.0) \\ &= 86.79 \text{ KPa.} \end{aligned}$$

$$\begin{aligned} \text{Distance from Baseplate Edge to the Web [ADIST]:} \\ &= (Bplen - WebIngh) / 2 \\ &= (715.0 - 664.2) / 2 \\ &= 25.4000 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Overturning Moment due To Bolt Tension [Mt]:} \\ &= Tforce \cdot Adist \\ &= 30.35 \cdot 25.4 \\ &= 0.77 \text{ Kg-m.} \end{aligned}$$

$$\begin{aligned} \text{Equivalent Bearing Pressure (f1):} \\ &= fc \cdot (Y - Adist) / Y \\ &= 86.79 \cdot (316.07 - 25.4) / 316.07 \\ &= 79.81 \text{ KPa.} \end{aligned}$$

$$\begin{aligned} \text{Overturning Moment due to Bearing Pressure [Mc]:} \\ &= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc) \\ &= (25.4^2 \cdot 285.0 / 6) \cdot (79.81 + 2 \cdot 86.79) \\ &= 0.79 \text{ Kg-m.} \end{aligned}$$

$$\begin{aligned} \text{Baseplate Required Thickness [Treq]:} \\ &= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))} \\ &= \sqrt{(6 \cdot \max(0.77, 0.79) / (285.0 \cdot 207.0))} \\ &= 0.8887 \text{ mm} \end{aligned}$$



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Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N1-Tube inlet

Nozl: 9 10:20am Sep 9,2026

Input, Nozzle Desc: N1-Tube inlet

From: 20

Pressure for Reinforcement Calculations	P	3206.070	KPa.
Temperature for Internal Pressure	Temp	230	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	230	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	619.15	mm
Design Length of Section	L	802.3959	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		400.80	mm
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		102.2600	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	150.0000	mm
Thickness of Pad	te	14.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm
Reinforcing Pad Width		13.8700	mm

The Pressure Design option was Design Pressure + static head.



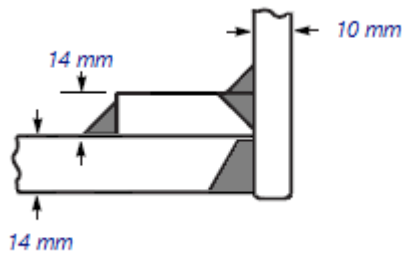
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Nozzle Calcs.: N1-Tube inlet

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Reinforcement CALCULATION, Description: N1-Tube inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	102.260 mm.
Actual Thickness Used in Calculation	10.000 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3206.07 \cdot 312.575 / (138 \cdot 1.0 - 0.6 \cdot 3206.07)$
 $= 7.3651 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3206.07 \cdot 54.13 / (118 \cdot 1.0 - 0.6 \cdot 3206.07)$
 $= 1.4952 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4921 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	216.5200 mm
Parallel to Vessel Wall, opening length	d	108.2600 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



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Nozzle Calcs.: N1-Tube inlet

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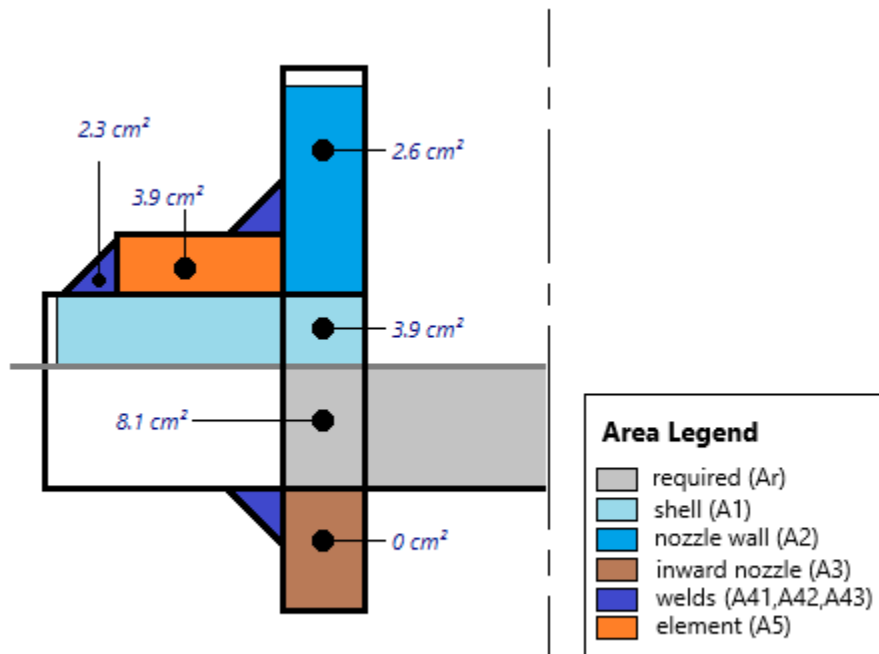


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	8.123	1.277	NA
Area in Shell	A1	3.861	9.225	NA
Area in Nozzle Wall	A2	2.589	3.061	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.295	2.295	NA
Area in Element	A5	3.884	3.884	NA
TOTAL AREA AVAILABLE	Atot	12.629	18.464	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness

Based on given Pad Thickness: 122.2600 14.0000 mm

Based on given Pad Diameter: 150.0000 0.0000 mm

Based on Shell or Nozzle Thickness: 122.2600 10.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (108.26 \cdot 7.3651 \cdot 1.0 + 2 \cdot 7.0 \cdot 7.3651 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 8.123 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 108.26(1.0 \cdot 11.0 - 1.0 \cdot 7.3651) - 2 \cdot 7.0$$

$$(1.0 \cdot 11.0 - 1.0 \cdot 7.3651) \cdot (1 - 0.855)$$



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Nozzle Calcs.: N1-Tube inlet

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$$= 3.861 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 27.5) \cdot (7.0 - 1.5) \cdot 0.8551$$

$$= 2.589 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 12.0^2 \cdot 1.0$$

$$= 2.295 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (150.0 - 122.26) \cdot 14.0 \cdot 1.0$$

$$= 3.884 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.4952 \text{ mm}$

Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb1 = 10.3651 \text{ mm}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 10.3651 \text{ mm}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb3 = 8.7300 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.73, \max(10.3651, 4.5)]$$

$$= 8.7300 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.4952, 8.73)$$

$$= 8.7300 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	37.3,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	257.7 N./mm ²	Passed
Occasional	:	11.6,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	21.5,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $tg = 10.0$, $tr = 1.495$, $c = 3.0 \text{ mm}$, $E* = 1.0$

Thickness Ratio = $tr \cdot E*/(tg - c) = 0.214$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C



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Nozzle Calcs.: N1-Tube inlet Nozl: 9 10:20am Sep 9, 2026

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 14.0, tr = 7.365, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.67$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
Min Metal Temp. at Required thickness (UCS 66.1) -37 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 10.0, tr = 1.495, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.214$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -37 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Weld Size Calculations, Description: N1-Tube inlet

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm
Intermediate Calc. for pad/shell Welds TminPad 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.9000 = $0.7 * t_{min}$	7.0700 = $0.7 * W_o$ mm
Pad Weld	5.5000 = $0.5 * t_{minPad}$	8.4840 = $0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= $\max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
= $\max(0, (8.1229 - 3.8614 + 2 \cdot 7.0 \cdot 0.8551 \cdot (1.0 \cdot 11.0 - 7.3651)) \cdot 138)$
= 6608.88 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= $(A2 + A5 + A4 - (W_i - Can / 707)^2 \cdot fr2) \cdot Sv$
= $(2.5888 + 3.8836 + 2.2951 - 0.0 \cdot 0.86) \cdot 138$
= 12337.28 Kgf

Weld Load [W2]:
= $(A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
= $(2.5888 + 0.0 + 0.8551 + (1.3168)) \cdot 138$
= 6699.10 Kgf

Weld Load [W3]:
= $(A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
= $(2.5888 + 0.0 + 2.2951 + 3.8836 + (1.3168)) \cdot 138$
= 14190.24 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= $(\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
= $(3.1416/2.0) \cdot 122.26 \cdot 10.0 \cdot 0.49 \cdot 118$
= 11323. Kgf



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Nozzle Calcs.: N1-Tube inlet

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Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 150.0 \cdot 12.0 \cdot 0.49 \cdot 138$$

$$= 19495. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 57.63) \cdot (10.0 - 3.0) \cdot 0.7 \cdot 118$$

$$= 10674. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 122.26 \cdot 14.0 \cdot 0.74 \cdot 138$$

$$= 27997. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 122.26 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 21997. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (19495 + 10674) = 30170 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (11323 + 27997 + 21997 + 0) = 61317 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (19495 + 21997 + 0) = 41493 \text{ Kgf}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 30169 \text{ Kgf, must exceed } W = 6608 \text{ Kgf} \quad \text{or } W1 = 12337 \text{ Kgf}$$

$$\text{Path 2-2} = 61316 \text{ Kgf, must exceed } W = 6608 \text{ Kgf} \quad \text{or } W2 = 6699 \text{ Kgf}$$

$$\text{Path 3-3} = 41492 \text{ Kgf, must exceed } W = 6608 \text{ Kgf} \quad \text{or } W3 = 14190 \text{ Kgf}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4052 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 6.0955 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.0955 mm

Input Echo, WRC297 Item 1, Description: N1-Tube inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	619.150	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm



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Nozzle Calcs.: N1-Tube inlet Nozl: 9 10:20am Sep 9,2026

Nozzle Diameter		102.260	mm
Nozzle Thickness		10.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	150.000	mm
Thickness of Reinforcing Pad	14.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	3206.07	KPa.
Radial Load	P	-764.84	Kgf
Circumferential Shear	Vc	509.89	Kgf
Longitudinal Shear	Vl	509.89	Kgf
Circumferential Moment	Mc	142.71	Kg-m.
Longitudinal Moment	Ml	142.71	Kg-m.
Torsional Moment	Mt	203.88	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 122.26 + 2 \cdot 1.65 \cdot \sqrt{325.075(14.0 - 3.0)}$$

$$= 319.594 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	650.150	mm
Nozzle Outside Diameter	(d) =	122.260	mm
Vessel Thickness used	(T) =	25.000	mm
Nozzle Thickness used	(t) =	7.000	mm
T / t	=	3.571	
d / t	=	17.466	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.959$$

Nr/P	=	0.114
Mr/P	=	0.100
M0/P	=	0.131
N0/P	=	0.235
MrD/Mc	=	0.156
NrDL/Mc	=	0.144
M0d/Mc	=	0.223
N0DL/Mc	=	0.088
MrD/Ml	=	0.133
NrDL/Ml	=	0.128
M0D/Ml	=	0.190
N0DL/Ml	=	0.233

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside



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Nozzle Calcs.: N1-Tube inlet

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Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	38	41	38	41

Outplane Stress Summary	25	51	75	18
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-14	14	14	-14
Inplane Pressure Stress	19	19	19	19

Inplane Stress Summary	10	25	44	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	26	52	76	18
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-24	24	24	-24
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	19	19	19	19

Outplane Stress Summary	5	35	57	-10
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	1	1	1	1
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Nozzle Calcs.: N1-Tube inlet Nozl: 9 10:20am Sep 9, 2026

Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-17	17	17	-17
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	38	41	38	41
Inplane Stress Summary	27	50	66	21

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)				
	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)				
	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	27	50	68	32

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)				
	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	27	31	27	31
Hoop Stress Summary	26	30	35	38

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)				
	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	39	-39	39	-39
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-96	96	96	-96
Axial Pressure Stress	13	13	13	13
Axial Stress Summary	-59	52	173	-98

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)				
	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12



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Nozzle Calcs.: N1-Tube inlet Nozl: 9 10:20am Sep 9,2026

Shear Stress Summary	15	15	8	8
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	91	61	173	138
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	27	31	27	31

Hoop Stress Summary	29	32	32	36
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	39	-39	39	-39
Axial Membrane (Mc)	-20	-20	20	20
Axial Bending (Mc)	-117	117	117	-117
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	13	13	13	13

Axial Stress Summary	-81	74	194	-120
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	12	12	12	12

Shear Stress Summary	8	8	15	15
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	111	76	196	160
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 122.26 + 2 \cdot 1.65 \cdot \sqrt{318.075(14.0 - 3.0)}$$

$$= 317.458 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 636.150 mm



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Nozzle Calcs.: N1-Tube inlet Nozl: 9 10:20am Sep 9,2026

Nozzle Outside Diameter (d) = 150.000 mm
 Vessel Thickness used (T) = 11.000 mm
 Nozzle Thickness used (t) = 20.870 mm
 T / t = 0.527
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.793$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.074
 Mr/P = 0.151
 M0/P = 0.044
 N0/P = 0.103
 MrD/Mc = 0.287
 NrDL/Mc = 0.102
 M0d/Mc = 0.090
 N0DL/Mc = 0.065
 MrD/Ml = 0.123
 NrDL/Ml = 0.071
 M0D/Ml = 0.040
 N0DL/Ml = 0.128

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	89	92	89	92
Outplane Stress Summary	83	91	140	73

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	56	-56	56	-56
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-57	57	57	-57
Inplane Pressure Stress	44	44	44	44
Inplane Stress Summary	43	44	168	-58

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	7	7	3	3

Vessel Stresses (N./mm²):



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Nozzle Calcs.: N1-Tube inlet Nozl: 9 10:20am Sep 9,2026

LONGITUDINAL PLANE (Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	84	92	168	132
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-41	41	41	-41
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	44	44	44	44

Outplane Stress Summary	20	71	114	-2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	56	-56	56	-56
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-132	132	132	-132
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	89	92	89	92

Inplane Stress Summary	9	165	290	-83
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	3	3	7	7
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	21	165	291	84
------------------------	----	-----	-----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	38	41	38	41	38	41	38	41
Circ. Pl (SUS)	-2	-2	6	6	-1	-1	3	3



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Nozzle Calcs.: N1-Tube inlet

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Circ. Q (SUS)	-11	11	29	-29	-10	10	24	-24
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	-1	-1	3	3	1	1	3	3
Long. Q (SUS)	-7	7	21	-21	-15	15	33	-33
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	38.0	41.0	38.0	41.0	38.0	41.0	38.0	41.0
Pm+Pl (SUS)	36.1	39.0	44.0	47.0	37.1	40.0	41.1	44.0
Pm+Pl+Q (Total)	26.1	50.6	73.1	18.2	27.2	50.3	66.4	32.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.00	138.00	Passed
Pm+Pl (SUS)	47.04	207.00	Passed
Pm+Pl+Q (TOTAL)	73.13	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	89	92	89	92	89	92	89	92
Circ. Pl (SUS)	-3	-3	15	15	-3	-3	11	11
Circ. Q (SUS)	-2	2	34	-34	-76	76	188	-188
Long. Pm (SUS)	44	44	44	44	44	44	44	44
Long. Pl (SUS)	-1	-1	9	9	2	2	10	10
Long. Q (SUS)	-1	1	113	-113	-25	25	57	-57
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	89.0	92.0	89.0	92.0	89.0	92.0	89.0	92.0
Pm+Pl (SUS)	86.0	89.0	104.0	107.0	86.0	89.0	100.0	103.0
Pm+Pl+Q (Total)	84.8	91.8	166.6	133.2	22.3	165.2	288.2	85.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	92.00	138.00	Passed
Pm+Pl (SUS)	107.02	207.00	Passed
Pm+Pl+Q (TOTAL)	288.20	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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Nozzle Calcs.: N1-Tube inlet

Nozl: 9 10:20am

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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	27	31	27	31	27	31	27	31
Circ. Pl (SUS)	-2	-2	6	6	1	1	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	13	13	13	13	13	13	13	13
Long. Pl (SUS)	-18	-18	22	22	-18	-18	22	22
Long. Q (SUS)	-57	57	135	-135	-78	78	156	-156
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	27.0	31.0	27.0	31.0	27.0	31.0	27.0	31.0
Pm+Pl (SUS)	30.6	34.5	37.2	39.2	33.5	37.5	36.4	37.5
Pm+Pl+Q (Total)	92.0	59.4	170.6	138.2	112.4	74.9	192.4	157.9

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	31.00	118.00	Passed
Pm+Pl (SUS)	39.16	177.00	Passed
Pm+Pl+Q (TOTAL)	192.39	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N1-Tube Outlet

Nozl: 10 10:20am Sep 9,2026

Input, Nozzle Desc: N1-Tube Outlet

From: 20

Pressure for Reinforcement Calculations	P	3200.001	KPa.
Temperature for Internal Pressure	Temp	230	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	230	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	619.15	mm
Design Length of Section	L	802.3959	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		450.80	mm
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		236.4800	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	450.0000	mm
Thickness of Pad	te	14.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm
Reinforcing Pad Width		94.7600	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



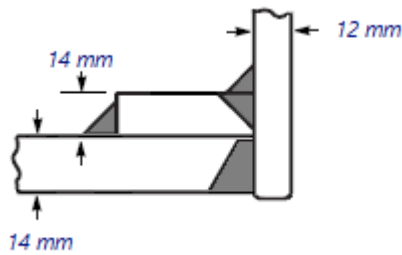
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Nozzle Calcs.: N1-Tube Outlet

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Reinforcement CALCULATION, Description: N1-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	236.480 mm.
Actual Thickness Used in Calculation	12.000 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3200.0 \cdot 312.575 / (138 \cdot 1.0 - 0.6 \cdot 3200.0)$
 $= 7.3510 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3200.0 \cdot 121.24 / (118 \cdot 1.0 - 0.6 \cdot 3200.0)$
 $= 3.3425 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6777 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	484.9600 mm
Parallel to Vessel Wall, opening length	d	242.4800 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



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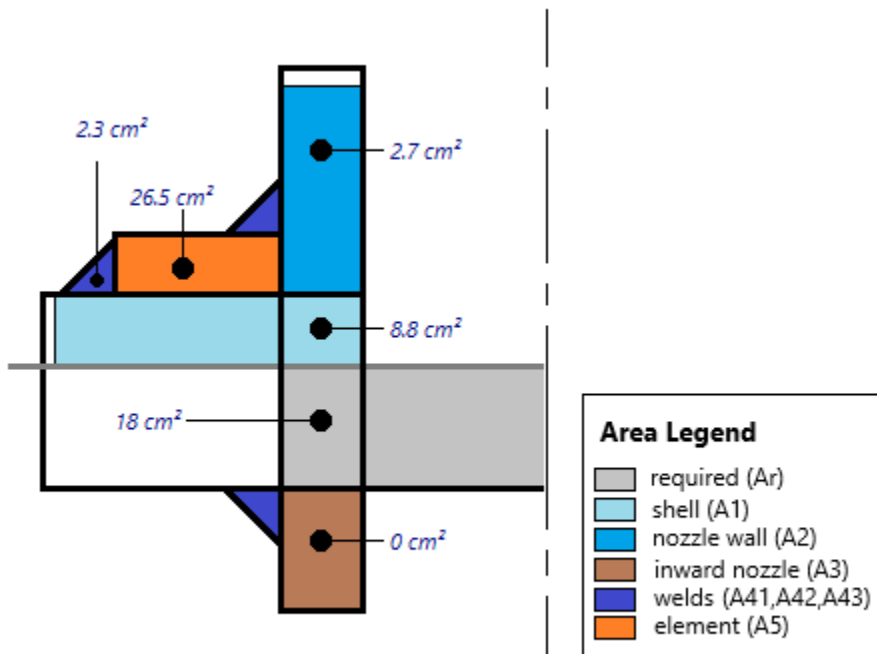


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	18.016	2.839	NA
Area in Shell	A1	8.753	20.829	NA
Area in Nozzle Wall	A2	2.661	3.914	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.295	2.295	NA
Area in Element	A5	26.533	26.533	NA
TOTAL AREA AVAILABLE	Atot	40.242	53.571	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness

Based on given Pad Thickness: 291.2485 14.0000 mm

Based on given Pad Diameter: 450.0000 2.2729 mm

Based on Shell or Nozzle Thickness: 296.3766 12.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (242.48 \cdot 7.351 \cdot 1.0 + 2 \cdot 9.0 \cdot 7.351 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 18.016 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 242.48(1.0 \cdot 11.0 - 1.0 \cdot 7.351) - 2 \cdot 9.0$$

$$(1.0 \cdot 11.0 - 1.0 \cdot 7.351) \cdot (1 - 0.855)$$



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$$= 8.753 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 27.5) \cdot (9.0 - 3.34) \cdot 0.8551$$

$$= 2.661 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 12.0^2 \cdot 1.0$$

$$= 2.295 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (450.0 - 260.48) \cdot 14.0 \cdot 1.0$$

$$= 26.533 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 6.3425 \text{ mm}$

Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb1 = 10.3510 \text{ mm}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 10.3510 \text{ mm}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb3 = 11.1026 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.103, \max(10.351, 4.5)]$$

$$= 10.3510 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(6.3425, 10.351)$$

$$= 10.3510 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	46.8,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	248.2 N./mm ²	Passed
Occasional	:	20.8,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	20.2,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $tg = 12.0$, $tr = 3.343$, $c = 3.0 \text{ mm}$, $E* = 1.0$

Thickness Ratio = $tr \cdot E*/(tg - c) = 0.371$, Temp. Reduction = $62 \text{ }^{\circ}\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $-3 \text{ }^{\circ}\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-48 \text{ }^{\circ}\text{C}$

Min Metal Temp. w/o impact per UG-20(f) $-29 \text{ }^{\circ}\text{C}$

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-46 \text{ }^{\circ}\text{C}$



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Nozzle Calcs.: N1-Tube Outlet

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Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 3.343$, $c = 3.0$ mm, $E* = 1.0$
 Thickness Ratio = $t_r * E* / (t_g - c) = 0.371$, Temp. Reduction = 62 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 14.0$, $t_r = 7.351$, $c = 3.0$ mm, $E* = 1.0$
 Thickness Ratio = $t_r * E* / (t_g - c) = 0.668$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 3.343$, $c = 3.0$ mm, $E* = 1.0$
 Thickness Ratio = $t_r * E* / (t_g - c) = 0.371$, Temp. Reduction = 62 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-46 °C
Gov. MDMT of the Reinforcement Pad	:	-37 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3200.00 / 5100.00 = 0.627$

Weld Size Calculations, Description: N1-Tube Outlet

Intermediate Calc. for nozzle/shell Welds	T_{min}	9.0000 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$5.5000 = 0.5 * T_{minPad}$	$8.4840 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (18.0163 - 8.753 + 2 \cdot 9.0 \cdot 0.8551 \cdot (1.0 \cdot 11.0 - 7.351)) \cdot 138)$
 $= 13825.26 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_r2) \cdot S_v$
 $= (2.6607 + 26.5328 + 2.2951 - 0.0 \cdot 0.86) \cdot 138$
 $= 44309.32 \text{ Kgf}$



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Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (2.6607 + 0.0 + 0.8551 + (1.693)) \cdot 138$$

$$= 7329.57 \text{ Kg}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (2.6607 + 0.0 + 2.2951 + 26.5328 + (1.693)) \cdot 138$$

$$= 46691.70 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 260.48 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 24123. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 450.0 \cdot 12.0 \cdot 0.49 \cdot 138$$

$$= 58486. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 125.74) \cdot (12.0 - 3.0) \cdot 0.7 \cdot 118$$

$$= 29944. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2.0) \cdot 260.48 \cdot 14.0 \cdot 0.74 \cdot 138$$

$$= 59648. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 260.48 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 46866. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (58486 + 29944) = 88430 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (24123 + 59648 + 46866 + 0) = 130638 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (58486 + 46866 + 0) = 105352 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 88430	Kgf, must exceed W = 13825 Kg	or W1 = 44309 Kg
Path 2-2 = 130637	Kgf, must exceed W = 13825 Kg	or W2 = 7329 Kg
Path 3-3 = 105352	Kgf, must exceed W = 13825 Kg	or W3 = 46691 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4750 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 28.7294 mm



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The Cut Length for this Nozzle is, Drop + Ho + H + T : 192.7294 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 242.48 mm

Input Echo, WRC297 Item 1, Description: N1-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	619.150	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	236.480	mm
Nozzle Thickness	12.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	450.000	mm
Thickness of Reinforcing Pad	14.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	3200.00	KPa.
Radial Load	P	-1912.10	Kgf
Circumferential Shear	Vc	1274.73	Kgf
Longitudinal Shear	Vl	1274.73	Kgf
Circumferential Moment	Mc	891.97	Kg-m.
Longitudinal Moment	Ml	891.97	Kg-m.
Torsional Moment	Mt	1274.24	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)} \\
 &= 260.48 + 2 \cdot 1.65 \cdot \sqrt{325.075(14.0 - 3.0)} \\
 &= 457.814 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	650.150	mm
Nozzle Outside Diameter	(d) =	260.480	mm
Vessel Thickness used	(T) =	25.000	mm



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Nozzle Calcs.: N1-Tube Outlet Nozl: 10 10:20am Sep 9,2026

Nozzle Thickness used (t) = 9.000 mm
 T / t = 2.778
 d / t = 28.942

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.043$$

Nr/P = 0.081
 Mr/P = 0.071
 M0/P = 0.062
 N0/P = 0.188
 MrD/Mc = 0.158
 NrDL/Mc = 0.108
 M0d/Mc = 0.159
 N0DL/Mc = 0.194
 MrD/Ml = 0.097
 NrDL/Ml = 0.072
 M0D/Ml = 0.097
 N0DL/Ml = 0.318

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-17	-17	17	17
Outplane Bending (ML)	-31	31	31	-31
Normal Pressure Stress	38	41	38	41
Outplane Stress Summary	7	50	103	22

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-31	31	31	-31
Inplane Pressure Stress	19	19	19	19
Inplane Stress Summary	0	36	69	-18

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	14	52	103	41
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-51	51	51	-51
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	19	19	19	19
Outplane Stress Summary	-25	54	97	-27

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-51	51	51	-51
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	38	41	38	41
Inplane Stress Summary	-3	76	110	-13

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	26	77	112	29

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-17	-17	17	17
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	49	53	49	53
Hoop Stress Summary	38	42	72	76



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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	42	-42	42	-42
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-151	151	151	-151
Axial Pressure Stress	24	24	24	24
Axial Stress Summary	-102	116	241	-146

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	13	13	13	13
Shear Stress Summary	16	16	9	9

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	144	120	242	223

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-10	-10	10	10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	49	53	49	53
Hoop Stress Summary	45	48	65	69

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	42	-42	42	-42
Axial Membrane (Mc)	-20	-20	20	20
Axial Bending (Mc)	-259	259	259	-259
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	24	24	24	24
Axial Stress Summary	-209	224	349	-253

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside



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Nozzle Calcs.: N1-Tube Outlet Nozl: 10 10:20am Sep 9, 2026

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	13	13	13	13

Shear Stress Summary	9	9	16	16
----------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	255	224	350	324

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 260.48 + 2 \cdot 1.65 \cdot \sqrt{318.075(14.0 - 3.0)}$$

$$= 455.678 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	636.150 mm
Nozzle Outside Diameter	(d) =	450.000 mm
Vessel Thickness used	(T) =	11.000 mm
Nozzle Thickness used	(t) =	103.760 mm
T / t	=	0.106
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.379$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.044
Mr/P	=	0.070
M0/P	=	0.020
N0/P	=	0.031
MrD/Mc	=	0.240
NrDL/Mc	=	0.094
M0d/Mc	=	0.071
N0DL/Mc	=	0.058
MrD/Ml	=	0.049
NrDL/Ml	=	0.043
M0D/Ml	=	0.015
N0DL/Ml	=	0.028

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	89	92	89	92
Outplane Stress Summary	93	88	131	68

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside



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Nozzle Calcs.: N1-Tube Outlet

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Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	65	-65	65	-65
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-46	46	46	-46
Inplane Pressure Stress	44	44	44	44

Inplane Stress Summary	62	26	170	-53
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	5	5	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Two * Max Shear Stress	94	89	170	122
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-68	68	68	-68
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	44	44	44	44

Outplane Stress Summary	-9	90	145	-28
-------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	65	-65	65	-65
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-231	231	231	-231
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	89	92	89	92

Inplane Stress Summary	-85	250	408	-182
------------------------	-----	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Shear (Vc)	0	0	0	0
---------------------	---	---	---	---



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Nozzle Calcs.: N1-Tube Outlet Nozl: 10 10:20am Sep 9,2026

Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	1	1	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	85	250	408	182

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	38	41	38	41	38	41	38	41
Circ. Pl (SUS)	-12	-12	22	22	-3	-3	7	7
Circ. Q (SUS)	-20	20	42	-42	-39	39	63	-63
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	-1	-1	5	5	-5	-5	15	15
Long. Q (SUS)	-19	19	43	-43	-40	40	62	-62
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	38.0	41.0	38.0	41.0	38.0	41.0	38.0	41.0
Pm+Pl (SUS)	26.1	29.1	60.0	63.0	35.0	38.0	45.1	48.1
Pm+Pl+Q (Total)	12.2	50.8	102.3	40.4	26.4	77.4	109.8	29.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.00	138.00	Passed
Pm+Pl (SUS)	63.03	207.00	Passed
Pm+Pl+Q (TOTAL)	109.81	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	89	92	89	92	89	92	89	92
Circ. Pl (SUS)	0	0	8	8	-9	-9	21	21
Circ. Q (SUS)	4	-4	32	-32	-166	166	296	-296
Long. Pm (SUS)	44	44	44	44	44	44	44	44
Long. Pl (SUS)	0	0	12	12	-5	-5	13	13
Long. Q (SUS)	19	-19	111	-111	-50	50	86	-86



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Nozzle Calcs.: N1-Tube Outlet

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Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	89.0	92.0	89.0	92.0	89.0	92.0	89.0	92.0
Pm+P1 (SUS)	89.0	92.0	97.0	100.0	80.0	83.0	110.0	113.0
Pm+P1+Q (Total)	93.5	88.3	167.1	123.1	86.1	249.0	406.1	183.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	92.00	138.00	Passed
Pm+P1 (SUS)	113.02	207.00	Passed
Pm+P1+Q (TOTAL)	406.06	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	49	53	49	53	49	53	49	53
Circ. P1 (SUS)	-12	-12	22	22	-5	-5	15	15
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	24	24	24	24	24	24	24	24
Long. P1 (SUS)	-18	-18	22	22	-18	-18	22	22
Long. Q (SUS)	-109	109	193	-193	-217	217	301	-301
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	13	13	13	13	13	13	13	13
Pm (SUS)	49.0	53.0	49.0	53.0	49.0	53.0	49.0	53.0
Pm+P1 (SUS)	37.3	41.3	71.4	75.3	44.2	48.2	64.5	68.4
Pm+P1+Q (Total)	143.6	118.3	239.6	222.9	255.8	223.6	347.9	324.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	53.00	118.00	Passed
Pm+P1 (SUS)	75.31	177.00	Passed
Pm+P1+Q (TOTAL)	347.90	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



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Nozzle Calcs.: N3-Shell inlet

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Input, Nozzle Desc: N3-Shell inlet

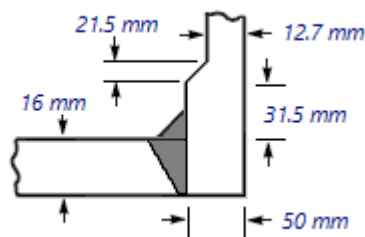
From: 50

Pressure for Reinforcement Calculations	P	4150.000	KPa.
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	619.15	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2003.98	mm
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		247.6000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.7000	mm
Hub Height of Integral Nozzle	h	31.5352	mm
Height of Beveled Transition	L'	21.5352	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N3-Shell inlet



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Nozzle Calcs.: N3-Shell inlet Nozl: 11 10:20am Sep 9,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	247.600 mm.
Actual Thickness Used in Calculation	12.700 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4150.0 \cdot 312.575 / (138 \cdot 1.0 - 0.6 \cdot 4150.0)$
 $= 9.5734 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4150.0 \cdot 126.8 / (138 \cdot 1.0 - 0.6 \cdot 4150.0)$
 $= 3.8836 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 37.3 / 0.5773$
 $= 64.6055 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	507.2000	mm
Parallel to Vessel Wall, opening length	d	253.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	32.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



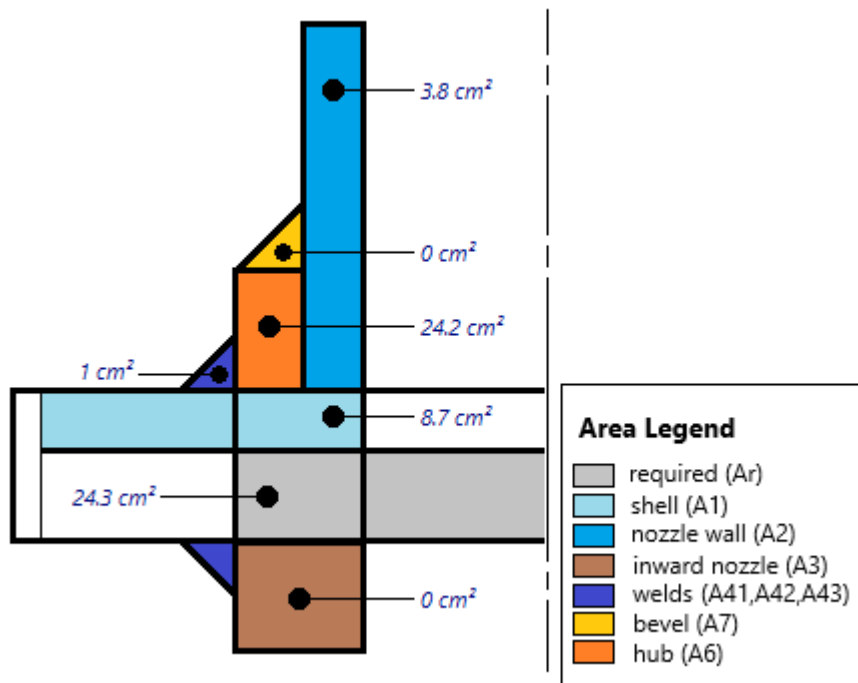
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Nozzle Calcs.: N3-Shell inlet

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	24.278	NA	NA
Area in Shell	A1	8.690	NA	NA
Area in Nozzle Wall	A2	3.781	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	24.229	NA	NA
TOTAL AREA AVAILABLE	Atot	37.700	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (253.6 \cdot 9.5734 \cdot 1.0 + 2 \cdot 47.0 \cdot 9.5734 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 24.278 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 253.6(1.0 \cdot 13.0 - 1.0 \cdot 9.573) - 2 \cdot 47.0$$

$$(1.0 \cdot 13.0 - 1.0 \cdot 9.5734) \cdot (1 - 1.0)$$

$$= 8.690 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 32.5)(9.7 - 3.88)1.0$$

$$= 3.781 \text{ cm}^2$$



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Nozzle Calcs.: N3-Shell inlet

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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(32.5, 250.0, 31.5)) \cdot (50.0 - 12.7) \cdot 1.0$$

$$= 24.229 \text{ cm}^2 \text{ Includes Bevel Area of } 0.704 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 6.8836 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 12.5734 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 12.5734 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.103, \max(12.5734, 4.5)]$$

$$= 11.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(6.8836, 11.1026)$$

$$= 11.1026 \text{ mm}$$

Available Nozzle Neck Thickness = 12.7000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 48.0,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 297.0 N./mm ²	Passed
Occasional	: 26.1,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 17.3,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govern. thk, tg = 16.0, tr = 9.573, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.736, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-29 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: N3-Shell inlet

Intermediate Calc. for nozzle/shell welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:



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Nozzle Calcs.: N3-Shell inlet Nozl: 11 10:20am Sep 9, 2026

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (24.2781 - 8.6899 + 2 \cdot 47.0 \cdot 1.0 \cdot (1.0 \cdot 13.0 - 9.5734)) \cdot 138) \\
 &= 26467.42 \text{ Kgf}
 \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2 + A4 - (W_i - \text{Can} / 1.707)^2 \cdot f_{r2}) \cdot S_v \\
 &= (28.0096 + 1. - 0.0 \cdot 1.0) \cdot 138 \\
 &= 40821.02 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\
 &= (28.0096 + 0.0 + 1. + (12.22)) \cdot 138 \\
 &= 58016.48 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\
 &= (28.0096 + 0.0 + 1. + 0.0 + (12.22)) \cdot 138 \\
 &= 58016.48 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 347.6 \cdot 10.0 \cdot 0.49 \cdot 138 \\
 &= 37648. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 150.3) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 138 \\
 &= 218599. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 347.6 \cdot (16.0 - 3.0) \cdot 0.74 \cdot 138 \\
 &= 73912. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (37648 + 218599) = 256246 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (37648 + 0 + 73912 + 0) = 111560 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Sonw + Tngw + Sinw) \\
 &= (37648 + 73912 + 0) = 111560 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 256246 Kgf, must exceed W = 26467 Kgf	or W1 = 40821 Kgf
Path 2-2 = 111560 Kgf, must exceed W = 26467 Kgf	or W2 = 58016 Kgf
Path 3-3 = 111560 Kgf, must exceed W = 26467 Kgf	or W3 = 58016 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5216 KPa.

The Drop for this Nozzle is : 53.3910 mm



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Nozzle Calcs.: N3-Shell inlet

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The Cut Length for this Nozzle is, Drop + Ho + H + T : 319.3910 mm

Input Echo, WRC297 Item 1, Description: N3-Shell inlet:

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	619.150	mm	
Shell Thickness	16.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	247.600	mm	
Nozzle Thickness	50.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-765 II		
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4150.00	KPa.
Radial Load	P	-1912.10	Kgf
Circumferential Shear	Vc	1274.73	Kgf
Longitudinal Shear	Vl	1274.73	Kgf
Circumferential Moment	Mc	891.97	Kg-m.
Longitudinal Moment	Ml	891.97	Kg-m.
Torsional Moment	Mt	1274.24	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.534) is > 0.5
check the limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (6.952) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 347.6 + 2 \cdot 1.65 \cdot \sqrt{319.075(16.0 - 3.0)}$$

$$= 560.136 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	638.150	mm
Nozzle Outside Diameter	(d) =	347.600	mm
Vessel Thickness used	(T) =	13.000	mm
Nozzle Thickness used	(t) =	47.000	mm
T / t	=	0.277	
d / t	=	10.000	



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Nozzle Calcs.: N3-Shell inlet

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$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.816$$

Nr/P	=	0.053
Mr/P	=	0.085
M0/P	=	0.028
N0/P	=	0.062
MrD/Mc	=	0.257
NrDL/Mc	=	0.092
M0d/Mc	=	0.077
N0DL/Mc	=	0.085
MrD/Ml	=	0.074
NrDL/Ml	=	0.049
M0D/Ml	=	0.022
N0DL/Ml	=	0.067

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	97	101	97	101
Outplane Stress Summary	93	100	152	80

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	56	-56	56	-56
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-65	65	65	-65
Inplane Pressure Stress	48	48	48	48
Inplane Stress Summary	37	56	184	-60

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	94	101	184	141

Vessel Stresses (N./mm²):



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Nozzle Calcs.: N3-Shell inlet

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CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-68	68	68	-68
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	48	48	48	48

Outplane Stress Summary	-7	93	155	-18
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	56	-56	56	-56
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-229	229	229	-229
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	97	101	97	101

Inplane Stress Summary	-83	267	403	-164
------------------------	-----	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	3	3	6	6
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	83	267	403	165

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	16	11	16

Hoop Stress Summary	8	13	28	33
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
---	-----------	-----------	--------------	--------------



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Nozzle Calcs.: N3-Shell inlet

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pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	2	3	15	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	8	13	28	33

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
hoop direction)				
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	16	11	16
Hoop Stress Summary	5	10	30	35

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
pipe cross-section)				
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-8	14	26	-8

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1



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Shear Stress Summary 0 0 1 1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	14	14	31	44

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	97	101	97	101	97	101	97	101
Circ. Pl (SUS)	-3	-3	15	15	-8	-8	18	18
Circ. Q (SUS)	-1	1	37	-37	-173	173	285	-285
Long. Pm (SUS)	48	48	48	48	48	48	48	48
Long. Pl (SUS)	-2	-2	12	12	-6	-6	18	18
Long. Q (SUS)	-9	9	121	-121	-50	50	86	-86
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	97.0	101.0	97.0	101.0	97.0	101.0	97.0	101.0
Pm+Pl (SUS)	94.0	98.0	112.0	116.0	89.0	93.0	115.0	119.0
Pm+Pl+Q (Total)	93.6	99.8	181.5	140.2	84.2	266.1	400.1	166.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	101.00	138.00	Passed
Pm+Pl (SUS)	119.02	207.00	Passed
Pm+Pl+Q (TOTAL)	400.15	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	16	11	16	11	16	11	16
Circ. Pl (SUS)	-3	-3	15	15	-6	-6	18	18
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-1	1	5	-5	-12	12	16	-16
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0



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Nozzle Calcs.: N3-Shell inlet

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Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	11.0	16.0	11.0	16.0	11.0	16.0	11.0	16.0
Pm+Pl (SUS)	8.0	13.0	26.0	31.0	5.0	10.0	29.0	34.0
Pm+Pl+Q (Total)	8.2	13.1	26.1	31.0	14.1	15.2	29.2	43.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16.00	138.00	Passed
Pm+Pl (SUS)	34.00	207.00	Passed
Pm+Pl+Q (TOTAL)	43.05	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: N4-Shell outlet

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Input, Nozzle Desc: N4-Shell outlet

From: 50

Pressure for Reinforcement Calculations	P	4156.067	KPa.
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	619.15	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2003.98	mm
User Entered Minimum Design Metal Temperature		-45.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		247.6000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.7000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	31.5352	mm
Height of Beveled Transition	L'	21.5352	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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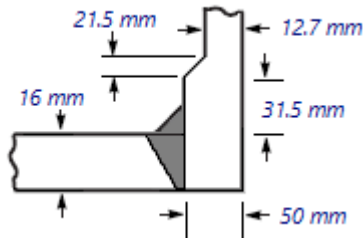
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Nozzle Calcs.: N4-Shell outlet

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Reinforcement CALCULATION, Description: N4-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	247.600 mm.
Actual Thickness Used in Calculation	12.700 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4156.07 \cdot 312.575 / (138 \cdot 1.0 - 0.6 \cdot 4156.07)$
 $= 9.5876 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4156.07 \cdot 126.8 / (138 \cdot 1.0 - 0.6 \cdot 4156.07)$
 $= 3.8893 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 37.3 / 0.5773$
 $= 64.6055 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	507.2000 mm
Parallel to Vessel Wall, opening length	d	253.6000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



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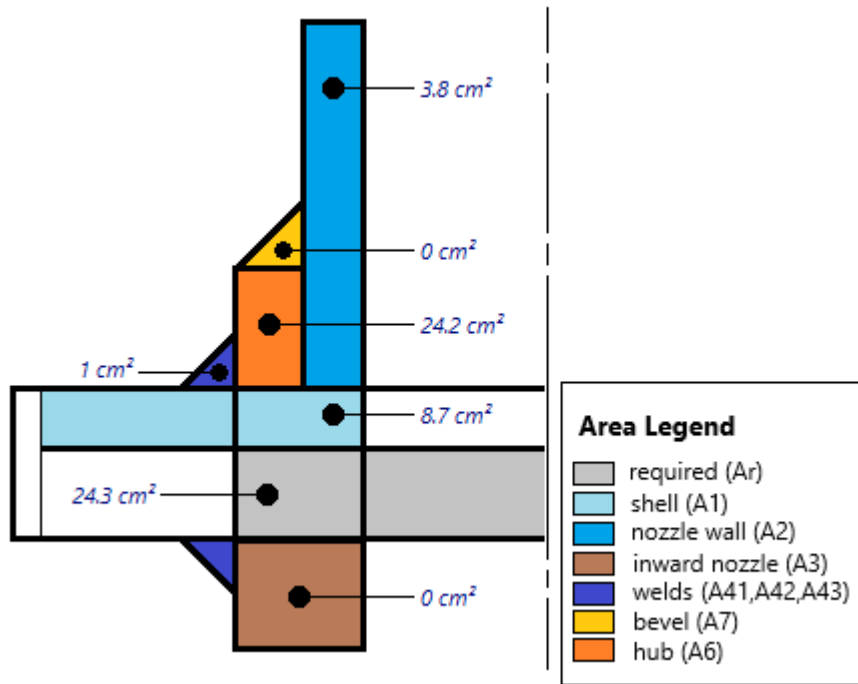
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Nozzle Calcs.: N4-Shell outlet

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	24.314	NA	NA
Area in Shell	A1	8.654	NA	NA
Area in Nozzle Wall	A2	3.777	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	24.229	NA	NA
TOTAL AREA AVAILABLE	Atot	37.660	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (253.6 \cdot 9.5876 \cdot 1.0 + 2 \cdot 47.0 \cdot 9.5876 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 24.314 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 253.6(1.0 \cdot 13.0 - 1.0 \cdot 9.588) - 2 \cdot 47.0$$

$$(1.0 \cdot 13.0 - 1.0 \cdot 9.5876) \cdot (1 - 1.0)$$

$$= 8.654 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 32.5)(9.7 - 3.89)1.0$$

$$= 3.777 \text{ cm}^2$$



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Nozzle Calcs.: N4-Shell outlet Nozl: 12 10:20am Sep 9, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(32.5, 150.0, 31.5)) \cdot (50.0 - 12.7) \cdot 1.0$$

$$= 24.229 \text{ cm}^2 \text{ Includes Bevel Area of } 0.704 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 6.8893 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 12.5876 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 12.5876 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.103, \max(12.5876, 4.5)]$$

$$= 11.1026 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(6.8893, 11.1026)$$

$$= 11.1026 \text{ mm}$$

Available Nozzle Neck Thickness = 12.7000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 48.1,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 296.9 N./mm ²	Passed
Occasional	: 26.2,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 17.3,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 12.7, tr = 3.889, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.401, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 16.0, tr = 9.588, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.738, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-29 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Nozzle Calcs.: N4-Shell outlet Nozl: 12 10:20am Sep 9, 2026

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -28 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 4156.07/5100.00 = 0.815

Weld Size Calculations, Description: N4-Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (24.3142 - 8.6538 + 2 \cdot 47.0 \cdot 1.0 \cdot (1.0 \cdot 13.0 - 9.5876)) \cdot 138)$
 $= 26550.30 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (28.0058 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 40815.73 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (28.0058 + 0.0 + 1. + (12.22)) \cdot 138$
 $= 58011.19 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (28.0058 + 0.0 + 1. + 0.0 + (12.22)) \cdot 138$
 $= 58011.19 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 347.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 37648. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 150.3) \cdot (50.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 218599. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 347.6 \cdot (16.0 - 3.0) \cdot 0.74 \cdot 138$
 $= 73912. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (37648 + 218599) = 256246 \text{ Kgf}$



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Nozzle Calcs.: N4-Shell outlet Nozl: 12 10:20am Sep 9,2026

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (37648 + 0 + 73912 + 0) = 111560 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (37648 + 73912 + 0) = 111560 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 256246 Kg, must exceed W = 26550 Kg or W1 = 40815 Kg
 Path 2-2 = 111560 Kg, must exceed W = 26550 Kg or W2 = 58011 Kg
 Path 3-3 = 111560 Kg, must exceed W = 26550 Kg or W3 = 58011 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5210 KPa.

The Drop for this Nozzle is : 53.3910 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.3910 mm

Input Echo, WRC297 Item 1, Description: N4-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	619.150	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	247.600	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-765 II	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4156.07	KPa.
Radial Load	P	-1912.10	Kgf
Circumferential Shear	Vc	1274.73	Kgf
Longitudinal Shear	Vl	1274.73	Kgf
Circumferential Moment	Mc	891.97	Kg-m.
Longitudinal Moment	Ml	891.97	Kg-m.
Torsional Moment	Mt	1274.24	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.534) is > 0.5
 check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (6.952) is < 10



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Nozzle Calcs.: N4-Shell outlet

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check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 347.6 + 2 \cdot 1.65 \cdot \sqrt{(319.075(16.0 - 3.0))}$$

$$= 560.136 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	638.150 mm
Nozzle Outside Diameter	(d) =	347.600 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	47.000 mm
T / t	=	0.277
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.816$$

Nr/P	=	0.053
Mr/P	=	0.085
M0/P	=	0.028
N0/P	=	0.062
MrD/Mc	=	0.257
NrDL/Mc	=	0.092
M0d/Mc	=	0.077
N0DL/Mc	=	0.085
MrD/M1	=	0.074
NrDL/M1	=	0.049
M0D/M1	=	0.022
N0DL/M1	=	0.067

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	97	102	97	102
Outplane Stress Summary	93	100	152	81

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	56	-56	56	-56
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-65	65	65	-65
Inplane Pressure Stress	48	48	48	48
Inplane Stress Summary	38	56	184	-59

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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Nozzle Calcs.: N4-Shell outlet Nozl: 12 10:20am Sep 9,2026

(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			5	5	5	5
Shear Stress Summary			6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress			94	101	184	141

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)			6	6	6	6
Outplane Bending (P)			18	-18	18	-18
Outplane Membrane (Mc)			-12	-12	12	12
Outplane Bending (Mc)			-68	68	68	-68
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			48	48	48	48
Outplane Stress Summary			-7	93	155	-18

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)			5	5	5	5
Inplane Bending (P)			56	-56	56	-56
Inplane Membrane (Mc)			-13	-13	13	13
Inplane Bending (Mc)			-229	229	229	-229
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			97	102	97	102
Inplane Stress Summary			-83	267	403	-164

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			5	5	5	5
Shear Stress Summary			3	3	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			83	267	403	164

Nozzle Stresses (N./mm²):



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Nozzle Calcs.: N4-Shell outlet Nozl: 12 10:20am Sep 9,2026

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	16	11	16
Hoop Stress Summary	8	13	28	33

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	2	3	15	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	8	13	28	33

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	16	11	16
Hoop Stress Summary	5	10	30	35

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Nozzle Calcs.: N4-Shell outlet Nozl: 12 10:20am Sep 9,2026

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-8	14	26	-8

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	14	14	31	44

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	97	102	97	102	97	102	97	102
Circ. Pl (SUS)	-3	-3	15	15	-8	-8	18	18
Circ. Q (SUS)	-1	1	37	-37	-173	173	285	-285
Long. Pm (SUS)	48	48	48	48	48	48	48	48
Long. Pl (SUS)	-2	-2	12	12	-6	-6	18	18
Long. Q (SUS)	-9	9	121	-121	-50	50	86	-86
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	97.0	102.0	97.0	102.0	97.0	102.0	97.0	102.0
Pm+Pl (SUS)	94.0	99.0	112.0	117.0	89.0	94.0	115.0	120.0
Pm+Pl+Q (Total)	93.6	100.8	181.5	141.2	84.2	267.1	400.1	165.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	102.00	138.00	Passed
Pm+Pl (SUS)	120.02	207.00	Passed
Pm+Pl+Q (TOTAL)	400.15	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:



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Nozzle Calcs.: N4-Shell outlet

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Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	16	11	16	11	16	11	16
Circ. Pl (SUS)	-3	-3	15	15	-6	-6	18	18
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-1	1	5	-5	-12	12	16	-16
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	11.0	16.0	11.0	16.0	11.0	16.0	11.0	16.0
Pm+Pl (SUS)	8.0	13.0	26.0	31.0	5.0	10.0	29.0	34.0
Pm+Pl+Q (Total)	8.2	13.1	26.1	31.0	14.1	15.2	29.2	43.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16.00	138.00	Passed
Pm+Pl (SUS)	34.00	207.00	Passed
Pm+Pl+Q (TOTAL)	43.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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ASME TS Calc:

Case: 1 10:20a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: U-Tube

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	4150.00 KPa.
Shell Maximum Operating Pressure	Psox,max	4150.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	16.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	619.150 mm
Shell Temperature for Internal Pressure	Ts	82.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	3200.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	3200.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	103.42 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	619.150 mm
Channel Design Temperature	TEMPC	230.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	302
Tube Wall Thickness	et	2.3200 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	1830.00 mm
Straight Tube Length (bet. inner tubsht faces) L		1740.00 mm
Design Temperature of the Tubes		230.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		108.00 N./mm ²
Tube Allowable Stress At Ambient		108.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	173.20 N./mm ²
Tube Pitch (Center to Center Spacing)	P	28.5750 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.3200 mm
Groove Weld Leg	ag	2.3200 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	284.072 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm



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ASME TS Calc:

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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	2089.6 cm ²
Perimeter of Tube Layout	Cp	1714.3 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature	T	230.00 °C
Tubesheet Material		SA-765 II

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03047
Tubesheet Allowable Stress at Temperature	S	136.80 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	90.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	8.0000 mm
Tubesheet Outside Diameter	A	800.000 mm
Dimension G for the Channel Side	Gc	664.301 mm
Area of the Untubed Lanes	AL	0.0 cm ²
Junction Stress Reduction option		Perform Plastic Calculation
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Channel Side Min. Design Pressure	Ptd,min	103.4195 KPa.

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	800.000 mm
Flange Inside Diameter	B	619.150 mm
Flange Face Outside Diameter	Fod	688.150 mm
Flange Face Inside Diameter	Fid	619.150 mm
Gasket Outside Diameter	Go	677.150 mm
Gasket Inside Diameter	Gi	651.150 mm
Small end Hub thk.	g0	16.0000 mm
Large end Hub thk.	g1	24.0000 mm
Gasket Factor,	m	2.75
Gasket Design Seating Stress	y	25.51 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	732.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	2.7500
Partition Gasket Design Seating Stress	yPart	25.51 N./mm ²
Partition Gasket Facing Sketch		Code Sketch 1a
Partition Gasket Column for Gasket Seating		Code Column II

Bolting Information:

Diameter of Bolt Circle	C	740.000 mm
Nominal Bolt Diameter	dB	25.4000 mm
Thread Series		TEMA Thread Series
Number of Bolts	n	40
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Alternate Flange Operating Bolt Load, Wm1	186712.89 Kg
Alternate Flange Seating Bolt Load, Wm2	103789.05 Kg
Alternate Flange Design Bolt Load, W	249385.36 Kg



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ASME TS Calc:

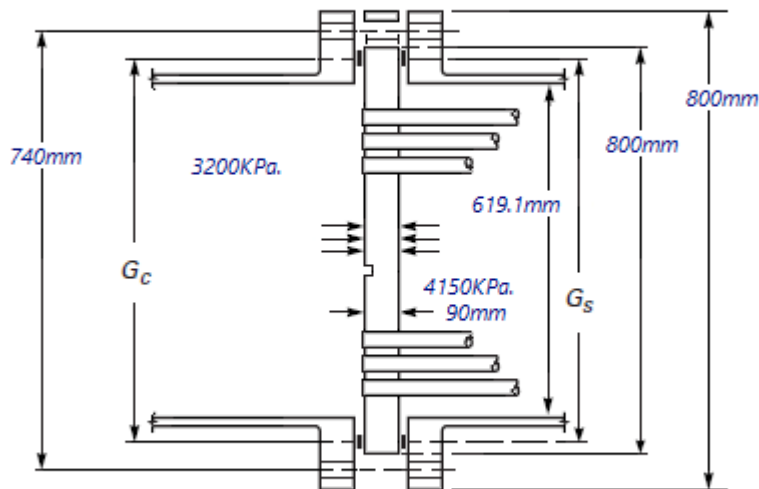
Case: 1 10:20a Sep 9, 2026

Note: The full bolt load $S_a \cdot A_b$, was used for all load cases.

Tubesheet Integral with
Tubesheet Extended as Flange

None
No

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with $C \leq 0.3\%$

Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with $C \leq 0.3\%$

Elastic Mod. at Design Temperature	230.0 °C	0.19046E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with $C \leq 0.3\%$

Elastic Mod. at Tubesht. Design Temp.	230.0 °C	0.19046E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with $C \leq 0.3\%$

Elastic Mod. at Design Temperature	230.0 °C	0.19046E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$\begin{aligned} &= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)} \\ &= (3200.0 \cdot (19.05/2 - 0)) / (108.0 \cdot 1.0 + 0.4 \cdot 3200.0) \\ &= 0.2789 + 0.0000 = 0.2789 \text{ mm} \end{aligned}$$



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ASME TS Calc:

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Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 230.00 °C
Elastic Modulus for Material 189879744.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.3200	19.05	2024.07	8.21	50.0000	0.0163147	113.40

EMAWP = $(2.167 / (D/T) - 0.0833) * B = 20479.4531$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.7377	19.05	2024.07	25.82	50.0000	0.0016494	82.40

EMAWP = $(4 * B) / (3 * (D/T)) = (4 * 82.3965) / (3 * 25.825) = 4253.853$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.7377 mm
Allowable Internal Pressure at Corroded thickness	29142.83 KPa.
Required Internal Design Pressure	3200.00 KPa.
Allowable External Pressure at Corroded thickness	20479.45 KPa.
Required External Design Pressure	4253.42 KPa.
Required Thickness due to Shell Side pressure	0.7377 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

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Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	664.306 mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 * ro + dt$
 $= 2 * 284.072 + 19.05$
 $= 587.194$ mm

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (28.575 - 19.05) / 28.575$
 $= 0.333$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$= Gs / Do$ (Configurations d, e, f)
 $= 664.3056 / 587.194$
 $= 1.1313$

Ratio [pc]:

$= Gc / Do$ (Configurations d)
 $= 664.3009 / 587.194$
 $= 1.1313$



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ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/90.0$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2t(Et/E) \cdot (St/S) \cdot (p); dt - 2t)$$

$$= \max(19.05 - 2 \cdot 2.32 (0.19046E+09/19046E+09) \cdot$$

$$(107/136) \cdot (0); 19.05 - 2 \cdot 2.32)$$

$$= 19.0500 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 28.575 / \sqrt{ (1 - 4 \cdot \min(0 ; 4 \cdot 587.194 \cdot 28.575) / (\pi \cdot 587.194^2)) }$$

$$= 28.5750 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (28.575 - 19.05) / 28.575$$

$$= 0.3333$$

$$h/p = 3.149606 ; \mu^* = 0.333333 ; E^* = 61916224 \text{ KPa.}$$

$$E^*/E = 0.325091 ; v^* = 0.334167$$

Note: As h/p (3.150) is > 2, data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / Do$$

$$= 800.0/587.194$$

$$= 1.3624$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (E \cdot \ln(K))$$

$$= (1 - 0.33) / 61916224 \cdot (0.19046E+09 \cdot \ln(1.36))$$

$$= 0.6334$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$$

$$= 587.194^2/16 \cdot [(1.131 - 1) \cdot (1.131^2 + 1) \cdot 4150.0 -$$

$$(1.131 - 1) \cdot (1.131^2 + 1) \cdot 3200.0]$$

$$= 6130836.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + W^* \cdot (Gc - Gs) / (2 \cdot \pi \cdot Do)$$

$$= 6130836 + 249385.4 \cdot (664.301 - 664.301) / (2 \cdot \pi \cdot 587.194)$$

$$= 6130836.0000 \text{ KPa} \cdot \text{mm}^2$$

Note: W^* is the maximum of the bolt loads between the shell and channel sides.

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M^*) - Do^2/32 \cdot F (Ps - Pt)) / (1 + F)$$

$$= ((6130836) - 587.194^2/32 \cdot 0.633 (4150.0 - 3200.0)) / (1 + 0.63)$$

$$= -215964.9062 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:



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$$\begin{aligned}
 &= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t) \\
 &= -215964.91 + 587.194^2/64 \cdot (3 + 0.334) \cdot (4150.0 - 3200.0) \\
 &= 16848560.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned}
 &= \max(|M_p|; |M_o|) \\
 &= \max(|-215964.90|; |16848560|) \\
 &= 16848560.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\
 &= 6 \cdot 16848560 / (0.3333 \cdot (90.0 - 0)^2) \\
 &= 37.4434 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned}
 &= 2 \cdot S \\
 &= 2 \cdot 136.8 \\
 &= 273.60 \text{ N./mm}^2
 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\
 &= 6 \cdot 16926600 / (0.3333 \cdot (33.3719 - 0)^2) \\
 &= 273.5933 \text{ N./mm}^2
 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned}
 &= h + C_{ats} + C_{atc} \\
 &= 33.3719 + 0 + 0 \\
 &= 33.3719 \text{ mm}
 \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned}
 &= \max(H_{reqB}; H_{reqS}) \\
 &= \max(33.3719; 3.1744) \\
 &= 33.3719 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned}
 &= |P_s - P_t| \\
 &= |4150.0 - 3200.0| \\
 &= 950.000 \text{ KPa}
 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned}
 &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\
 &= 4 \cdot 0.333 \cdot 90.0 / 587.19 \cdot \min(0.8 \cdot 136.798; 0.533 \cdot 208.0) \\
 &= 22.365 \text{ N./mm}^2
 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned}
 &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\
 &= 1 / (4 \cdot 0.333 \cdot 109.439) \cdot (4 \cdot 2089.6 / 1714.3) \cdot |4150.0 - 3200.0| + 0 \\
 &= 3.1744 \text{ mm}
 \end{aligned}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [At]:

$$\begin{aligned}
 &= \pi (d_o - t) t \\
 &= \pi (19.05 - 2.32) 2.32 \\
 &= 1.219 \text{ cm}^2
 \end{aligned}$$

Tube Weld Size Results per UW-20.3:



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ASME TS Calc:

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Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.32 (19.05 - 2.32) 107.998$$

$$= 1342.830 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.32 (19.05 + 0.67 \cdot 2.32) 108.0$$

$$= 1342.8300 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 107.998 / 107.998$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.32 (19.05 + 0.67 \cdot 2.32) 108.0$$

$$= 909.5944 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.32 \cdot (19.05 - 2.32) 1.0 \cdot 1.0)} - 0.75 \cdot 19.05 \right)$$

$$= 2.772 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3920 mm

Minimum Required Groove Weld Leg [agr]: 1.3920 mm

Tube-Tubesheet Jt allowable, 1342.83 is >= tube strength 1342.83 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	37.4 <=	273.6 N./mm ²	Ok
Tubesheet shear stress	3.9 <=	109.4 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	33.372	90.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.392	2.320	Ok
Tube-Tubesheet Groove Weld Leg :	1.392	2.320	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses				Case Type	Pass/Fail
		Bend	Allwd	Shear	Allwd			
D1uc	61.105	4.760	126	274	13	109	Ps+Pt D1	Ok
D2uc	70.448	6.327	168	274	17	109	Ps+Pt D2	Ok
D3uc	33.372	1.413	37	274	4	109	Ps+Pt D3	Ok
D4uc	10.845	0.154	4	274	...	109	Ps+Pt D4	Ok
D1c	72.105	4.760	164	274	15	109	Ps+Pt-c D1	Ok
D2c	81.448	6.327	218	274	20	109	Ps+Pt-c D2	Ok



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D3c	44.372	1.413	49	274	4	109	Ps+Pt-c D3	Ok
D4c	21.845	0.154	5	274	...	109	Ps+Pt-c D4	Ok

Max: 81.4484 6.327 mm 0.795 0.180 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	81.448	90.000	Ok
Tube thickness :	0.738	2.320	Ok
Tube-tubesheet fillet weld leg :	1.392	2.320	Ok
Tube-tubesheet groove weld leg :	1.392	2.320	Ok

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5348.80	1.000	5348.72	1.000
Tubesheet Shear Stress	23641.94	1.000	23641.95	1.000
Tube Pressure Stress	29142.82	1.000	20479.44	1.000
Minimum MAWP	5348.80		5348.72	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	7002.90	1.000	7002.90	1.000
Tubesheet Shear Stress	27169.97	1.000	27169.96	1.000
Tube Pressure Stress	29142.82	1.000	22164.86	1.000
Minimum MAPnc	7002.90		7002.90	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure UCS-66.3 (c):

= tubesheet thickness/4

= 90.0/4

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ASME TS Calc:

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$$= 22.500 \text{ mm}$$

Thickness Ratio = 0.593, Temperature Reduction per Fig. UCS 66.1 = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -4 °C

Min Metal Temp. at Required thickness (UCS 66.1) -27 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:

$$= \max(\text{pt/Tubeside MAPnc}, \text{ps/Shellside MAPnc}), \text{ must be } \leq 1$$
$$= \max(3200.0/7002.9, 4150.0/7002.9)$$
$$= 0.593$$

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Problems/Failures Summary:

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: A standard flange seems to be governing the Shellside MAWP!**

**** Warning: A standard flange seems to be governing the Channelside MAWP!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: